

Operations research models for railway rolling stock planning

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Operations Research Models for Railway Rolling Stock Planning

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Operations Research Models for Railway Rolling Stock Planning

PROEFSCHRIFT

ter verkrijging van de graad van doctor aan de
Technische Universiteit Eindhoven, op gezag van de
Rector Magnificus, prof.dr.ir. C.J. van Duijn, voor een
commissie aangewezen door het College voor
Promoties in het openbaar te verdedigen
op woensdag 12 april 2006 om 16.00 uur

door

Gábor Maróti

geboren te Szombathely, Hongarije

Dit proefschrift is goedgekeurd door de promotoren:

prof.dr.ir. A.M.H. Gerards

en

prof.dr. L.G. Kroon

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The final words are addressed to my family; I am writing those in Hungarian.

Legvégül következnek azok, akiknek a legtöbb köszönettel tartozom: édesanyám, édesapám, testvérem, nagybátyám. Tiszta szívükből támogattak, amikor évekre Hollandiába költöztem, még ha fájt is kicsit a távolság. Hiányukat alig-alig pótolta a telefon és a rövidke hazai vakáció. Most, hogy elkészült ez a kis könyv, nekik ajánlom, és tudom: ők a legbüszkébbek és ők örülnek legjobban a világon. Remélem, ezzel törleszthetek valamit szeretetükből és törődésükből. Anyu, Apu, Zoli, Jóska: köszönöm.

Amstelveen, February 2006

Gábor Maróti

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Chapter 1

Introduction

More than hundred eighty years passed since that magnificent September day in 1825 when for the first time in history a steam-locomotive hauled a passenger train. The legendary *Locomotion* was driven by George Stephenson personally. Hundreds of enthusiastic people got in open coal waggons in Darlington and a great celebration started two hours later in Stockton as the train arrived there. This ride proved the concept of steam-hauled passenger railways. Within a couple of years, the horses were utterly released from their duties of pulling heavy waggons between Darlington and Stockton.

Railway transportation looked in those by-gone years quite differently from our understanding of railways today. There was no timetable at all; the numerous railway operators could run their trains whenever the trajectory was free; they literally fought for the right of using the tracks. There was no safety system; collision was only avoided by the low speed of the trains (and later, when they became faster, by sheer luck). The carriages had no springs; the passengers must have felt relieved after the 12-mile journey. Nevertheless, the Stockton and Darlington Railway was a financial success.

The pioneers of passenger railways would be quite astonished to see what their dreams have evolved into. Railways are now part of our everyday life. Trains operate according to carefully set-up timetables, safety has highest priority and comfortable carriages make long journeys easily bearable. Railways gained large social importance, too. Once stand-alone small railway lines grew to large companies and became a solid pillar of economy. For over a century, the development level of a country was directly measured by the density of its railway network. Till today, passenger

and freight railway transportation play an important role in the economy of many countries.

For many years, railway companies did not have to face much competition in public passenger and freight transportation. In the past decades, this changed drastically. The railways lost a large part of their market share to automobiles. Recently, air traffic took over many middle- and long-distance train travellers. In addition, a directive of the European Union required opening the national railway market in the 90's. Till then, most state-owned railway companies in Europe had been the only ones to provide railway services in their countries; after liberalising the market, they had to compete for the customers. These developments urge the railway companies to attract more customers by raising their service level and to cut their costs by working more efficiently. Improving their planning process contributes to reach both of these goals.

Railway companies are nearly inexhaustible sources of planning problems. Till recently, all of them have been dealt with manually; many are still handled without automation and optimisation. Railway applications attracted soon the attention of mathematical research. Many of the problems are of a combinatorial character and suitable for operations research methods. Conversely, problems of railway practise have an influence on operations research by showing interesting and useful directions to extend existing methods and to explore new ones. In the last decade, more and more computer-aided tools turned out to improve the railway planning process significantly. Besides intensive research, the virtually exponential increase of computational power contributes a lot to these successful applications. Nonetheless, comparing the number of existing operations-research-based planning tools to the plenty of railway problems indicates that there shall be enough railway-related research topics for a long time. Also, mathematical research shall certainly continue on railway optimisation topics that have not been addressed so far. In any case, it shall be exciting to see to what extent railway planning can be automated and optimised in the coming years and decades.

1.1 Topic of this Thesis

This thesis focuses on planning problems that arise at the major Dutch passenger railway operator *Nederlandse Spoorwegen* (NS). Therefore infrastructure management and freight railway transportation are not considered.

Planning the railways for years, months, weeks or days ahead leads to substantially different problems; in this regard railway planning problems can be *strategic*, *tactical*, *operational* and *short-term*.

Another way to classify railway planning problems is based on their target: they concern the *timetable*, the *rolling stock* and the *crew*. Timetabling answers the questions which locations are to be connected by direct trains and when the trains have to run. Rolling stock planning determines how many locomotives and passenger carriages are needed and how to use them for trains. Crew scheduling of most passenger railway operators concerns the questions how many train drivers and conductors are needed and how to assign them to the trains. These three topics include problems of very different characteristics. Timetabling determines the railway network while crew and rolling stock scheduling allocate the available resources. But the requirements in crew and rolling stock planning differ substantially, too. For example, rolling stock units are bound to the tracks so they can block the way through a station for each other. Crew has much more flexibility as train drivers and conductors can just walk from one platform to another. However, crew scheduling has to respect wishes of the employees, sometimes at the cost of efficiency. A well-known recent example of that is the case of the cost-efficient crew schedules of NS in 2001 which turned out to be unacceptable for train drivers and conductors.¹ The employees' wishes result in very complex requirements on the crew schedules; the rules in rolling stock planning are usually simpler.

Of the wide spectrum of railway planning problems, this thesis deals with rolling stock planning in the tactical, operational and short-term phase. Here we give a brief overview of rolling stock planning; for the sake of completeness we also include strategic planning.

Strategic planning determines the amount of rolling stock needed in the future. In the other planning stages, the available rolling stock is to be assigned to the timetable services. Tactical, operational and short-term planning take different levels of detail of reality into account, so the requirements that the schedules have to fulfil differ for the planning phases. Tactical planning produces the basic shape of the weekly schedule, operational planning refines this for the actual calendar weeks. Short-term planning modifies these schedules to comply with some requirements that are not dealt with in earlier planning phases. Moreover, short-term planning supervises the execution of the schedules.

¹This example is commonly referred to as 'rondje om de kerk' (around the church) since in the cost-efficient schedules the daily workload of the employees often contains only trains on a single trajectory back and forth.

The three major objectives in rolling stock planning are *service quality*, *operational costs* and *robustness*. A good service quality means that trains have enough seat capacity to cover the passenger demand. Also, rolling stock on inter-city trains with many long-distance passengers is expected to provide more comfort than regional trains. A higher service quality encourages more travellers to use the train instead of their cars. When running the trains, railway operators have rolling stock related expenses such as electricity or fuel consumption and maintenance costs; efficient schedules minimise these expenses. Everyday railway operations have to face with disruptions and delays; robust rolling stock schedules are less affected by them. Robustness of the schedules can be increased when the number of possible sources for delays is kept low and spreading of delays is prevented as much as possible. Hence the rolling stock schedules can also contribute to raising the punctuality of the railway system. Of course, these criteria contradict one another; the operators have to find a good balance of them.

Strategic and tactical planning only consider these criteria. In operational and short-term planning, however, time is often too short to look for a schedule that matches best the objective criteria above. More important in such cases is to come up *quickly* with a solution that fulfils the requirements and that ensures an acceptable level of service quality, efficiency and robustness.

1.2 Research Questions

This thesis deals with tactical, operational and short-term rolling stock planning problems of passenger railway operators. The main goals of this research are the following.

1. Identify tactical, operational and short-term rolling stock planning problems and develop operations research models for describing them.
2. Analyse the considered models, investigate their computational complexity and propose solution methods.
3. Investigate to what extent solutions of the models give rise to solutions of the original railway planning problems. This includes testing the solution methods on real-life instances.

The focus of this thesis is limited to solution approaches that fall into the following two groups. We formulate models as integer linear programs and solve them by commercial MIP solvers, making use of various techniques to speed-up the solution

process. Besides this, we propose flow-type heuristic algorithms and study their behaviour on real-life instances.

Operations research is a broad field; this thesis does not consider many other powerful solution approaches that are used in rolling stock planning models in the literature. In particular, column generation, Lagrangian relaxation, advanced local search and genetic algorithms are not discussed here.

The models of this thesis arise from rolling stock planning problems of NS. As a consequence, our methods and results are to some extent tailored to these problems. Nonetheless, we believe that many of the solution approaches can be applied for rolling stock planning problems of other operators as well.

1.3 Outline of this Thesis

Chapter 2 describes the railway planning process at NS in detail. We also give a brief overview of operations research publications that address railway optimisation.

Chapter 3 deals with tactical rolling stock circulations. We provide two models for this problem. First we describe the ‘Composition Model’. With appropriate fine-tuning, it can be solved to near optimality even for the hardest instances of NS within a couple of hours. This part of Chapter 3 is based on the paper

FIOOLE, P.J., KROON, L.G., MARÓTI, G., SCHRIJVER, A. (2004)
A Rolling Stock Circulation Model for Combining and Splitting of Passenger Trains. CWI Research Report PNA-E0420, Center for Mathematics and Computer Science, Amsterdam, The Netherlands.
 To appear in *European Journal of Operational Research*.

Also in Chapter 3, we describe the ‘Job Model’ which is an alternative model for determining tactical rolling stock circulations. We compare the two models on instances of NS.

In Chapter 4 we give two models for the maintenance routing problem that arises in short-term planning: the ‘Interchange Model’ and the ‘Transition Model’. The Interchange Model is designed to take as much details of reality into account as possible. Besides complexity investigations, we propose a heuristic solution approach and report our computational results on instances of NS. This first part of Chapter 4 is based on the paper

MARÓTI, G., KROON, L.G. (2004). *Maintenance Routing for Train Units: the Scenario Model*. CWI Research Report PNA-E0414, Center for Mathematics and Computer Science, Amsterdam, The Netherlands. Revised version with the title *Maintenance Routing for Train Units: the Interchange Model* to appear in *Computers and Operations Research*.

The Interchange Model requires a large amount of input data which may be inaccessible in real-life applications. This motivates the conceptually much simpler Transition Model. We discuss the computational complexity of the Transition Model and compare it with the Interchange Model based on computational results. This second part of Chapter 4 follows the paper

MARÓTI, G., KROON, L.G. (2005) *Maintenance Routing for Train Units: the Transition Model*. *Transportation Science*, 39(4):518–525.

Chapter 5 is devoted to operational rolling stock circulations. We describe a two-phase approach that is used currently at NS for operational rolling stock planning. In this thesis we only study the second phase. We analyse the complexity of this problem and propose a heuristic solution method. Operational planning problems can also be formulated as instances of tactical rolling stock planning models. In Chapter 5, we compare the heuristic algorithm with the Composition Model on instances of NS.

In Chapter 6 we draw some conclusions and indicate possible directions for further research on the topics considered in this thesis.

This thesis is supplied with three appendices. Throughout the thesis we need terms for describing the railway planning and execution process. We provide our understanding of railway terminology in Appendix A. The notion of *b*-transshipments is used in Chapter 5; Appendix B gives the most important definitions and theorems related to *b*-transshipments. Finally, the models in Chapters 3, 4 and 5 require a number of parameters and variables; all these notations are collected in Appendix C.

Chapter 2

Planning Railways in the Netherlands

The smooth traffic on the Dutch railway system relies on the cooperation of several companies. In this chapter we first give an overview of these companies and discuss their responsibilities and tasks. Next, we have a closer look at the structure of the major passenger operator *Nederlandse Spoorwegen* (NS) and we describe the planning process at NS in detail, with special emphasis on rolling stock planning. This detailed picture of the rolling stock planning process enables us to point out where the problems considered in this thesis arise and why they are relevant. We also indicate which parts of the planning process have been addressed in earlier research.

2.1 Dutch Railway Companies and Their Responsibilities

Despite the liberalisation of the Dutch railway market in 1995, the state remained an important factor in it. In particular, it owns the railway infrastructure as well as all shares of the main passenger railway operator NS. (A railway operator is a company that runs passenger or freight trains.)

Managing the infrastructure and operating the trains, formerly both carried out by NS, are now strictly split. The latter became the main task of NS, while infrastructure management was carried out by independent state-owned companies that had been parts of NS until 1995. In 2003, these infrastructure management companies were united into *ProRail*.

2.1.1 Infrastructure Management by ProRail

Railway infrastructure includes the tracks, overhead lines, bridges, fly-overs, switches and safety devices between and inside the stations. Being responsible for infrastructure management, ProRail has three major tasks.

ProRail carries out capacity investigations and advises on building new trajectories, on doubling existing tracks and on redesigning the layout of the stations. These investigations include forecasting the passenger demand several years or even decades ahead. The construction works themselves are organised by ProRail as well. Moreover, ProRail schedules and coordinates maintenance of the railway infrastructure.

The second important task of ProRail is to divide the infrastructure capacity among the railway operators. This is done by checking the proposed timetables of the operators and by either approving them or requiring some modifications. As the Dutch railway system is heavily used, in particular in the densely populated Western part of the country, the railway operators' requests for time windows on the trajectories are often contradicting.

The third task of ProRail is traffic control. In case of smaller disturbances and delays, ProRail decides which train may first enter a railway trajectory or a station. Moreover, emergency scenarios are worked out for handling large-scale disturbances quickly. For example, if an important trajectory becomes unavailable due to a malfunctioning switch or a broken overhead line, the corresponding emergency scenario gives a guideline to the traffic control to what extent the traffic can be carried out on the still intact infrastructure, which trains should be cancelled, and so on.

Of the three parts of railway planning (such as timetabling, rolling stock scheduling and crew scheduling), ProRail coordinates the timetables during the entire planning horizon: from early strategic planning until real-time operations. Rolling stock and crew are planned by the individual operators.

2.1.2 Railway Operators in the Netherlands

Until the middle of the 90's, NS was the only Dutch railway operator. In 1995, the Dutch market was opened for other operators and since that time, a number of peripheral railway lines has been tendered. *Connexxion*, *NoordNed* and *Syntus* achieved the right to operate some of the train lines in the Eastern and Northern part of the Netherlands. However, NS remained the largest operator. Besides passenger transportation, freight traffic also contributes to the heavy utilisation of the Dutch railway network. The largest player on the Dutch market is *Railion*, a former part of

NS and now merged with the German company *DB Cargo*. Other important freight train operators in the Netherlands are *ACTS*, *Rail4Chem* and *ERS*.

According to the contract between NS and the Dutch state, NS may exclusively exploit the core of the railway system for passenger transportation until 2015. NS obtains a premium if it manages to increase its share of the morning peak traffic. In exchange, NS guarantees a certain service level, measured for example by the number of operated trains and by their punctuality. NS has to pay a fine if the punctuality does not reach a certain percentage. In this context, punctuality is defined as the percentage of trains that have at most 3 minutes delay on arrival at one of the larger stations. The contract also binds the possibility of raising the ticket prices to punctuality performance. An agreed price raise may take place when the average punctuality over a period of 12 months reaches a certain value.

In the future, further lines currently exploited by NS shall be tendered. Figure 2.1 shows a map of the Dutch passenger railway network and indicates the operators exploiting them in 2005.

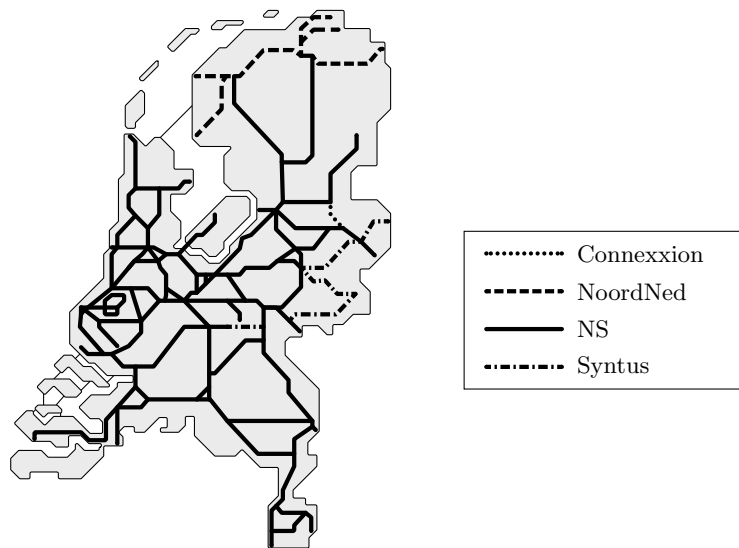


Figure 2.1: The Dutch passenger railway network in 2005.

2.1.3 The Structure of Nederlandse Spoorwegen¹

NS is the largest passenger operator in the Netherlands. In 2004, NS ran about 2,800 passenger carriages with a total capacity of 265,000 seats on a rail network of about 2,800 km. With more than 20,000 employees NS is one of the largest employers in the Netherlands.

The timetable of a day contains about 4,700 services, operating on over 100 train lines. These timetable services connect 388 stations. About 1,000,000 passenger journeys on a working day contributed to over 14 milliard passenger-kilometres for the whole year. In 2004, NS reached a 3-minutes arrival punctuality of 86.0%. An internationally accepted performance measure is the 5-minutes arrival punctuality: it was 92.6%, in Europe only eclipsed by the Swiss Railways.

In order to see the value of these punctuality data, it is worth mentioning that the track utilisation of the Dutch railway network is the highest in Europe. In 2002, NS realised a ratio of 49.4 train kilometres per network kilometres. Switzerland, having the second busiest network in Europe, reached a ratio of about 42 train kilometres per network kilometres (Poort (2002)).

NS is divided into several parts. In this thesis we only refer to three of them that are directly involved in rolling stock planning.

NS Reizigers or NSR (NS Passengers) is responsible for operating the trains themselves. It plans the timetable and schedules the rolling stock as well as the crew. Furthermore, NSR is responsible for carrying out these plans. The problems that we consider in this thesis arise in rolling stock planning and operation at NSR.

NS Commerce (NS Commerce) is the marketing and sales department of NS. It connects the company to the customers. It is responsible among others for predicting the passenger demand. Based on these forecasts, NS Commerce decides which stations are to be connected by direct trains.

NedTrain is responsible for the quality of the rolling stock. By carrying out regular preventive maintenance, inside and outside cleaning and safety checks, NedTrain keeps the rolling stock available for trains. NedTrain also refurbishes rolling stock units in order to adjust them to the changing demands and to provide more comfort to the passengers.

Further parts of NS manage the stations and the real estate around the stations, operate international trains, support the entire NS Group in human resource management and legal affairs, and so on.

¹The facts on NS have been collected from NS Intranet (2005).

2.2 Planning Process at NSR

The steps of the railway planning process can be classified in several ways. A common criterion is the length of the planning horizon. Based on this, one usually distinguishes strategic, tactical, operational and short-term planning phases (see Anthony (1965)). This hierarchy in time also means that the products of a planning phase serve as input for later phases where the earlier decisions may be revised according to updated information.

Strategic planning has a planning horizon of several years or even decades, the main focus is on capacity planning.

Tactical planning allocates the available capacity with a planning horizon of 2 months up to a year.

Operational planning sets up the fully detailed plans, the planning horizon varies from 3 days to 2 months. Mainly, it concerns adjusting the tactical plans to the forthcoming weeks.

Short-term planning comprises problems that arise when the actual train operations take place or just before that. It covers planning steps with a planning horizon of at most 3 days.

These planning phases are discussed in detail later in this chapter.

Planning steps are also grouped based on their target: they belong to *timetabling*, *rolling stock scheduling* or *crew scheduling*.

Finally, a third way to classify planning steps is to distinguish *central* and *local planning*. Central planning steps affect the entire railway network, such decisions are made at the Logistics Department of NSR in Utrecht. Local planning steps have an impact only on a single station. For example, platform assignment is a local planning step. Local planning steps related to rolling stock are carried out by shunting planners and by the *Transportbesturingsorganisatie*² (TBO), which is a department of NSR. Local crew planning is arranged by planners located at the 29 crew depots.

In the next few sections, we first describe the rolling stock and human resources of NSR. Subsequently, we give an overview of each planning phase in detail by gathering planning steps of timetabling, rolling stock scheduling and crew scheduling that belong to the planning phases. Thereafter, we have a closer look at the shunting

²Transport Control Organisation.

process. Finally, we point out some features that distinguish NSR from other railway companies. These special features explain why operations research models and methods in the literature are not (or hardly) applicable to instances of NSR.

2.2.1 Rolling Stock and Human Resources

NSR operates most trains by electrical and diesel *units*. A unit consists of a number of carriages, it is supplied with an engine and has driver’s cabins on both ends. The number of carriages in a unit only indicates its length and seat capacity for first and second class passengers. Units cannot be split up in everyday operations.

Units are available in several *types*. Units of the same type have identical technical characteristics such as length or seat capacity. When functioning properly, they are only distinguished by the number of kilometres they travelled since their last preventive maintenance check. Some types are *compatible*: units of compatible types can be combined to form a longer train, while units of incompatible types cannot. For example, “Koploper-3” and “Koploper-4” form a pair of compatible types (see Figure 2.2). These two types contain units with 3 and units with 4 carriages and are mainly used for inter-city services.

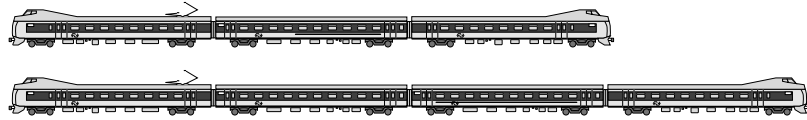


Figure 2.2: “Koploper” units with 3 carriages and with 4 carriages.

Shorter units have higher costs per seat, while longer units may lead to inefficient rolling stock usage. Having different unit lengths enables one to create compositions whose capacity better matches the passenger demand. For example, “Koploper” units can form a train with 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14 or 15 carriages.³

Currently, NSR uses about 590 units of 17 types. The length of the units ranges from 2 to 6 carriages, adding up to over 2,000 carriages in the units. There are single-deck and double-deck units. Some types are designed for inter-city or inter-regional trains, other types are more appropriate for regional trains. Each type is compatible with at most one other type. The largest type of NSR is the “Mat64-2”: NSR uses 167 pieces of these two-carriage units for regional trains. Note that NSR also operates

³Due to safety regulations and platform lengths, trains of NSR may contain no more than 15 carriages.

about 830 locomotive-hauled railway carriages and 120 locomotives. However, the planning problems considered in this thesis arise when scheduling the units.

The crew scheduling problems of NSR concern train drivers and conductors, they are the most important operational human resources for NSR. In 2005, NSR employed about 2,800 drivers and about 3,100 conductors divided into 29 crew depots. Each driver and conductor starts and finishes the daily duty at his/her depot.

2.2.2 Strategic Planning

Strategic planning concerns decision making several years in advance. It sets the target performance and service quality. Moreover, strategic planning makes sure that there are enough resources to achieve these goals. The most visible strategic decision is to determine the basic shape of the timetable by setting the train lines.

Stochastic considerations play an important role in forecasting the future passenger demand as well as the long-term availability of the required resources. Strategic planning entirely belongs to central planning.

Strategic Timetabling

The passenger demand is estimated for the forthcoming years. At this very first point of the planning process, the *origin-destination matrix* is determined: it specifies the estimated number of passengers on a day between each pair of stations.

Based on these forecasts, the train lines are determined. A *train line* is a series of trains that directly connect given stations. In the Netherlands, train lines are of three types: regional train lines with trains calling at every station along their path; inter-regional train lines where trains do not call at the smallest stations; and inter-city train lines where trains usually call only at major stations. Train lines have given frequencies, e.g. once or twice per hour.

Example. *The 3000 line is an inter-city train line, connecting Nijmegen (Nm) to Den Helder (Hdr). Trains of the 3000 line call at Nijmegen, Arnhem (Ah), Ede-Wageningen (Ed), Utrecht (Ut), Duivendrecht (Dvd), Amsterdam Amstel (Asa), Amsterdam Centraal (Asd), Alkmaar (Amr), Den Helder as well as at each station between Alkmaar and Den Helder (see Figure 2.3). The 3000 line is operated twice an hour in both directions.*

Determining line systems means balancing several conflicting criteria. For instance, long train lines provide convenient service for many passengers. On the other

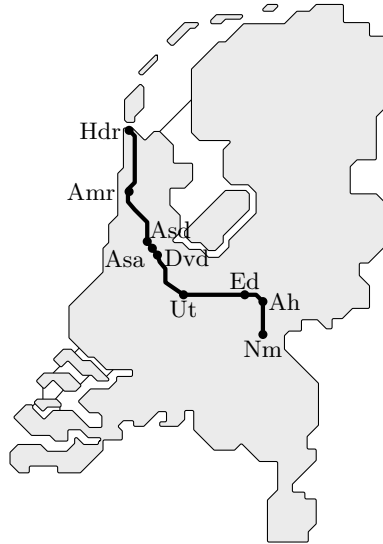


Figure 2.3: The 3000 line connecting Nijmegen (Nm) to Den Helder (Hdr).

hand, they are sensitive to delays and disturbances and they often lead to less efficient rolling stock schedules. Line planning is carried out by NS Commercie and NSR together.

One of the difficulties of line planning is the fact that passenger behaviour is very difficult to model: It has a strong stochastic character. Moreover, demand for seats depends on the line system: appealing new train connections may attract passengers who otherwise would have taken their cars.

Usually, the line system does not change much for years. In 2007, however, deeper changes are expected due to three on-going infrastructure development projects that will have been finished by that time: the High Speed Line, connecting Amsterdam to Rotterdam and further to Brussels and Paris; the “Utrechtboog” that allows direct train traffic between Schiphol Airport and Utrecht; and doubling the tracks between Amsterdam and Utrecht.

Strategic Rolling Stock Planning

Strategic rolling stock planning aims at determining the number of units needed to cover passenger demand in the forthcoming years. It has the longest horizon among all planning steps of NSR. It takes years until newly ordered units are actually delivered. The expected service time of the units amounts to decades. Strategic

decisions on rolling stock concern large amounts of money and they determine the train traffic and the service quality for years.

The basic decisions are purchasing or leasing new rolling stock and refurbishing existing units. Redundant units may be sold or disassembled. Another problem is to set the strategy for regular preventive maintenance. More frequent maintenance probably results in more reliable rolling stock, decreasing the disturbances caused by defect units. On the other hand, it also leads to higher maintenance costs and it requires a higher number of units in use.

Strategic Crew Planning

Crew planning on the strategic level aims to provide enough train drivers and conductors for the forthcoming years. This can be achieved by hiring new crew members or by internal trainings. Train drivers and conductors start and finish their daily duties at their home depot. Thus strategic crew planning also involves decisions on the depots themselves: on opening or closing depots and on dividing the work among the depots. Firing employees is no option at the moment, as crew members have a work guarantee until 2010. Similarly to rolling stock planning, it takes a few years until strategic crew planning measures have an effect: the training to become a conductor takes about a year and to become a driver takes about two years.

2.2.3 Tactical Planning

Tactical planning at NSR refers to a planning horizon ranging from 2 months to 1 year. The output of tactical planning consists of the *tactical timetable*, the *tactical rolling stock schedule* and the *tactical crew schedule*; they are planned for a generic week of the year. Tactical planning has a bit of an operational character too, since its products take many details of reality into account.

Tactical Timetabling

Like many European railway operators, NSR has a cyclic timetable, the cycle time is 60 minutes. The timetable of every hour is basically the same, the daily timetable is obtained by carrying out this pattern repeatedly.

Tactical timetable planning is carried out by central planners. It takes the previously determined line plan and estimated passenger demand as input. Moreover, the specification contains marketing aspects like providing appealing train connections to the passengers.

The first timetabling step is to create the *Basic One-Hour Pattern* which describes the departure and arrival times of the trains assuming that the timetable of each hour is identical. Then the one-hour pattern is extended to a one-day timetable by distinguishing peak hours, off-peak hours and night hours. Finally, adjusting the one-day timetable to the different demands on different days of the week yields a timetable for a generic week⁴.

In the timetabling phase, the rolling stock and crew is not considered explicitly yet. The limited infrastructure capacity of stations is taken into account as follows. Based on a draft timetable, local planners set up the *Platform Occupation Charts* that assign the timetable services to the platforms. Difficulties in creating the Platform Occupation Charts are reported to the central planners who update the draft timetable accordingly.

We have seen in Section 2.1.1 that ProRail supervises and coordinates the timetables of the railway operators. To that end, NSR sends its timetable of a generic week to ProRail and modifies it if ProRail requests seriously motivated changes. Timetabling receives feed-back also from rolling stock planning at later stages of the planning process, leading to slight adjustments of the timetable. However, these modifications are undesirable because the new timetable has again to be checked with ProRail.

Tactical Rolling Stock Planning

Once the timetable is known, the rolling stock has to be scheduled to carry out the timetable services. The rolling stock schedules have two parts: the centrally planned *tactical rolling stock circulations* that describe the assignment of available rolling stock to the timetable services and the locally created *tactical shunting plans* that specify the train movements inside the stations. (We discuss the shunting process in Section 2.2.6.)

The first step is to assign the available rolling stock to the so-called *line groups*: A line group is the collection of interconnected lines. For example, the 2100, 2400 and 2600 lines connecting Amsterdam (Asd) to Vlissingen (Vs), Dordrecht (Ddr) and The Hague (Gvc) form a line group (see Figure 2.4). The 3000 line in itself forms another line group. According to the current planning strategy of NSR, rolling stock is bound to line groups as much as possible: a daily workload of a unit preferably contains only trains that belong to the same line group. The reason for this decomposition is that otherwise the rolling stock scheduling problems would be far too large. Besides this

⁴Passenger demand on Tuesday, Wednesday and Thursday are very similar, therefore these three days get nearly identical schedules in tactical planning.

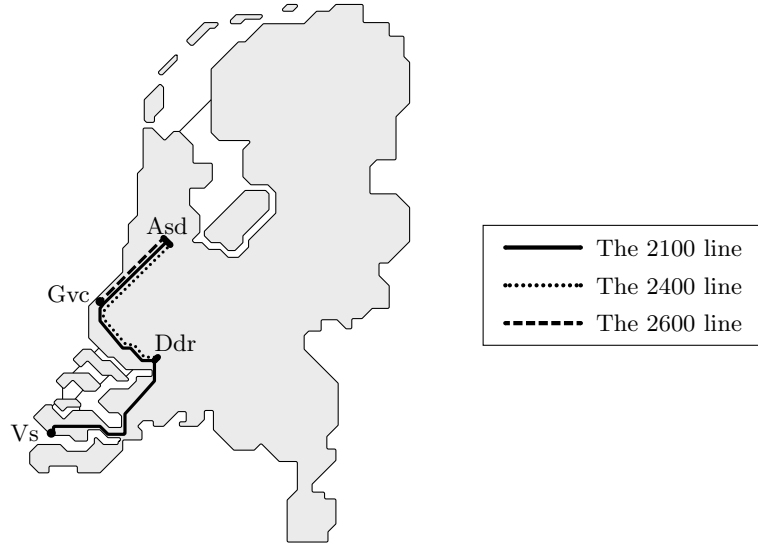


Figure 2.4: The 2100, 2400 and 2600 lines connecting Amsterdam (Asd) to Vlissingen (Vs), Dordrecht (Ddr) and The Hague (Gvc).

pragmatic reason, the decomposition also adds to the robustness of rolling stock plan. If trains in a line group suffer from disturbances, the delays do not spread to another line group easily. In the subsequent stages of the planning process, central rolling stock planners consider the line groups to be fairly independent. Of course, train movements that belong to different line groups may conflict with each other inside the stations due to the limited infrastructure capacity. Resolving these conflicts is left to local planners.

The next step is to determine the rolling stock circulation for every line group by specifying the composition for each train. Usually, the rolling stock schedules are determined independently for each day of the week. Yet, the planners make sure that the circulations of consecutive days can be attached to each other by taking care of the number of units per type that spend the night at the stations.

Central planners have only very restricted information about shunting possibilities at the stations. Therefore local planners receive the draft rolling stock circulations and create detailed shunting plans. Local planners may request changes if the draft rolling stock circulations lead to unsolvable shunting problems. After a small number of iterations, the rolling stock circulations and the shunting plans are accepted by both central and local planners.

The major criteria in railway planning are efficiency, service quality and reliability. In tactical rolling stock planning, they refer to cutting operational costs by reducing carriage-kilometres, to providing enough seat capacity on the timetable services and to keeping the number of composition changes low. Further details about tactical planning are given in Chapter 3 that deals with the tactical rolling stock circulation problem.

In the tactical phase, units are not considered as individuals yet. Instead, *duties* are determined for anonymous units: A duty is the daily workload of a unit. The duties are assigned to individual units in the short-term planning phase, a couple of days before actually carrying out the duties. The tactical shunting plans determine the successive duty of each duty that is to be carried out by the same particular unit on the next day. However, these night transitions are not very binding; they serve much more as a capacity check so that the rolling stock schedules of two consecutive days fit together.

Tactical Crew Planning

Crew scheduling is divided into two parts. First, it consists of generating the duties for train drivers and conductors, where a duty is a daily workload of an anonymous employee. Thereafter, the duties on different days are composed to rosters. Generating the duties is a central planning task, while rostering is carried out by local planners at the crew depots.

When creating the daily duties, the main objective is to minimise the number of needed duties subject to a large number of requirements. Examples of natural restrictions include the requirement that every train should have a driver and a given number of conductors and that the driver of a train must know both the trajectory and the rolling stock type. Other requirements are based on agreements with trade unions. They include restrictions on the duties themselves: a duty may not be longer than 9 hours, it must contain a meal break longer than 30 minutes, and so on. Finally, there are global constraints on the schedule. For example, there is a bound on the average time duration of the duties per depot and the workload is to be distributed among the depots as fairly as possible. Note that crew scheduling takes the rolling stock schedules as input. One of the reasons for this is that the number of needed conductors depends of the length of the train.

The second step in crew planning is *rostering*: the daily duties are composed to chains of duties, each chain is to be assigned to a crew member. Duties have different characteristics like early duties, night duties or long duties. The chain must fulfil complex requirements, making sure that the employees get enough rest. For

example, a sequence of at least 3 night duties must be followed by a rest period of at least 48 hours. Another rule requires that each employee has a fully free weekend at least once in every 3 consecutive weeks.

Crew rostering is one of the few tactical planning steps whose products may not be changed in the operational planning phase. In short-term planning, however, the crew rosters often need adjustments.

2.2.4 Operational Planning

In this thesis, the term operational planning is used for planning tasks with horizons of 3 days up to 2 months. In operational planning, the generic week plan (i.e. the result of tactical planning) is adjusted to the specific demands of the particular weeks. Reasons for such adjustments can be the need for extra trains because of cultural or sports events, national feasts, and so on. Another reason is the unavailability of railway tracks because of maintenance. In particular, the unavailability of tracks may require large-scale adjustments. At NSR, most effort in operational planning is spent for these larger adjustments. The usual duration of infrastructure maintenance works is 1–2 days (mostly in the weekend), but they may also take a couple of weeks.

The operational planning steps themselves are similar to those in tactical planning, but the main focus differs. Tactical plans are intended to minimise objective criteria that directly translate to operational costs and service quality. Since the horizon of typical operational planning problems is relatively short, efficiency is no longer the main objective. Much more important is to come up with the adjusted plans quickly and to make sure those plans can be carried out smoothly. As we have seen in the previous section, several rounds of information exchange between central planners, local planners and ProRail are required to make them agree on the tactical plans. This agreement is also necessary for the operational modifications. Therefore the tactical plans have to be modified in such a way that central and local planners can agree on the adjustments quickly.

The products of operational planning are the *operational timetable*, the *operational rolling stock schedule* and the *operational crew schedule*. A couple of days before the operational plans are to be carried out, they are transferred to the traffic control offices of ProRail and to TBO (cf. Section 2.2.5). From then on, they are responsible for adjusting the operational plans to the real-time events.

Operational Timetabling

The tactical timetable is updated by inserting or deleting trains and by modifying the arrival and departure times of existing ones. In case of infrastructure maintenance, the updated timetable must obey the altered infrastructure availability. Also, the operational timetable has to be approved by ProRail.

Operational Rolling Stock Planning

Similarly to the tactical rolling stock schedules, the operational rolling stock schedules have two parts: the centrally planned *operational rolling stock circulations* and the locally created *operational shunting plans*.

Central planners take the shunting possibilities into account in a similar way as they do in tactical planning. The draft operational rolling stock circulations are sent to the local planners who try to create the corresponding detailed shunting plans and require modifications if necessary. Similarly to tactical planning, the operational rolling stock schedules consider duties for anonymous units.

Usually, minor modifications of the tactical plans are handled easily. Much more effort is needed in case of infrastructure maintenance. Then planners at NSR apply the following two-phase method. First, they set up an *intermediate plan* that meets requirements except for the *initial inventories*. The initial inventory of a station is the number of units per type that start there at the beginning of the planning horizon. So, it can happen that 3 units are available at a station prior to the infrastructure maintenance, while the intermediate plan requires 4 departing units at the beginning of the planning horizon; then the station has an *off-balance*. A quick planning process requires that the intermediate plan is very likely agreed on by local planners. To achieve this, the intermediate plan contains as few underway composition changes as possible. The second phase resolves the off-balances by additional modifications. This two-phase method is discussed in detail in Chapter 5.

Operational Crew Planning

Operational crew planning means adjusting the duties in the tactical crew schedule by central planners. Tactical crew rosters were created to fulfil complex rules during the forthcoming months and the employees know their working times several weeks ahead. Changing the rosters would make their working times unpredictable. Therefore the crew rosters may not change in operational planning. Instead, the operational crew planners try to modify the daily duties of the drivers and conductors, while maintaining the number of duties and their characteristics such as early starting

duties, night duties, and so on. Thereby the modified duties remain compatible with the crew rosters. Again, in contrast to tactical planning with the number of duties as the main objective, operational planning is much more a feasibility problem.

2.2.5 Short-term Planning

Short-term planning is the last phase of railway planning, it amounts to making decisions with a horizon of at most 3 days. In particular, it includes real-time reacting to the latest developments. The operational plans are transferred to the traffic control offices and to TBO 3 days before carrying them out. From that point on, the traffic control offices, being part of ProRail, are responsible for the timetable, while TBO is responsible for adjusting and executing the operational rolling stock and crew schedules. So, most short-term planning steps are local. Yet, there are some aspects that require central coordination.

In short-term planning problems, there is not much time for computations in order to get the best possible solutions. Mostly, any feasible solution which keeps the railway system moving is good.

Short-term Timetabling

The most important task is delay and disruption management. The infrastructure capacity has been divided between the operators. However, the operators may not be able to run their trains on time due to disturbances. The traffic control offices, which are part of ProRail, decide in which order the delayed trains may enter the stations and the trajectories.

Short-term Rolling Stock Planning

Short-term rolling stock planning is carried out by TBO which is divided into a central and a local part. The central part of TBO is called *Materieelregelcentrum*⁵ (MRC). The planning tasks are shared between the parts of TBO as follows.

The operational plan describes daily duties but not the assignment of duties to particular units. This is done by local planners of TBO. They can do this arbitrarily as long as each duty gets a unit of the type it was planned for.

When carrying out the operational rolling stock schedules, delays or disruptions may occur. Then TBO must react to provide rolling stock for the timetable services. However, lack of instantly available rolling stock at the right place often may lead

⁵Rolling Stock Management Centre.

to additional delays or even cancellation of trains. A special form of disruption management is to keep track of type mismatches. Time to time, local planners are forced to fill duties with a “wrong” type. Central planners at MRC have a global view of the rolling stock circulations and they try to correct the mismatches by sending instructions to the local planners. As in earlier phases, each change of the rolling stock plan must be approved by local planners.

An important issue in short-term rolling stock management is maintenance. Units need regular preventive check-ups. Each unit must undergo a small safety inspection every 48 hours. Since such inspections can take place basically at every shunting yard, it is part of the shunting process. However, units also need more involved maintenance checks, say every 30,000 km: Units that travelled that far since their previous maintenance check must go to a maintenance facility. This is arranged by maintenance routing planners at MRC. So, maintenance routing is a central planning step. On every day, the units with the highest number of kilometres since their last maintenance check are listed, these units are called *urgent*. They must undergo maintenance in the forthcoming 3 days. Maintenance routing planners modify the operational rolling stock schedules so that the urgent units are routed to the maintenance facilities. Being that close to executing the operational plans, there is no time to come up with completely new rolling stock schedules. Instead, small and localised exchanges are applied, for example by swapping two units of the same type. Maintenance routing is discussed in detail in Chapter 4.

Short-term Crew Planning

Short-term crew planning is a task of TBO, too. The main task is to handle disruptions. Each train must be supplied with a driver and a given number of conductors. If this is not possible because of disturbances of the railway system, the train has to be delayed or even cancelled.

2.2.6 Shunting

Shunting refers to all train movements inside the stations. As one could see in the previous sections, shunting is an important issue in every phase of rolling stock planning. This thesis does not deal with the shunting problem explicitly. Nonetheless, shunting considerations play an important role in each model of the subsequent chapters. The two most significant elements of shunting are routing trains through a station and positioning temporarily not used units to the shunting yards.

In the daytime, the passing trains need a free platform to let the passengers get in and out of the train. Moreover, they must be routed through the stations in such a way that they do not block the paths of each other. Sometimes, the compositions must be adjusted to the passenger demand by coupling units to the trains or uncoupling units from them. This happens often just before and just after peak hours when the number of passengers increases or decreases. Since the timetable of NSR is very dense, the stations provide only a restricted capacity to change the compositions. On the other hand, composition changes are potential sources for delays, therefore they are to be avoided as much as possible.

At night, the units are to be stored in such a way that the start-up on the next morning requires the smallest possible number of shunting movements. Implicitly, it includes assigning the units arriving in the evening to rolling stock duties starting next morning. Moreover, units have to be cleaned every day and they must undergo a small technical check on every second day.

The shunting process is arranged by local planners located at several major stations. Solving the shunting problem itself is quite a challenging task, even for a single day and for a single middle-sized station. Therefore, central planners must rely on the knowledge of the local planners throughout the entire planning process.

2.2.7 Information Flow between the Planning Phases

As we have seen in the previous sections, the planning steps interact in a quite complex way. We give an intuitive overview of the most significant interactions in Figure 2.5. Rectangles indicate central planning tasks and ovals indicate local planning tasks. The arrows show the information flows. Dashed arrows are drawn when the influence of a planning step to another is possible but undesirable.

Note that reality is more complex than the figure indicates. Occasionally, planning tasks may have an influence on each other even if no according arc is drawn in Figure 2.5. For example, tactical and operational crew planners may request changes in the timetable if they are unable to create the crew duties. However, any feed-back of these kinds is much more an exception than a rule.

2.2.8 Special Features of NSR

Heavily Utilised Railway System

The most peculiar property of the Dutch railway network is its heavy workload. In 2004, NSR realised 115,000,000 train-kilometres on a network of 2,800 kilometres.

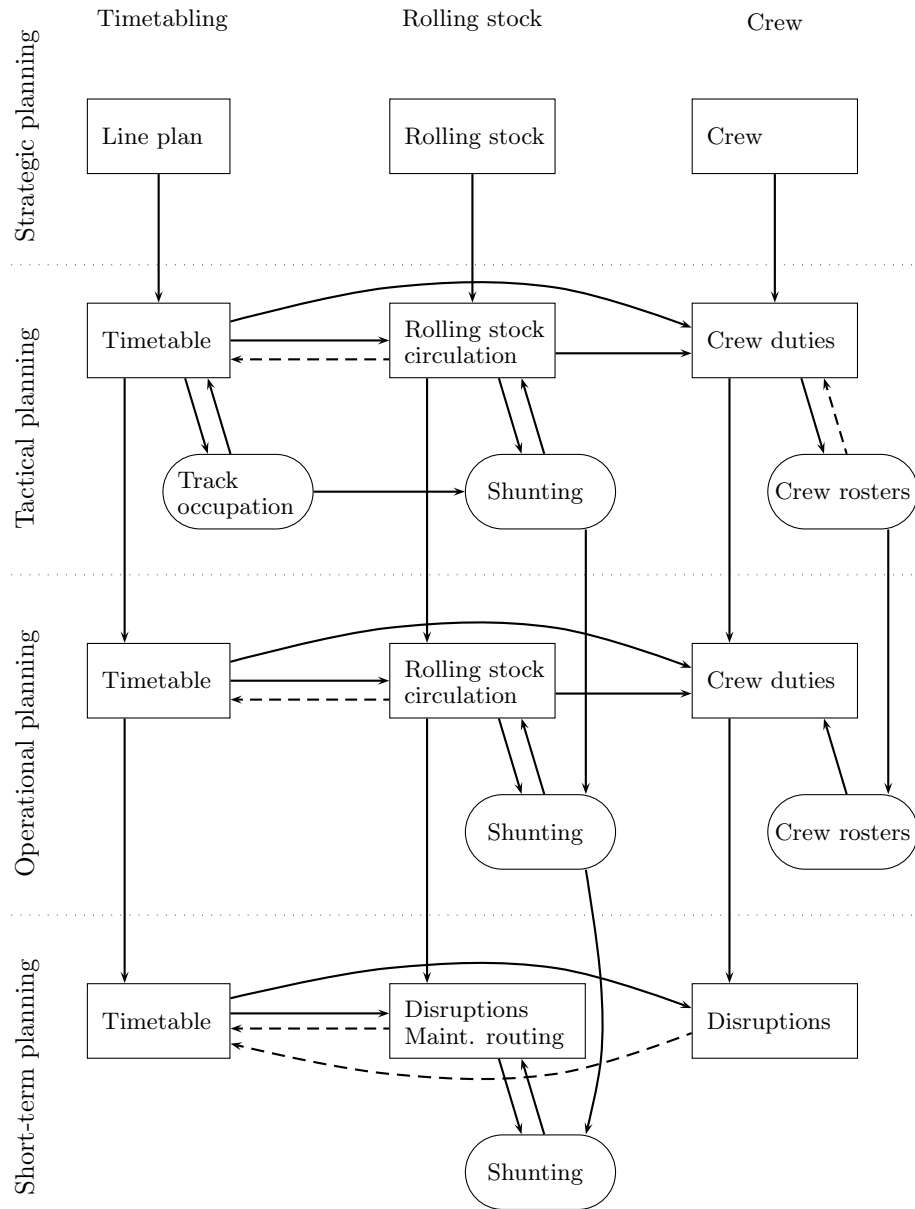


Figure 2.5: Information flow between the planning steps. Rectangles are central planning tasks, ovals are local planning tasks. Arrows indicate the information flow. Dashed arrows represent possible but undesirable influence.

Some tracks between major cities are so heavily used that often the 3-minutes security headway between the trains prevents planners to insert additional trains.

A consequence of this is that stations have a large throughput. When a train arrives at a station, it may depart soon as another timetable service, we call this a *turn-around*. Also, the arriving units may be put to a shunting yard and used again only a couple of hours later. In any case, the arrival platform must be freed up as soon as possible, allowing another train to arrive. Heavy traffic often restricts the possibilities to adjust the compositions during the turn-arounds. In fact, it is preferred that no composition change happens at all. However, this may result in either an inefficient schedule or high seat shortages.

A busy timetable also leads to short turn-around times making sure that the platforms become free as soon as possible. Also, all rolling stock is needed during peak hours: a long turn-around simply occupies rolling stock without using it to carry passengers. The turn-around times at NSR range from about 5 minutes to about 30 minutes.

Rolling Stock

Unlike most railway operators, NSR mainly uses units instead of locomotive-hauled carriages. Having a driver's seat on both ends of units allows a fast and easy turn-around process in case of direction changes. Also, the shunting process for self-propelled units is easier than for locomotive-hauled carriages.

Units containing different numbers of carriages can be combined to compositions of various lengths in order to match passenger demand. However, multiple types in a train force one to consider the order of the units in the composition, not only their number. For example, '334' and '343' are both compositions with two "Koploper" units of length 3 and with one unit of length 4, but they provide different possibilities for composition changes. The unit of length 4 lies on the right hand side of composition '334', it may be possible to uncouple it, resulting in a composition '33'. Composition '343', however, contains the unit of length 4 in the middle, therefore much more effort is required to uncouple it; in practise it is virtually impossible.

Successor Trains

We mentioned in Section 2.2.3 that the line system is divided into line groups and that a unit is basically bound to a line group during a day. The decomposition into line groups turns rolling stock planning into subproblems of tractable size. Furthermore,

a rolling stock schedule with units bound to line groups is expected to be less sensitive to disturbances and delays.

A unit that arrives at a station in a train can continue its daily duty in several trains departing from that station. By the short turn-around times and the line groups, arriving units usually go over to the earliest departing train that belongs to the same line group. We call it the *successor train* of the arriving train. The arriving train itself is the *predecessor* of its successor train.

We emphasise that it may be possible to adjust a composition during a short turn-around. Units can be uncoupled and placed to a shunting yard, or other units that have been stored at the station may be added to the train before departure. The layout of the station determines in most cases at which end of the composition this is possible.

Some in-coming trains have no successor trains. This happens mostly to trains that arrive in the evening. Their units are supposed to go directly to the shunting yards, to be used a couple of hours later or the next day. Similarly, early departing trains do not have predecessors.

At NSR successor trains are specified as early as at the beginning of the tactical planning phase. In many versions of the rolling stock circulations problem in the literature and at other railway companies, determining the successor trains is part of the problem.

Splitting and Combining Trains

Trains have usually at most one successor and at most one predecessor. However, NSR operates some lines where timetable services are split and combined. For example, the 1600 line contains services from Enschede to Amsterdam and from Enschede to Schiphol: the trains from Enschede are split in Amersfoort, the front part departing to Amsterdam, the rear part departing to Schiphol. On the way back, trains from Amsterdam and Schiphol arrive at Amersfoort to be combined and to depart a couple of minutes later towards Enschede (see Figure 2.6).

Note that we only speak about splitting if the successor trains depart within a couple of minutes after their predecessor train arrived. We distinguish splitting from the case when units are simply uncoupled during a turn-around. When a train is split, the exact composition of both split parts must be carefully identified. However, the order of uncoupled units on the shunting yards does not matter much as the shunting crew has usually enough time later to carry out necessary shunting movements. A similar restriction applies for combining trains.

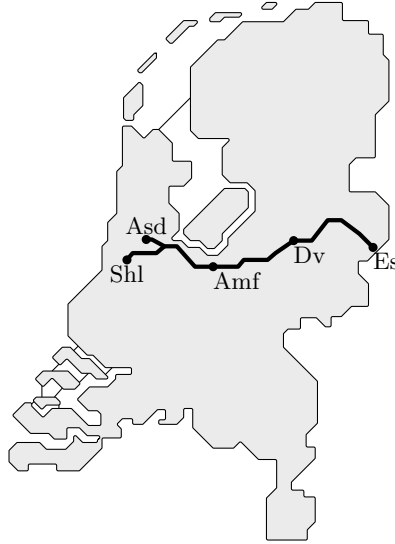


Figure 2.6: The 1600 line.

Splitting and combining occurs very often in the so-called Noord-Oost line group, an interconnected system of inter-city lines of NSR. Largely due to the complications caused by splitting and combining, the Noord-Oost line group is considered to be the hardest instance among all rolling stock circulation problems of NSR.

2.3 Operations Research in Railway Planning

Optimisation of railway transportation has always been an intensively investigated topic in operations research⁶. Surveys such as by Assad (1980) and by Cordeau et al. (1998) consider the entire spectrum of railway planning, from infrastructure related questions through problems at passenger and freight operators. Since this thesis deals with problems that arise at NSR, we restrict ourselves here to research questions of passenger railway operators. Even then, the literature list is not intended to be complete. An exhaustive overview of the topic falls out of scope of this thesis. We also refer to a recent survey of Huisman et al. (2005) on operations research methods in passenger railway transportation, with a special focus on planning problems of NSR.

⁶Moreover, one of the first applications of the famous Max Flow Min Cut theorem was to investigate how to destroy the enemy's railway system as efficiently as possible (see Schrijver (2003) p. 168).

In the next few sections, we list a number of publications and successful operations research applications in railway planning. We structure the literature overview in the same way as we classified the planning steps in the previous sections.

2.3.1 OR Methods for Strategic Planning

Timetabling

The most important research problem here is line planning. Goossens (2004) provides an overview of this topic, focusing mainly on efficiency improvement for railway operators. Bussieck (1998), Scholl (2001) and Scholl (2005) describe methods to optimise customers' convenience of the line system, by providing as much direct connections as possible or by minimising the number of transfers.

Rolling Stock and Crew

Although long-term rolling stock and crew management concern significant amounts of money and have major consequence for the service quality for years or decades, they have received less attention in the operations research literature than tactical or operational planning. A reason for this is the difficulty to forecast the future demand for resources.

Strategic planning models must be able to deal with uncertain factors. Therefore the natural approach is stochastic optimisation. Recently, Van Dijk (2003) developed a model for strategic crew management. The model makes suggestions for hiring or internal training of employees, taking among others the fluctuating number of employees and their unavailability due to training, vacation or illness into account.

2.3.2 OR Methods for Tactical Planning

Among all phases of the railway planning process, tactical planning attracted the most attention in the operations research literature. Cyclic (and non-cyclic) timetabling, rolling stock circulations and crew scheduling are discussed in many publications. The research resulted in several software packages that became indispensable planning tools of the railway companies.

Timetabling

Cyclic timetabling is an intensively researched topic. Peeters (2003) describes this problem in detail with an extensive list of references. Another important and difficult

problem is to evaluate the quality of a timetable. One of the major objectives is robustness: the timetable should allow high overall punctuality even when there are disturbances. Robustness is a requirement at the network level that is very difficult to express in terms of the decision variables used in timetabling models. Instead, simulation and stochastic analysis tools are used to compare given timetables.

ProRail and NSR make use of the software package *DONS*⁷ (see Hooghiemstra (1996), Hooghiemstra et al. (1999), Schrijver and Steenbeek (1994) and Zwaneveld et al. (1996)). ProRail uses DONS primarily to analyse the effect of infrastructure extensions on timetables. Within NSR, DONS is used to determine the basic shape of the timetables in the near future.

The simulation software *SIMONE* has been developed for ProRail and NSR to analyse the sensitivity of timetables to disturbances and delays. Vromans and Kroon (2004) and Vromans et al. (2005) describe quantitative methods to evaluate and optimise the robustness of timetables. State-of-the-art results in this topic are summarised in Vromans (2005).

Rolling Stock

The first step in tactical rolling stock planning at NSR is to allocate the rolling stock to the line groups. Abbink et al. (2004) describe a model for this problem. The main criterion is to minimise the seat shortages during the morning peak hours and to minimise the number of different rolling stock types per line group. This model is currently used at NSR to divide the rolling stock among the line groups.

Rolling stock circulation has been addressed in quite a few papers. For example, Schrijver (1993), Ben-Khedher et al. (1998), Brucker et al. (1998), Cordeau et al. (2000), Cordeau et al. (2001a), Alfieri et al. (2002), Lingaya et al. (2002) and Peeters and Kroon (2003) deal with this problem. Chapter 3 is devoted to tactical rolling stock planning, we discuss these papers there more deeply.

Crew

Crew scheduling, especially creating crew duties, belongs to the most successful applications of operations research methods. Intensive research has lead to a major break-through in the last couple of years.

The duty generation problem in crew scheduling can be formulated as a generalised set covering problem, the decision variables express whether or not a candidate

⁷Designer Of Network Schedules.

duty is selected. The solution method can be based on the linear programming relaxation or on Lagrangian relaxation. Integer solutions may be obtained by a branch-and-price method, described for example by Barnhart et al. (1998b). Alternatively, heuristic algorithms may be used. The algorithm described by Caprara et al. (1999) turned out to be especially useful in practise. Since 2000, the software package *TURNI* has been used at NSR to create duties for the crew. TURNI implements a column generation framework and makes use of the heuristic algorithm of Caprara et al. (1999). With appropriate fine-tuning, TURNI is capable to produce duties that are accepted both by the management of NSR and by the employees. More details about results with TURNI can be found in Kroon and Fischetti (2001) and in Abbink et al. (2005).

Most other major European railway companies use crew scheduling software, too. The best known packages are CARMEN (Kohl (2003), used among others at the German Railways) and TRACKS II (Fores et al. (2001), used at several railway companies in the United Kingdom). We also mention that similar crew scheduling problems arise at bus and airline companies.

In contrast to crew duties, generating crew rosters fully automatically is still a challenge. Kohl and Karish (2004) provide an overview on crew rostering methods, mainly used in airline industry. Traditionally, most European public transportation companies apply a special variant of rosters, so-called cyclic rosters. Recent publications on cyclic crew rostering include Caprara et al. (1998) and Sodhi and Norris (2004).

2.3.3 OR Methods for Operational Planning

We already mentioned that tactical rolling stock and crew planning at NSR have a certain operational flavour. Therefore, most publications on rolling stock and crew scheduling that we cited in Section 2.2.3 are relevant here, too.

2.3.4 OR Methods for Short-term Planning

A number of papers in the operations research literature deals with short-term railway planning problems. However, very few successful decision support tools have been developed so far. Some recent works on dispatching public transportation systems include Suhl and Mellouli (1999), Shen and Wilson (2001) and Suhl et al. (2001). Although not part of the railway planning process, it is worthwhile to mention the timetable information system applications described by Schulz et al. (2002) and by Wagner and Willhalm (2003).

Maintenance routing is one of the most studied short-term planning problems in the literature. Maintenance routing problems with specifications that differ from the problems at NSR have been described (mostly for airline applications) by Feo and Bard (1989), Clarke et al. (1997), Barnhart et al. (1998a), Gopalan and Talluri (1998), Talluri (1998) and Anderegge et al. (2003). Moreover, the model of Lingaya et al. (2002) for operational rolling stock planning also incorporates maintenance. The maintenance routing problem is discussed in detail in Chapter 4.

2.3.5 OR Methods for Shunting

Several papers in the literature address the shunting problem. Winter (1999), Winter and Zimmermann (2000) and Blasum et al. (2000) focus on dispatching trams, Gallo and di Miele (2001) consider the shunting problem for buses. Tomii et al. (1999) and Tomii and Zhou (2000) describe a genetic algorithm to solve related problems. Freling et al. (2005) and Lentink et al. (2003) seek for solution methods that can cope with several practical issues of shunting problems at NSR. A more extensive description of the topic is given by Lentink (2006). Despite promising computational results, solving shunting problems remains a challenging task, even for a single day and for a single middle-sized station.

2.4 Further Aspects

In the previous sections we listed a number of planning tasks that need to be carried out and that are addressed to some extent by operations research techniques. Yet, there are other aspects that can have a large influence on the entire planning process or at least on some parts of it.

Redesigning the Planning Process

The careful reader must have realised that the planning process described in the previous sections contains planning steps whose results will barely (or never) be used when actually carrying out the plans. For example, the tactical rolling stock schedules are not carried out as they are; instead, they are transformed to the operational schedules. The operational schedules contain fully detailed shunting plans but these are hardly ever carried out without major changes, too. New shunting plans must often be created in the short-term planning phase in order to react to the latest developments. Moreover, several factors that appear as late as in the short-term

phase may affect the shunting plans. Such factors are maintenance routing, cleaning and technical inspections of the units.

Planners should not need to create fully detailed rolling stock plans one year in advance with the knowledge that a lot of the details must be revised later. Therefore tactical planning and partly operational planning needs reliable tools to estimate shunting capacity of the stations.

Filling in the fine details as late as possible is the basic idea of the project *Herontwerp* (Redesign) of NSR. It aims at restructuring the planning process so that early stages only focus on capacity planning but not dealing with all the details. Besides the mentioned example of the shunting plans, *Herontwerp* addresses the whole spectrum of planning questions at NSR. An important part of the project is to develop the required capacity checking tools.

Information Systems

Computer-based decision support tools but also human planners need the appropriate input data. Currently, central planners and local planners do not have direct access to each other's data. Unifying the databases, being a part of *Herontwerp*, is an on-going project at NSR.

Short-term rolling stock planners are often faced with the problem that they cannot get the exact location of a physical unit instantly. This may have several consequences, such as slowing down the maintenance routing process. The units have recently been supplied with GPS receivers to locate them and with GSM devices to transmit their position and status information.

An important point in using computers as decision support tools is to display the output in such a way that the planners can interpret the results easily. For example, a recently developed application for visualisation of automatically generated rolling stock circulations turned out to be crucial for the acceptance of the system and its results by the planners.

Chapter 3

Tactical Rolling Stock Circulations

In tactical planning, the rolling stock schedule is created for a generic week of the year so that the schedule respects every practical requirement and so that it can be carried out repeatedly. The schedule is composed of the rolling stock circulations and the shunting plans. In this chapter, we discuss the rolling stock circulation problem that arises in tactical planning.

Till recently, the tactical rolling stock circulations of NSR were created manually. The models and results below show that the problem can be successfully treated by operational research methods.

The chapter is organised as follows. First we formulate the problem and give a literature overview. Then we describe two ways of modelling the tactical rolling stock circulation problem. The first approach is the ‘Composition Model’ which is highly related to network flows. The second approach is the ‘Job Model’ which is a set-covering type model. We present computational results for both models on instances of NSR.

3.1 Literature Overview

The tactical rolling stock circulation problem has been addressed in quite a few papers in the literature. However, in many cases the problem formulation substantially differs from the specifications of NSR. NSR uses units of several types. The restricted shunting possibilities enforce one to consider the order of these different types in the

compositions. However, models in the literature were often set up for locomotive-hauled railway carriages whose order may not matter. Also, the problem specification of NSR determines the successor trains described earlier. This is not always the case in tactical rolling stock planning problems in the literature. An additional peculiarity is that the objective function includes seat-shortage kilometres and the number of composition changes besides the usual criterion of carriage-kilometres. Finally, no models were developed so far to handle splitting and combining of trains.

We first discuss research results chronologically that were developed to solve instances of NSR. Naturally, these are the papers most related to the topic of this chapter.

Schrijver (1993) considers the problem of minimising the number of units needed to satisfy the passenger seat demand for a single train line and a single day. The method computes for every trip the number of units per type to be used, not considering the order of the units in the compositions. Seat shortages and the number of composition changes do not appear in the objective function. The integer programs are solved by commercial MIP solvers.

Groot (1996) and later Alfieri et al. (2002) describe an integer programming model to determine the rolling stock circulation for multiple rolling stock types on a single line and a single day, considering also the order of the units in the compositions. The objective is to minimise the number of units or the carriage-kilometres such that the given passenger demand is satisfied; seat shortages are not allowed and the number of composition changes is not minimised. The solution methodology is to decompose the problem into subproblems and to use their solutions to reduce the size of the original problem, so that it becomes tractable by commercial MIP solvers. The approach is tested on real-life case-studies of NSR.

Van Montfort (1997) presents a model for computing circulations of locomotive-hauled railway carriages. The objective is to minimise the number of carriages and the number of carriage-kilometres subject to given demands. The model deals with first and second class carriages and uses a simple but realistic way to take the order of the carriages into account. The model is solved by commercial MIP solvers.

Peeters and Kroon (2003) describe a model with exactly the same specifications as we are considering in this chapter. They apply Dantzig-Wolfe decomposition as solution technique. The resulting large linear program is first solved by column generation, then branch-and-price is applied to obtain integer solutions. With appropriate fine-tuning, the approach is capable to solve real-life instances of NSR within a couple of seconds. However, the model strongly uses the fact that trains have at most

one predecessor and one successor train. That is, splitting and combining does not occur.

The papers below deal with problems whose specification differs from ours.

Brucker et al. (1998) consider the problem of routing railway carriages through a railway network. The carriages should be used in timetable services or empty trains such that each timetable service can be operated with at least a given number of carriages, thereby satisfying the passenger demand. The order of the carriages is not considered. The objective is to minimise a non-linear cost function. The solution approach is based on local search techniques such as simulated annealing.

Ben-Khedher et al. (1998) study the allocation of rolling stock units to the French TGV trains. Shunting is not an issue since trains may consist of at most two units. However, the rolling stock circulation must be adjusted to the latest demand from the seat reservation system. Therefore, this problem has an operational or even short-term planning flavour. The objective is to maximise the expected profit for the company. To reach that goal, various operations research techniques are applied such as stochastic optimisation, branch-and-bound and column generation.

Cordeau et al. (2000) present a locomotive and carriage assignment problem. The authors formulate the problem as a large integer program and use Benders decomposition to solve it. Computational experiments show that optimal solutions can be found quickly. In a subsequent paper, Cordeau et al. (2001a) extend their model with various aspects such as maintenance of rolling stock. They propose a heuristic branch-and-bound algorithm for the extended model, solving the linear programming relaxations by column generation. Neither of these models considers the order of the carriages in the compositions. The difficulty of the problem in these publications is partly caused by the fact that finding successor trains is part of the problem.

Lingaya et al. (2002) describe a model for operational management of locomotive-hauled railway carriages. They explicitly take the order of the carriages in the trains into account and assume that the successor trains have already been specified. Several real-life aspects, such as maintenance, are considered. So, this model solves similar problems that arise in operational and short-term planning at NSR. Seat shortages and the number of composition changes are not minimised. Also, the model does not handle splitting and combining of trains. The solution approach is based on a Dantzig-Wolfe reformulation solved by column generation.

3.2 Obtaining Real Instances

Having seen a literature overview of the previous section, we turn back to rolling stock circulation problems of NSR and describe where and how instances of this problem arise.

In tactical planning, we consider *trips* rather than timetable services. A trip is a sequence of train movements that has to be carried out without composition changes. If the train line admits composition changes underway, then a timetable service consists of several trips. The notion of successor and predecessor trains extends to trips. The tactical rolling stock circulation problem amounts to assigning units to the trips and to creating duties for anonymous units. The problem is solved separately for line groups as described and motivated in Section 2.2.3. Mostly, rolling stock circulation instances only consider a single day. This second decomposition reduces the size of the problem further, allowing a quicker solution process. Moreover, there is no real need to compute rolling stock circulations for more than one day for the following reasons.

There are hardly trains at night, so then almost all units are put to the shunting yards. Most shunting yards have enough capacity to change the compositions at night. Therefore, by slightly optimistic assumptions on the shunting process, the rolling stock circulations on two consecutive days are only connected by the number of units (per type) that start a day at a station and the number of units that finished the previous day at that station. We call these numbers the *initial* and *final inventories*, referring to the beginning and the end of a day. Once these inventories are known, the rolling stock circulation problem automatically decomposes into subproblems, each of them corresponding to a single day.

Empty trains may be necessary to carry over units to a station where they are needed. We assume that all possible empty trains are listed in the timetable as trips without passenger demand. That is, the models below cannot invent new empty trains. Note that empty trains occur in tactical rolling stock circulations quite rarely.

In our computations, we pay special attention to the so-called ‘Noord-Oost’ line group, an interconnected system of inter-city lines. In what follows, we shall time to time illustrate concepts and requirements by this particular instance. In the next section we give some more details about the Noord-Oost line group.

3.2.1 Noord-Oost Line Group

The Noord-Oost line group is a system of inter-city lines that form the backbone of the Dutch railway network. It is considered to be the most difficult rolling stock cir-

culuation problem of NSR. The timetable of the involved lines is more or less periodic with a period of one hour. It contains 167 timetable services on a workday, divided into 665 trips, according to the possibilities to change the compositions underway.

The Noord-Oost line group consists of lines 500, 700, 1600, 1700 and a small number of additional trains. These lines connect Amsterdam (Asd), Schiphol (Shl), Rotterdam (Rtd) and The Hague (Gvc) in the Western part of the Netherlands to Leeuwarden (Lw), Groningen (Gn) and Enschede (Es) in the Northern and Eastern part of the country. Utrecht (Ut), Amersfoort (Amf), Deventer (Dv) and Zwolle (Zl) are important underway stations. (See Figures 3.1 and 3.2.)

The lines are operated by units of the “Koploper” type which are available with 3 or 4 carriages each. In a concrete problem instance, there are about 50 units of length 3 and 35 units of length 4. The maximal length of a train varies from 8 to 15 carriages. Typically these upper bounds are 12 or 15 carriages per trip which allows 15 and 30 different compositions, respectively. On average, the manually created rolling stock schedule allocates about 6.8 carriages (2.0 units) to a trip.

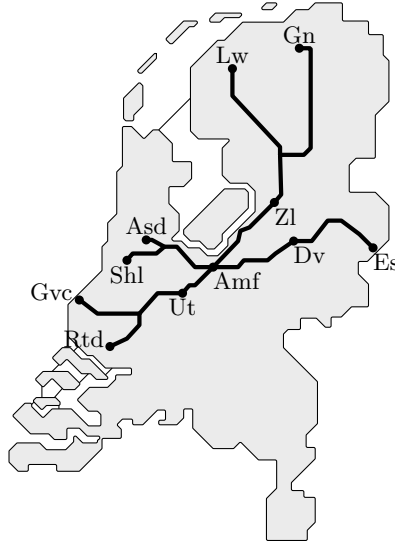


Figure 3.1: The Noord-Oost lines on the map of the Netherlands.

Almost all trains in the Noord-Oost are split and combined at certain locations. A train in the 1600 line arriving in Amersfoort from Enschede is split into two parts. The front part continues to Amsterdam, while the rear part continues to Schiphol. On the way back, the train from Amsterdam is connected to the rear of the train

from Schiphol. Trains in the 1700 line are split and combined in Utrecht. Trains in the 500 line are split and combined in Utrecht and in Zwolle and trains in the 700 line are split and combined in Amersfoort and in Zwolle (see Figure 3.2).

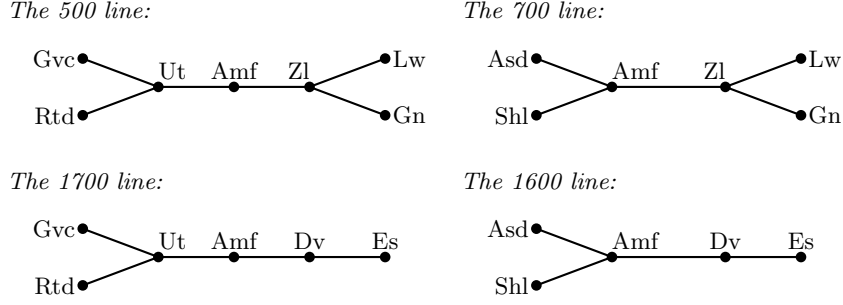


Figure 3.2: The Noord-Oost lines schematically.

3.3 Assumptions on the Shunting Process

Before specifying the rolling stock circulation problem itself, we describe a simplified model for the shunting used by central rolling stock planners in tactical (and operational) planning.

The basic assumption is that time is the most important bottleneck in the shunting process: roughly speaking, any composition change is possible at a shunting yard if the shunting crew has enough time to carry it out. This assumption may look a bit too rough to be realistic. But combined with assumptions on the composition changes as listed in Section 3.4, it captures the major issues of the shunting process that arise when setting up and carrying out the daily rolling stock duties.

We introduce the concept of *inventory* which is the number of units per type that are stored at a given moment at a given station. Units in the inventory of a station are available to be added to any trip departing from there.

Consider an arriving trip t which has the successor trip t' . A composition change between trip t and t' means that the ‘main part’ of the composition goes over from t to t' , while some units may be coupled or uncoupled. In this simplified shunting model, only coupled or uncoupled units increase and decrease the inventory; units that travel through a station with a short stop are not part of the inventory.

Uncoupled units can be used later for other trips. In order to provide enough time for necessary shunting, uncoupled units do not increase the inventories immediately,

but only a certain *re-allocation time*, say 30 minutes, later. We denote the re-allocation time after arriving trip t by $\varrho(t)$.

These assumptions are illustrated schematically in Figure 3.3. Station s is represented by a time-line. Consider an arriving trip t and its successor trip t' . Arrows to the departure node and from the arrival node indicate the units added to the trip from the inventory or detached from the trip and added to the inventory.

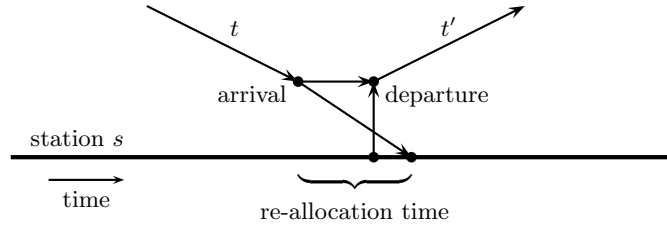


Figure 3.3: Interaction between trip t , its successor trip t' and the inventory.

A trip t that departs from a station and has no predecessor gets its units from the inventory of the station. We assume that units that have been stored at the station prior to the departure of t can be joined to a composition of any order. Similarly, units that arrived in a trip t without any successor trip go to the inventory of the arrival station. Again, the re-allocation time $\varrho(t)$ must be respected when uncoupled units increase the inventory. We illustrate these latter cases in Figure 3.4.

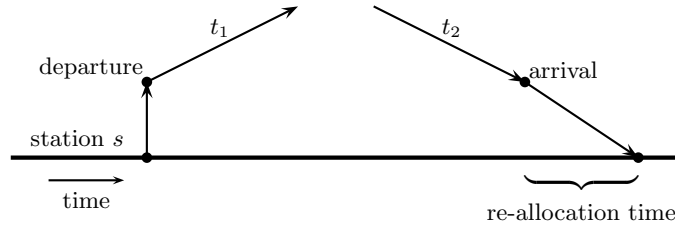


Figure 3.4: Interaction between trips without predecessor or successor trips and the inventory.

We emphasise that sometimes this simple model of the shunting process is too optimistic: the re-allocation time may not be enough to carry out the composition changes. In such cases, central rolling stock planners receive feed-back from local planners and adjust their plans to meet the requirements of the local planners.

3.4 Composition Changes in Practise

In this section, we briefly discuss some requirements that apply to changing a composition between a trip and its successor(s) in all instances of NSR. The correctness of the models later in this chapter shall rely on properties of real-life problem instances that we list here.

Let us focus first on the case when each trip has at most one successor and at most one predecessor. If a trip has a successor, the time between the corresponding arrival and departure ranges from about 5 to 30 minutes. This does not allow complex composition changes. Yet, coupling units to the arriving composition or uncoupling units from it might be possible. The most important restriction in practise is the following:

When a composition is changed between a trip and its successor, then
either coupling or uncoupling can take place, but *not both*. (3.1)

This restriction implies that if a trip and its successor are served by the same number of units, then they have identical compositions.

In practise, every arriving trip has a well-defined *coupling-side* and *uncoupling-side*, each of them is either the front or the rear of the arriving train. Units may be coupled only on the coupling-side and may be uncoupled only from the uncoupling-side. The coupling- and uncoupling-sides depend on the lay-out of the station. Note that in exceptional cases, shunting may be allowed on both sides of a train.

Example. *Figures 3.5 and 3.6 indicate a trip that arrives at a station and its successor trip which is carried out without a direction change. In such cases, coupling is usually allowed only at the front, while uncoupling is usually allowed only at the rear of the arriving train. Accordingly, the composition of the white and the grey unit becomes a composition of a white, a grey and a black unit (Figure 3.5) or a single grey unit (Figure 3.6).*

Units that have just been uncoupled or that are just to be coupled need to be stored at the station. However, it might be difficult to find storage space for a large number of units instantly. This motivates the rule that

at most two units may be coupled or uncoupled at once. (3.2)

In addition, the lay-out of the station may impose further constraints. For example, the storage tracks at station Deventer have a limited length, therefore at most 6 carriages may be coupled or uncoupled at Deventer at once. When using units of

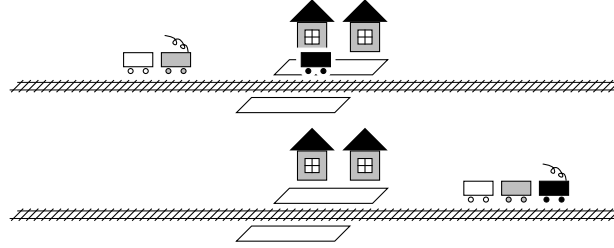


Figure 3.5: Coupling the black unit at the front of the arriving train.

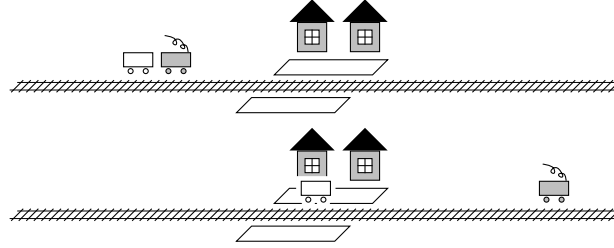


Figure 3.6: Uncoupling the white unit at the rear of the arriving train.

type “Koploper” with 3 or 4 carriages, Deventer allows coupling or uncoupling one or two units of length 3, but only one unit of length 4.

In case of splitting and combining, trips may have two successors and predecessors. In case of combining, the timetable prescribes which predecessor arrives earlier, thereby also which predecessor becomes the front part of the combined train. Similarly, the departure order of the successors is also pre-defined when a train is split. Coupling and uncoupling units when trains are split or combined is undesirable since every extra shunting operation may lead to delays; some stations do not permit such extra operations at all.

Some rolling stock circulations of NSR prescribe an entire composition exchange between a trip t and its successor t' : the units on trip t go into the inventory of the station, while trip t' receives new units from the inventory. Entire composition exchanges are fully accepted and often used in operational planning. However, they are undesirable in tactical planning and used only exceptionally. Therefore, we do not allow composition exchanges in our tactical rolling stock circulation models.

3.5 Problem Formulation

Now we can formulate the tactical rolling stock circulation problem at NSR. The input consists of the following elements.

- The number of available units per type as well as the wished initial and final inventories of the stations. (Later in this section we also discuss slightly less specified input.)
- The timetable that specifies the departure and arrival times of the trips as well as their successor and predecessor trips. Trips may have zero, one or two predecessors and successors. Empty trains where the rolling stock circulation may choose from are also explicitly given in the timetable.
- Upper and lower bounds are given on the length of a composition on each trip. The upper bounds are determined by the lengths of the platforms along the trip. The lower bounds are set according to the passenger demand.
- The list of allowed composition changes between a trip and its successor. We discussed details of the allowed composition changes in Section 3.4.
- The estimated numbers of first and second class passengers for each trip. We explain in Section 3.7 how the passenger demand is computed.

The output contains the compositions assigned to the trips and the duties for the units. Recall that a duty of a unit describes which trips that unit serves in, also specifying the position of the unit in the composition. The output must satisfy the following *primary* constraints.

- Each composition change between a trip and its successor complies with the specifications.
- The initial and final inventories at the stations are equal to the wished inventories.
- The compositions assigned to the trips obey the lower and upper bounds in the problem specification.
- The value of the objective function (described in Section 3.6) is as small as possible.

Example. Consider the rolling stock circulation problem in Figure 3.7. The figure shows the time-space diagram between stations s_1 and s_2 as well as four trips: trips with numbers 101 and 105 from s_1 to s_2 and their successor trips with numbers 102 and 106 back to s_1 . The wished initial inventory of station s_1 is 3, the initial inventory of station s_2 is 1. The wished final inventories are equal to the wished initial inventories.

Then a possible rolling stock schedule is to assign two units to trip 101 and to uncouple one of these units at s_2 , to assign one unit to trip 105 and to couple a unit to it before departing as trip 106. Such a rolling stock schedule is desirable if the passenger demand of trips 101 and 106 is much higher than on trips 102 and 105. The paths of the units through the time-space diagram are indicated by bold lines in the figure. We omitted the time periods from these paths when the units are part of the inventories of the stations.

The rolling stock schedule described above is in fact feasible because the unit that was located at s_2 at the beginning is available to be coupled to the arrived trip 105. Observe that if the time difference between the arrival of trip 101 and the departure of trip 106 is greater than the re-allocation time, the shunting crew has a choice which unit should be coupled to the arriving trip 105.

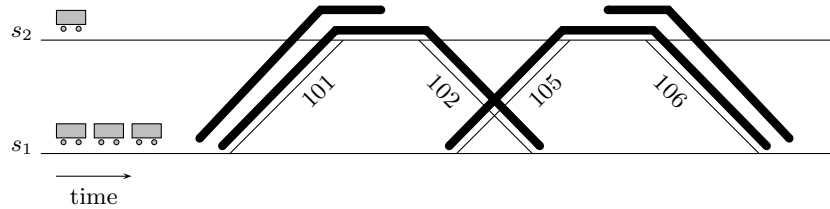


Figure 3.7: Example of a rolling stock circulation.

Besides these primary constraints, the instances may have additional *secondary* constraints like the following three restrictions.

1. The *continuity requirement* states that if a timetable service admits underway composition changes (i.e. it is divided into several trips), at least one unit should follow the complete route of that service. It assures that passengers can travel along the entire route of the timetable service without changing seats underway.

2. Most instances prescribe the initial and final inventories explicitly. These are specified such that stations can satisfy the storage demand of all line groups. Sometimes, usually for line groups that are planned first, the initial and final inventories can be chosen arbitrarily. In such cases, it may make sense not to specify the inventories but to require that for each station and for each type, the initial inventory equals the final inventory. We call such a circulation a *cyclic solution*. A reason for using cyclic solutions is that the expected number of passengers on Tuesdays, Wednesdays and Thursdays are similar and that the tactical timetables on these days are identical. Therefore, repeating an appropriate cyclic solution three times immediately yields a rolling stock schedule for these three days.
3. Stations have limited storage capacity. Therefore the instances may have an explicit upper bound on the number of units (or on their total physical length) that can be stored at a station at the same time.

Throughout the whole chapter, we use the following notations. Let $\mathcal{M}$ be the set of rolling stock types. For each $m \in \mathcal{M}$, let n_m denote the number of available units of type m , c_m the number of carriages in a unit of type m and ℓ_m its physical length in metres. The set of service classes is denoted by $\mathcal{C}$. In instances of NSR, there are two classes: first and second class. A unit of type m has $\kappa_{m,c}$ seats of class c .

The set of stations is denoted by $\mathcal{S}$. The wished initial and final inventory of type m at station s is denoted by $i_{s,m}^0$ and $i_{s,m}^\infty$. Let $\mathcal{T}$ be the set of trips. A trip t is characterised by its departure station $s_d(t)$, arrival station $s_a(t)$, departure time $\tau_d(t)$ and arrival time $\tau_a(t)$. The length of the trip in kilometres is given by the parameter d_t . Trip t must receive compositions with at least $\mu_t^{\min}$ and at most $\mu_t^{\max}$ carriages in total. The passenger demand per service class is $\delta_{t,c}$.

If trip t has one successor, let $\sigma(t)$ denote the successor of trip t . The re-allocation time after trip t is $\varrho(t)$. If trip t has two successors, then they are denoted by $\sigma^1(t)$ and $\sigma^2(t)$. The set $\mathcal{T}_0$ denotes the set of trips that have no predecessor trips, while $\mathcal{T}_\infty$ denotes the set of trips without successor trips.

3.6 Objective Criteria

Creating an appropriate rolling stock circulation means finding a balance between several objectives such as minimising the number of carriage-kilometres (efficiency), the amount of seat shortages (service quality) and the number of composition changes

(robustness). The models described in this chapter allow to make such a trade-off. The objectives are explained below in detail.

The operational cost of rolling stock depends on traction power, but also on maintenance: after a certain number of kilometres, each unit is directed to a maintenance facility for preventive check-up. These two factors are approximately proportional to the carriage-kilometres.

The input of the tactical rolling stock circulation problem contains the estimated number of first and second class passengers for every trip. Good service quality means, among others, that all passengers, in particular first class passengers, have a seat during their journey. Outside rush hours, rolling stock capacity is usually sufficient for this. But during peak hours there may be more passengers than seats. Our measure for the seat shortages is the sum over all trips of the expected number of passengers without a seat, multiplied by the length of the trip. We call this measure *seat-shortage kilometres*.

Compositions can be modified at certain stations by coupling units to a trip or by uncoupling units from it. Changing the composition of a trip requires a number of shunting movements between the platform area and the shunting area of a station. This may lead to disturbances of the train operations. Moreover, there is always the risk that coupling of the units does not work properly, causing additional delays.

Thus a smaller number of composition changes may increase the robustness of the railway system. Moreover, the shunting costs are also related to the crew: each shunting movement requires a train driver. On the other hand, changing the compositions between two trips may allow to use the rolling stock more efficiently and also to decrease seat shortages.

Note that the quality of the solutions can be measured by several other characteristics. For example, the largest seat shortage on a trip should not be too high (e.g. at most 20% of the passengers of a trip during rush hours). Seat shortages in the first class can be penalised more heavily than in the second class. Another criterion is to keep the number of composition changes during rush hours small. The objective function and the constraints in the models below can easily be modified in order to take these additional criteria into account.

3.7 Passenger Demand

In tactical rolling stock circulations, the estimated number of passengers per trip is part of the input. Here, we discuss briefly how these numbers are obtained.

Throughout the whole year, conductors report the number of first and second class passengers of each trip. Moreover, NSR employs or hires people to count passengers and to enquire them about their journey. These counts serve as basis for both the origin-destination matrix in line planning (see Section 2.2.2) and the passenger demand used in tactical planning.

For each trip in the one-week timetable, the mean and the standard deviation of the observations are computed. The passenger demand is set to the mean plus the standard deviation. Tactical rolling stock planners try to allocate at least that many seats to each trip. Then assuming that the observations have a normal distribution, the allocated seat capacity is sufficient with a probability of at least 85%. This definition of the passenger demand has been used for many years at NSR. One must bear in mind that the conductors' counts are somewhat unreliable. Experience shows that the reported counts tend to be significantly too high.

In our models, we implemented this as follows. Here we only consider one of the service classes, for sake of simpler notations. Suppose that passenger counts $d_1, \dots, d_k$ were observed for trip t with relative frequencies $\pi_1, \dots, \pi_k$ and that the standard deviation is s . Suppose that c seats are allocated for the trip. Then the seat shortage on trip t is

$$\max \left(\sum_{i=1}^k \pi_i d_i + s - c, 0 \right). \quad (3.3)$$

However, there are other possible measures for the seat shortages. A natural candidate is the expected number of seat shortages:

$$\sum_{i=1}^k \pi_i \cdot \max(d_i - c, 0) \quad (3.4)$$

and eventually imposing the constraint that with a probability of 85%, there is no seat shortage:

$$\sum_{i: d_i \leq c} \pi_i \geq 85\%. \quad (3.5)$$

Replacing the traditional seat shortage measure (3.3) by (3.4) and eventually by (3.5) in the models may lead to rolling stock circulations with quite differently distributed seat shortages among the trips. Since decision makers have no unanimous opinion when comparing circulations with different seat shortage structures, the alternative measures for seat shortages have not been accepted (yet).

3.8 The Composition Model

In this section, we describe a model that fits to the specifications of NSR and that can be solved in a couple of hours even for the largest instances of NSR (and in a couple of minutes for most instances) by commercial integer programming software. First, we reformulate the problem slightly and specify the input formally. Subsequently, we describe the basic model for the case when no splitting and combining is allowed. Then we extend this basic model to handle splitting and combining of trains. Finally, we report our computational results. Note that further computational results for the Composition Model are given in Sections 3.11.5 and 5.5 where alternative methods for generating rolling stock circulations are compared to the Composition Model.

3.8.1 Observation About the Duties

The output of the rolling stock circulation problem contains the duties of the rolling stock units. However, one does not need to compute the duties explicitly. The assumptions on the shunting process allow one to reformulate the problem:

The rolling stock circulation problem amounts to determining the composition of each trip such that after the departure of a trip from a station, the inventory of each rolling stock type is non-negative.

Indeed, the composition of a trip and of its successor(s) unambiguously determine the number of coupled and uncoupled units per type. Whenever units are to be coupled to a trip, the inventory of the given station can satisfy this demand because the inventory is non-negative after the departure. Then the rolling stock duties themselves can be specified by matching the units entering the inventory of a station to the units of the same type leaving the inventory. In practise, the matching can be created in a Last-In-First-Out manner or in any other way which is convenient for the shunting crew.

3.8.2 Integer Programming Model

In the basic model, we do not allow splitting and combining of trains. We need some more notations.

A *composition* of units is an ordered sequence of elements of $\mathcal{M}$. Taking again the example of units with 3 and 4 carriages, the strings ‘334’ and ‘343’ represent compositions containing two units of length 3 and one unit of length 4. When assigning compositions to trips, we assume that the right-hand side of the string corresponds to the front of the train. The number of units in composition p is denoted by $|p|$. Let

$\nu(p)_m$ denote the number of units of type m in composition p . For a composition p , let $\nu(p)$ be a vector in $\mathbb{Z}^{\mathcal{M}}$ with entries $\nu(p)_m$ for each rolling stock type $m \in \mathcal{M}$.

The set of compositions that are allowed for trip t is denoted by $\mathcal{P}_t$. Members of the set $\mathcal{P}_t$ must comply with the lower bound $\mu_t^{\min}$ and with the upper bound $\mu_t^{\max}$ required for the train length on trip t .

Let $\mathcal{G}_t$ denote the set of pairs of compositions (p, p') such that $p \in \mathcal{P}_t$, $p' \in \mathcal{P}_{\sigma(t)}$ and such that the composition change from p to p' after trip t is allowed. Thus the shunting possibilities are encoded in the sets $\mathcal{G}_t$.

Whenever a composition may be changed between trip t and its successor $\sigma(t)$, we assume that the units may be uncoupled from t , while units can be coupled to $\sigma(t)$. Finally, we assume that for a trip without a predecessor (i.e. for $t \in \mathcal{T}_0$), all units serving t must be coupled just before t . Similarly, for a trip without a successor (i.e. for $t \in \mathcal{T}_\infty$), all units must be uncoupled just after t . These assumptions comply with Figures 3.3 and 3.4.

The main decision variables in the model are the following:

$$\begin{aligned} X_{t,p} &\in \{0, 1\} && \text{whether composition } p \text{ is used for trip } t \text{ } (X_{t,p} = 1) \text{ or not} \\ &&& (X_{t,p} = 0). \\ Z_{t,p,p'} &\in \{0, 1\} && \text{whether trip } t \text{ has composition } p \text{ and trip } \sigma(t) \text{ has compo-} \\ &&& \text{position } p' \text{ } (Z_{t,p,p'} = 1) \text{ or not } (Z_{t,p,p'} = 0). \end{aligned}$$

The variable $N_{t,m}$ denotes the number of units of type m that are used on trip t . We also use the variables $C_{t,m}$ and $U_{t,m}$: $C_{t,m}$ for the number of units of type m that are to be coupled to right before trip t and $U_{t,m}$ for the number of units of type m that are to be uncoupled right after trip t .

The variable $I_{t,m}$ denotes the inventory of type m at station $s_d(t)$ immediately after the departure of trip t . The variables $I_{s,m}^0$ and $I_{s,m}^\infty$ denote the number of units of type m stored at station s at the beginning and at the end of the day.

Now, the basic part of the *Composition Model* reads as follows.

$$\text{Minimise } F(N, X, N) \tag{3.6}$$

subject to

$$\sum_{p \in \mathcal{P}_t} X_{t,p} = 1 \qquad \forall t \in \mathcal{T} \tag{3.7}$$

$$X_{t,p} = \sum_{\substack{p' \in \mathcal{P}_{\sigma(t)} : \\ (p,p') \in \mathcal{G}_t}} Z_{t,p,p'} \quad \forall t \in \mathcal{T} \setminus \mathcal{T}_\infty, p \in \mathcal{P}_t \quad (3.8)$$

$$X_{\sigma(t),p'} = \sum_{\substack{p \in \mathcal{P}_t : \\ (p,p') \in \mathcal{G}_t}} Z_{t,p,p'} \quad \forall t \in \mathcal{T} \setminus \mathcal{T}_\infty, p' \in \mathcal{P}_{\sigma(t)} \quad (3.9)$$

$$N_{t,m} = \sum_{p \in \mathcal{P}_t} \nu(p)_m X_{t,p} \quad \forall t \in \mathcal{T}, m \in \mathcal{M} \quad (3.10)$$

$$C_{\sigma(t),m} = \sum_{\substack{(p,p') \in \mathcal{G}_t : \\ \nu(p')_m > \nu(p)_m}} (\nu(p')_m - \nu(p)_m) \cdot Z_{t,p,p'} \quad \forall t \in \mathcal{T} \setminus \mathcal{T}_\infty, m \in \mathcal{M} \quad (3.11)$$

$$U_{t,m} = \sum_{\substack{(p,p') \in \mathcal{G}_t : \\ \nu(p)_m > \nu(p')_m}} (\nu(p)_m - \nu(p')_m) \cdot Z_{t,p,p'} \quad \forall t \in \mathcal{T} \setminus \mathcal{T}_\infty, m \in \mathcal{M} \quad (3.12)$$

$$C_{t,m} = N_{t,m} \quad \forall t \in \mathcal{T}_0, m \in \mathcal{M} \quad (3.13)$$

$$U_{t,m} = N_{t,m} \quad \forall t \in \mathcal{T}_\infty, m \in \mathcal{M} \quad (3.14)$$

$$\begin{aligned} I_{t,m} = I_{s_d(t),m}^0 &- \sum_{\substack{t' \in \mathcal{T} : s_d(t') = s_d(t), \\ \tau_d(t') \leq \tau_d(t)}} C_{t',m} \\ &+ \sum_{\substack{t' \in \mathcal{T} : s_a(t') = s_d(t), \\ \tau_a(t') \leq \tau_d(t) - \varrho(t')}} U_{t',m} \quad \forall t \in \mathcal{T}, m \in \mathcal{M} \end{aligned} \quad (3.15)$$

$$I_{s,m}^\infty = I_{s,m}^0 - \sum_{\substack{t \in \mathcal{T} : \\ s_d(t) = s}} C_{t,m} + \sum_{\substack{t \in \mathcal{T} : \\ s_a(t) = s}} U_{t,m} \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.16)$$

$$I_{s,m}^0 = i_{s,m}^0 \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.17)$$

$$I_{s,m}^\infty = i_{s,m}^\infty \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.18)$$

$$X_{t,p} \in \{0, 1\} \quad \forall t \in \mathcal{T}, p \in \mathcal{P}_t \quad (3.19)$$

$$N_{t,m}, C_{t,m}, U_{t,m}, I_{t,m} \in \mathbb{R}_+ \quad \forall t \in \mathcal{T}, m \in \mathcal{M} \quad (3.20)$$

$$I_{s,m}^0, I_{s,m}^\infty \in \mathbb{R}_+ \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.21)$$

$$Z_{t,p,p'} \in \mathbb{R}_+ \quad \forall t \in \mathcal{T} (p, p') \in \mathcal{G}_t \quad (3.22)$$

We describe the objective function (3.6) in detail in Section 3.8.4. Constraints (3.7) state that, for each trip, exactly one allowed composition is used. Constraints (3.8) and (3.9) guarantee correct composition changes between consecutive trips. Constraints (3.10) link the compositions of a certain trip to the numbers of units

of each type used on this trip. Constraints (3.11) – (3.14) specify the numbers of coupled and uncoupled units. Constraints (3.15) describe the inventory of the stations: The value $I_{t,m}$ is the number of units that are stored at the station at the begin of the day increased and decreased by one for each unit that is uncoupled or coupled until the departure time of trip t . Recall that uncoupled units increase the inventory only after the re-allocation time has elapsed, while coupled units decrease the inventory immediately. Constraints (3.16) set the variables for final inventories. Constraints (3.17) – (3.18) make sure that the initial and final inventories comply with the problem specifications. Finally, constraints (3.19) – (3.22) describe the variable domains.

One easily verifies the following lemma.

Lemma 3.1. *In any feasible solution of the model (3.6) – (3.22), the variables $U_{t,m}$, $C_{t,m}$, $I_{t,m}$, $I_{s,m}^0$, $I_{s,m}^\infty$ and $Z_{t,p,p'}$ are integral.*

Proof. The variables $I_{s,m}^0$ and $I_{s,m}^\infty$ have fixed integer values. Then the binary variables $X_{p,m}$ uniquely determine the value of $U_{t,m}$, $C_{t,m}$, $I_{t,m}$ and $Z_{t,p,p'}$. ■

Note that the composition changes could be modelled without the variables $Z_{t,p,p'}$ but using additional linear constraints on the variables $X_{t,p}$:

$$X_{\sigma(t),p'} \leq \sum_{\substack{p \in \mathcal{P}_t: \\ (p,p') \in \mathcal{G}_t}} X_{t,p} \quad \forall t \in \mathcal{T} \setminus \mathcal{T}_\infty, p' \in \mathcal{P}_{\sigma(t)}.$$

That is, a composition p' can be assigned to trip $\sigma(t)$ only if trip t has a composition p that may go over to p' . Such a model is described by Alfieri et al. (2002). However, the projection of the linear relaxation of their model onto the space of the variables $X_{t,p}$ strictly contains the projection of the linear relaxation of the Composition Model onto the variables $X_{t,p}$. When applying a branch-and-bound procedure, the Composition Model works with better lower bounds. Moreover, according to Lemma 3.1, one never has to branch on a variable $Z_{t,p,p'}$. Thus we can expect that the number of nodes in the branch-and-bound tree becomes smaller. On the other hand, the Composition Model has to solve larger linear programs. State-of-the-art software such as CPLEX, however, can cope with the linear programs that appear in our applications.

Observe that the model (3.6) – (3.22) can be interpreted as a single-commodity network flow problem with side constraints. Indeed, consider the *transition graph* defined on the node set

$$\{(t,p) \mid t \in \mathcal{T}, p \in \mathcal{P}_t\}$$

and with arc set

$$\left\{ \left((t, p), (\sigma(t), p') \right) \mid t \in \mathcal{T} \setminus \mathcal{T}_0, (p, p') \in \mathcal{G}_t \right\}.$$

Then the core of the model is formed by constraints (3.8) – (3.9), describing a network flow problem in the transition graph. As we shall see soon, the objective function is linear for this network flow problem.

3.8.3 Adding Secondary Constraints

The model (3.6) – (3.22) implements all primary constraints of the problem formulation. Secondary constraints can be added to the model as follows.

Cyclic rolling stock circulations can be formulated by replacing the constraints (3.17) and (3.18) by the following

$$n_m = \sum_{s \in \mathcal{S}} I_{s,m}^0 \quad \forall m \in \mathcal{M}, \quad (3.23)$$

$$I_{s,m}^0 = I_{s,m}^\infty \quad \forall s \in \mathcal{S}, m \in \mathcal{M}. \quad (3.24)$$

That is, for each type, the initial inventories add up to the number of available units and the initial inventories are equal to the final inventories. According to the following lemma, the variables $I_{s,m}^0$, $I_{s,m}^\infty$ and $I_{t,m}$ may still be defined as continuous.

Lemma 3.2. *Consider the model (3.6) – (3.16), (3.19) – (3.22), (3.23), (3.24) for the problem of finding cyclic rolling stock circulations and suppose that the model has a feasible solution. Then it has an integral optimal solution.*

Proof. Consider an optimal solution and suppose that $I_{s_1,m}^0$ is not integral in this solution for a station s_1 and a type m . Then the variable $I_{s_1,m}^\infty$ as well as of the variables $I_{t,m}$ with $s_d(t) = s_1$ have the same fractional part as $I_{s_1,m}^0$ has. Moreover, (3.23) implies that $I_{s_2,m}^0$ is not integral for another station s_2 as well. Again, the variable $I_{s_2,m}^\infty$ as well as the variables $I_{t,m}$ with $s_d(t) = s_2$ have the same fractional part as $I_{s_2,m}^0$. Let $\varepsilon = \min\{I_{s_1,m}^0; 1 - I_{s_2,m}^0\}$. Decreasing the variables $I_{s_1,m}^0$, $I_{s_1,m}^\infty$ as well as the variables $I_{t,m}$ with $s_d(t) = s_1$ by ε and at the same time increasing the variables $I_{s_2,m}^0$, $I_{s_2,m}^\infty$ as well as the variables $I_{t,m}$ with $s_d(t) = s_2$ by ε and keeping all other variables unchanged yields another feasible solution of the model. It is still optimal since the objective function (3.6) only depends on the variables $N_{t,m}$, $X_{t,p}$ and $Z_{t,p,p'}$. Moreover, the number of variables with an integer value has increased. Repeating this rounding step, one obtains an integral optimal solution. ■

The total physical length of the units stored at the stations at any moment is expressed by the variables $I_{s,m}^0$, $I_{s,m}^\infty$ and $I_{t,m}$. Limitation on the storage capacity of stations can be taken into account by the constraints

$$\sum_{m \in \mathcal{M}} \ell_m \cdot I_{t,m} \leq W_s \quad \forall t \in \mathcal{T} : s_d(t) = s \quad (3.25)$$

and by similar constraints for $I_{s,m}^0$ and $I_{s,m}^\infty$ where ℓ_m is the length of a unit of type m and W_s is the total storage capacity of station s . Note that the variables $I_{s,m}^0$ must be integral if constraints (3.25) are inserted into the model. Indeed, otherwise two stations with storage capacity of 10 metres each could “share” a unit of length 20 metres. Once the variables $I_{s,m}^0$ are integral, the integrality of the variables $I_{s,m}^\infty$ and $I_{t,m}$ follows automatically.

For the continuity requirement, we need additional notations. Suppose that a timetable service consists of trips $t_1, \dots, t_k$ where $t_{i+1} = \sigma(t_i)$ for each $i = 1, \dots, k-1$ (see Figure 3.8). The continuity requirement states that at least one unit should serve in each of the trips $t_1, \dots, t_k$. Without loss of generality we may assume that the train carries out these trips without changing the riding direction. At the stations underway, units may be coupled or uncoupled at the left- or the right-hand side of the train. We index the units in a composition increasingly from the left: the left-most unit has index 1, the second unit from the left has index 2, and so on.

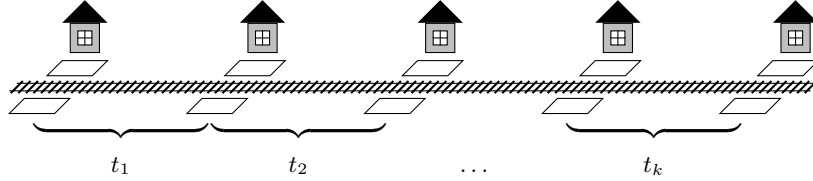


Figure 3.8: A timetable service composed of trips $t_1, \dots, t_k$.

For $i = 1, \dots, k-1$, let α_i^R (and β_i^R) denote the total number of units (irrespective of their types) that are uncoupled from (coupled to) the right-hand side of the composition after trip t_i . Similarly, let α_i^L (and β_i^L) denote the number of uncoupled (coupled) units on the left-hand side of the composition after trip t_i . Let γ_i be the number of units that serve on trip t_i . Clearly, $\gamma_i = \gamma_{i-1} - \alpha_{i-1}^L - \alpha_{i-1}^R + \beta_{i-1}^L + \beta_{i-1}^R$ for every $i = 2, \dots, k$. These parameters are uniquely determined by the variables $X_{t,p}$.

Suppose that a particular unit serves in trips $t_1, \dots, t_k$, let ℓ_i be its index in the composition on trip t_i . Then the numbers ℓ_i satisfy the following inequalities:

$$\ell_i \in \mathbb{Z} \quad \forall i = 1, \dots, k, \quad (3.26)$$

$$1 \leq \ell_i \leq \gamma_i \quad \forall i = 1, \dots, k, \quad (3.27)$$

$$\ell_i = \ell_{i-1} - \alpha_{i-1}^L + \beta_{i-1}^L \quad \forall i = 2, \dots, k, \quad (3.28)$$

$$\ell_i \leq \gamma_i - \alpha_i^R \quad \forall i = 1, \dots, k-1. \quad (3.29)$$

Indeed, ℓ_i is an integer between 1 and γ_i . The indices $\ell_1, \dots, \ell_k$ directly depend on coupling and uncoupling at the left-hand side as described by (3.28). Finally, (3.29) holds since the unit is not uncoupled after trip t_i on the right-hand side.

Conversely, suppose that the system (3.26) – (3.29) with variables $\ell_1, \dots, \ell_k$ has a solution. Then the unit of index ℓ_1 on trip t_1 serves in trips $t_2, \dots, t_k$ with indices $\ell_2, \dots, \ell_k$. That is, the continuity constraint is equivalent with the feasibility of the system (3.26) – (3.29).

The values ℓ_i are determined by ℓ_1 . Therefore (3.26) – (3.29) can be reformulated as follows.

$$\ell_1 \in \mathbb{R} \quad (3.30)$$

$$1 \leq \ell_1 \leq \gamma_1 \quad (3.31)$$

$$1 \leq \ell_1 - \sum_{j < i} \alpha_j^L + \sum_{j < i} \beta_j^L \leq \gamma_i - \alpha_i^R \quad \forall i = 2, \dots, k-1 \quad (3.32)$$

$$1 \leq \ell_1 - \sum_{j < k} \alpha_j^L + \sum_{j < k} \beta_j^L \leq \gamma_k \quad (3.33)$$

The variables $\ell_2, \dots, \ell_k$ are expressed in terms of ℓ_1 . Moreover, there are only integral lower and upper bounds on ℓ_1 . Therefore ℓ_1 may be chosen continuous.

3.8.4 Objective Function

As was mentioned before, the objective function contains three major elements: (i) carriage-kilometres, (ii) seat-shortage kilometres and (iii) the number of composition changes. Given the above defined decision variables, these elements can be computed easily.

For the total number of carriage-kilometres CKM, the following holds:

$$\text{CKM}(N) = \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}} d_t \cdot c_m \cdot N_{t,m}$$

where d_t is the length of trip t and c_m is the number of carriages in a unit of type m .

The total number of seat-shortage kilometres SKM satisfies

$$\text{SKM}(X) = \sum_{c \in \mathcal{C}} \sum_{t \in \mathcal{T}} \sum_{p \in \mathcal{P}_t} d_t \cdot s_{t,p,c} \cdot X_{t,p}.$$

Here $\mathcal{C}$ is the set of service classes and $s_{t,p,c}$ denotes the anticipated number of seat shortages of class c when composition p is used for trip t . The shortage is computed by comparing the forecasted number of passengers to the capacity of the compositions as described in Section 3.7.

As stated in (3.1), coupling and uncoupling at the same time is not allowed. Therefore, the total number of composition changes CCH can be determined by adding all the variables $Z_{t,p,p'}$ with $\nu(p) \neq \nu(p')$. Thus

$$\text{CCH}(Z) = \sum_{t \in \mathcal{T}} \sum_{\substack{(p,p') \in \mathcal{G}_t : \\ \nu(p) \neq \nu(p')}} Z_{t,p,p'}.$$

The objective function $F(X, Z, N)$ is a non-negative linear combination of these criteria, the weight factors reflecting the relative importance. In order to control the elements of the objective function more explicitly, we may also introduce upper bounds on CKM, SKM and CCH.

3.8.5 New Integer Decision Variables

The variables $X_{t,p}$ describe the exact composition for each trip t . Now it turns out that, if new binary decision variables are introduced to describe the length of a train, then the integrality constraint for the variables $X_{t,p}$ can be dropped. By this reformulation, the number of binary decision variables decreases. For example, in the Noord-Oost instance, about 9,900 binary variables $X_{t,p}$ are replaced by about 5,700 new binary variables.

The new decision variables determine the *number* of units per type that is allocated to the trips, without specifying the exact order in the train. For a trip t , we define the set of vectors in $\mathbb{Z}_+^{\mathcal{M}}$

$$\mathcal{B}_t = \{\nu(p) \mid p \in \mathcal{P}_t\}.$$

Recall that $\nu(p)$ describes the numbers of units per type in composition p . We introduce the binary variables

$Y_{t,b} \in \{0, 1\}$ whether combination $b \in \mathcal{B}_t$ is used for trip t ($Y_{t,b} = 1$) or
not ($Y_{t,b} = 0$).

Their connection with the already defined variables $X_{t,p}$ is described by

$$Y_{t,b} = \sum_{p \in \mathcal{P}_t : b = \nu(p)} X_{t,p} \quad \forall t \in \mathcal{T} \ b \in \mathcal{B}_t. \quad (3.34)$$

The variables $Y_{t,b}$ correspond to the higher-level decisions like capacity allocation, while the variables $X_{t,p}$ fill in the fine details of the solution.

Theorem 3.3. *Consider the model (3.6) – (3.22) extended by the binary variables $Y_{t,b}$ and by the constraints (3.34). Replace the constraint (3.19) in this model by $0 \leq X_{t,p} \leq 1 \ \forall t \in \mathcal{T}$ and $p \in \mathcal{P}_t$. Then this relaxed mixed integer program has an integral optimal solution whenever it has a feasible solution.*

Proof. Let $(X, Y, N, C, U, I^0, I^\infty, I, Z)$ be an optimal solution of the relaxed problem: the variables $Y_{t,b}$ are binary, while the variables $X_{t,p}$ may have fractional values. Consider the transition graph defined in Section 3.8.2. Set the capacity of an arc to 0 if the corresponding value $Z_{t,p,p'}$ is zero and set the capacity to 1 otherwise. The values $Z_{t,p,p'}$ form a network flow in this graph. Thus there exists an integer valued network flow $\hat{Z}_{t,p,p'}$ with the same amount of flow. Define the values $\hat{X}_{t,p}$ according to the constraints (3.8) – (3.9).

We shall prove that if we fix the binary values of the variables $Y_{t,b}$ in the relaxed model, then the values of the variables $N_{t,m}$, $C_{t,m}$, $U_{t,m}$, $I_{s,m}^0$, $I_{s,m}^\infty$ and $I_{t,m}$ as well as the objective criteria CKM, SKM and CCH are uniquely determined even if $X_{t,p}$ and $Z_{t,p,p'}$ may have fractional values. Then it follows immediately that $(\hat{X}, Y, N, C, U, I^0, I^\infty, I, \hat{Z})$ is an integral feasible (and optimal) solution of the relaxed model.

Indeed, the value of the variables $I_{t,m}^0$ and $I_{t,m}^\infty$ is fixed by the constraints. Moreover, one easily derives that for each trip t and rolling stock type m , we have

$$N_{t,m} = \sum_{b \in \mathcal{B}_t} b_m Y_{t,b}. \quad (3.35)$$

Consider now any trip t . The integrality of the variables $Y_{t,b}$ implies that a variable $X_{t,p}$ can have a positive value only if the composition p contains as many units of each type as the variables $N_{t,m}$ indicate. Then (3.8) and (3.9) imply that all members of the set

$$\{(p, p') \in \mathcal{G}_t \mid Z_{t,p,p'} > 0\}$$

have the same differences $\nu(p')_m - \nu(p)_m$ for every $m \in \mathcal{M}$ (namely $N_{\sigma(t),m} - N_{t,m}$). Therefore

$$C_{t,m} = \max \{N_{\sigma(t),m} - N_{t,m}, 0\}$$

and

$$U_{t,m} = \max \{N_{t,m} - N_{\sigma(t),m}, 0\}.$$

Then it follows that the variables $I_{t,m}$ are also uniquely determined.

Clearly, the number of carriage-kilometres CKM only depends on the variables $Y_{t,b}$. Moreover, the number of seat-shortage kilometres SKM is also determined by the variables $Y_{t,b}$. Indeed, all compositions that correspond to a certain combination $b \in \mathcal{B}_t$ have the same capacity and thus the same amount of seat shortages (denoted by $s_{t,b}$) on trip t . Therefore

$$\text{SKM} = \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{B}_t} d_t \cdot s_{t,b} \cdot Y_{t,b}.$$

Finally, each trip has a contribution

$$\sum_{\nu(p) \neq \nu(p')} Z_{t,p,p'}$$

to the number of composition changes CCH. This contribution is 1 if $N_{t,m} \neq N_{\sigma(t),m}$ for some rolling stock type m and 0 otherwise. Then (3.35) implies that CCH is determined by the variables $Y_{t,m}$. $\blacksquare$

3.8.6 Splitting and Combining Trains

In order to handle splitting and combining trains, we use an extension of the basic model. First, we describe splitting trains.

Let $\mathcal{T}^s$ be the set of trips after which the corresponding train is split into two trains. Then for each trip $t \in \mathcal{T}^s$, there are two trips $\sigma^1(t)$ and $\sigma^2(t)$ that take place immediately after splitting trip t . Recall that the departure order of the trips $\sigma^1(t)$ and $\sigma^2(t)$ is determined by the timetable.

Let $\mathcal{G}_t^s$ be the set of 3-tuples (p, p_1, p_2) such that $p \in \mathcal{P}_t$, $p_1 \in \mathcal{P}_{\sigma^1(t)}$, $p_2 \in \mathcal{P}_{\sigma^2(t)}$ and p is the concatenation of the strings p_1 and p_2 . We introduce a variable

$$Z_{t,p,p_1,p_2}^s \in [0, 1] \tag{3.36}$$

for each $t \in \mathcal{T}^s$ and $(p, p_1, p_2) \in \mathcal{G}_t^s$. These variables are linked to the other variables similarly to the constraints (3.8) – (3.9):

$$X_{t,p} = \sum_{p_1, p_2 : (p, p_1, p_2) \in \mathcal{G}_t^s} Z_{t,p,p_1,p_2}^s \quad \forall p \in \mathcal{P}_t, \quad (3.37)$$

$$X_{\sigma^1(t), p_1} = \sum_{p, p_2 : (p, p_1, p_2) \in \mathcal{G}_t^s} Z_{t,p,p_1,p_2}^s \quad \forall p_1 \in \mathcal{P}_{\sigma^1(t)}, \quad (3.38)$$

$$X_{\sigma^2(t), p_2} = \sum_{p, p_1 : (p, p_1, p_2) \in \mathcal{G}_t^s} Z_{t,p,p_1,p_2}^s \quad \forall p_2 \in \mathcal{P}_{\sigma^2(t)}. \quad (3.39)$$

It may be possible to modify the compositions further by allowing to couple or uncouple units when the train is split. This can easily be modelled by modifying the sets $\mathcal{G}_t^s$. The variables $C_{t,m}$ and $U_{t,m}$ as well as the objective function should then also be adjusted as follows. Assuming that units may be uncoupled from the arriving trip $t \in \mathcal{T}^s$ and may only be coupled to $\sigma^1(t)$, the following constraints are needed for each $m \in \mathcal{M}$:

$$U_{t,m} = \sum_{\substack{(p, p_1, p_2) \in \mathcal{G}_t^s : \\ \nu(p_1)_m + \nu(p_2)_m > \nu(p)_m}} (\nu(p_1)_m + \nu(p_2)_m - \nu(p)_m) \cdot Z_{t,p,p_1,p_2}^s, \quad (3.40)$$

$$C_{\sigma^1(t), m} = \sum_{\substack{(p, p_1, p_2) \in \mathcal{G}_t^s : \\ \nu(p)_m > \nu(p_1)_m + \nu(p_2)_m}} (\nu(p)_m - \nu(p_1)_m - \nu(p_2)_m) \cdot Z_{t,p,p_1,p_2}^s, \quad (3.41)$$

$$C_{\sigma^2(t), m} = 0. \quad (3.42)$$

Furthermore, the objective function must be adjusted by adding the following term to the number of composition changes:

$$\sum_{t \in \mathcal{T}^s} \sum_{\substack{(p, p_1, p_2) \in \mathcal{G}_t^s : \\ \nu(p) \neq \nu(p_1) + \nu(p_2)}} Z_{t,p,p_1,p_2}^s.$$

Combining two trains can be described in a similar way. One defines the set $\mathcal{T}^c$ of trips t that have two predecessors t_1, t_2 as well as the sets $\mathcal{G}_t^c$ of 3-tuples (p, p_1, p_2) such that $p \in \mathcal{P}(t)$, $p_1 \in \mathcal{P}(t_1)$, $p_2 \in \mathcal{P}(t_2)$ and p is the concatenation of the strings p_1 and p_2 . Then one introduces the variables Z_{t,p,p_1,p_2}^c for $t \in \mathcal{T}^c$ and $(p, p_1, p_2) \in \mathcal{G}_t^c$.

Then Lemma 3.1 extends to the following lemma.

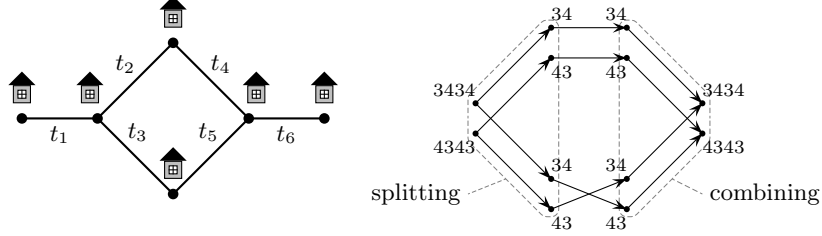


Figure 3.9: A pair of splitting and combining events and the corresponding composition graph. The composition is reversed after trip t_3 .

Lemma 3.4. *In any feasible solution of the model (3.6) – (3.22), (3.36) – (3.39) and eventually (3.40) – (3.42), the variables $U_{t,m}$, $C_{t,m}$, $I_{s,m}^0$, $I_{s,m}^\infty$, $I_{t,m}$, $Z_{t,p,p'}$ and Z_{t,p,p_1,p_2}^s have integral values.*

Also in the case of splitting and combining we can introduce the decision variables $Y_{t,b}$ described in Section 3.8.5. Unfortunately, Theorem 3.3 does not extend. The following example shows that introducing the variables $Y_{t,b}$ and relaxing the variables $X_{t,p}$ may lead to a feasible model which does not have any feasible integral solution.

Example. Figure 3.9 indicates trips $t_1, \dots, t_6$. Trip t_1 is split into trips t_2 and t_3 . The successors of t_2 and t_3 are t_4 and t_5 , they are combined to trip t_6 . The composition is unchanged after trip t_2 but reversed after trip t_3 .

Trips t_1 and t_6 can have composition ‘3434’ or ‘4343’ and trips t_2, t_3, t_4, t_5 can have composition ‘34’ or ‘43’. Then we have $\mathcal{P}_{t_1} = \mathcal{P}_{t_6} = \{3434, 4343\}$, $\mathcal{P}_{t_2} = \mathcal{P}_{t_3} = \mathcal{P}_{t_4} = \mathcal{P}_{t_5} = \{34, 43\}$, $\mathcal{G}_{t_1}^s = \{(3434, 34, 34), (4343, 43, 43)\}$, $\mathcal{G}_{t_2} = \{(34, 34), (43, 43)\}$, $\mathcal{G}_{t_3} = \{(34, 43), (43, 34)\}$ and $\mathcal{G}_{t_6}^c = \{(3434, 34, 34), (4343, 43, 43)\}$. Setting the variables $X_{t_i,p}$ with $i = 1, \dots, 6$ and $p \in \mathcal{P}_{t_i}$ to $\frac{1}{2}$ and setting also all variables Z, Z^s and Z^c to $\frac{1}{2}$ yields a fractional solution of the Composition Model. There is, however, no integral solution.

So we required all variables $Y_{t,b}$ and $X_{t,p}$ to be integral. However, when we tested the model with $X_{t,p}$ relaxed to rational, we always found fully integral optimal solutions. Therefore we gave the variables $Y_{t,b}$ higher priority than the variables $X_{t,p}$ (see Section 3.10.1).

3.8.7 Special Structure of the Instances

In order to get a tighter linear description of the problem, we can exploit the special structure of the instances as follows. In the Noord-Oost line group, splitting and combining events often appear in pairs: Whenever a train after trip t_1 is split in Utrecht (Ut), one part goes to The Hague (Gvc) (trip t_2), the other part goes to Rotterdam (Rtd) (trip t_3). These parts turn back to Utrecht (trips t_4 and t_5) where they are combined again and leave Utrecht (trip t_6) together. When a train is split or combined in Utrecht, no units can be coupled or uncoupled. However, coupling and uncoupling is allowed in Rotterdam and The Hague. (See Figure 3.10.)

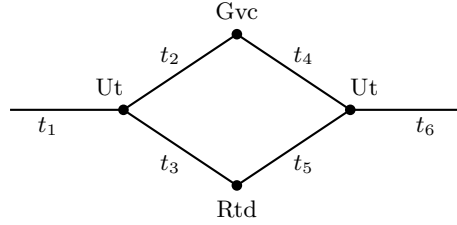


Figure 3.10: Splitting and combining in Utrecht.

The basic idea is to define binary decision variables

$$Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}}$$

that have value 1, if for each $i = 1, \dots, 6$, trip t_i has composition p_i . Of course, we only have such a new variable if the compositions p_i comply with all composition change rules. That is, if $p_i \in \mathcal{P}(t_i)$, $(p_1, p_2, p_3) \in \mathcal{G}_{t_1}^s$, $(p_2, p_4) \in \mathcal{G}_{t_2}$, $(p_3, p_5) \in \mathcal{G}_{t_3}$ and $(p_6, p_4, p_5) \in \mathcal{G}_{t_6}^c$. We call such a 6-tuple $(p_1, \dots, p_6)$ a *splitting-combining scenario*.

Then we do not need the variables $X_{t,p}$, $Y_{t,b}$ and $Z_{t,p,p'}$ that are attached to the trips t_2 , t_3 , t_4 and t_5 , thus we delete them from the model. We also delete the variables Z^s (and Z^c) we used to describe splitting (and combining) at trips t_1 (and t_6). Instead, we introduce the following constraints to describe the composition changes:

$$X_{t_1, p} = \sum_{p_2, \dots, p_6} Z_{t_1, p, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad \forall p \in \mathcal{P}_{t_1}, \quad (3.43)$$

$$X_{t_6, p} = \sum_{p_1, \dots, p_5} Z_{t_1, p_1, p_2, p_3, p_4, p_5, p}^{\text{Ut}} \quad \forall p \in \mathcal{P}_{t_6}. \quad (3.44)$$

These are analogous with constraints (3.8) – (3.9) and (3.37) – (3.39).

We still need the variables $I_{t_i,m}$ for $i = 1, \dots, 6$ to ensure non-negative inventories at any moment. Moreover, in order to keep track of the number of (un)coupled units in Rotterdam and The Hague, we use integer variables $U_{t_2,m}$, $U_{t_3,m}$, $C_{t_4,m}$ and $C_{t_5,m}$ and express them as weighted sums of the variables Z^{Ut} as follows.

$$U_{t_1,m} = C_{t_2,m} = C_{t_3,m} = 0 \quad \forall m \in \mathcal{M} \quad (3.45)$$

$$U_{t_4,m} = U_{t_5,m} = C_{t_6,m} = 0 \quad \forall m \in \mathcal{M} \quad (3.46)$$

$$U_{t_2,m} = \sum_{\substack{p_1, \dots, p_6 : \\ \nu(p_2)_m > \nu(p_4)_m}} (\nu(p_2)_m - \nu(p_4)_m) \cdot Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad \forall m \in \mathcal{M} \quad (3.47)$$

$$C_{t_4,m} = \sum_{\substack{p_1, \dots, p_6 : \\ \nu(p_4)_m > \nu(p_2)_m}} (\nu(p_4)_m - \nu(p_2)_m) \cdot Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad \forall m \in \mathcal{M} \quad (3.48)$$

$$U_{t_3,m} = \sum_{\substack{p_1, \dots, p_6 : \\ \nu(p_3)_m > \nu(p_5)_m}} (\nu(p_3)_m - \nu(p_5)_m) \cdot Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad \forall m \in \mathcal{M} \quad (3.49)$$

$$C_{t_5,m} = \sum_{\substack{p_1, \dots, p_6 : \\ \nu(p_5)_m > \nu(p_3)_m}} (\nu(p_5)_m - \nu(p_3)_m) \cdot Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad \forall m \in \mathcal{M} \quad (3.50)$$

The variables Z^{Ut} appear in the objective function with weights that indicate the carriage-kilometres, seat-shortage kilometres and the number of composition changes for the deleted trips $t_2, \dots, t_5$. It follows immediately that the model modified in such a way still describes the same rolling stock circulation problem. One might wonder whether the integrality of the variables Z^{Ut} can always be relaxed. Unfortunately, this cannot always be the case as the following example shows.

Example. Suppose that penalties for composition changes are the only non-zero weighted terms in the objective function. Suppose that trip t_1 has composition ‘3333’ (i.e. 4 units with 3 carriages) and trip t_6 has composition ‘33’. Moreover, suppose that the other parts of the model enforce uncoupling one unit each both at The Hague and at Rotterdam, that is $U_{t_2,3} = U_{t_3,3} = 1$. Then there is only one splitting-combining scenario that realises these requirements: assigning composition ‘33’ to both t_2 and t_3 and uncoupling one unit each at both The Hague and Rotterdam (see Figure 3.11(a)). This solution pays twice the composition change penalty.

However, there is a cheaper fractional solution: We have a variable Z_1^{Ut} (with somewhat sloppy notation) of a splitting-combining scenario which assigns composition '333' to trip t_2 and composition '3' to t_3 , t_4 and t_5 (see Figure 3.11(b)). Moreover, we have a variable Z_2^{Ut} of another splitting-combining scenario which assigns composition '333' to trip t_3 and composition '3' to t_2 , t_4 and t_5 (see Figure 3.11(c)). These latter solutions pay the penalty for the composition changes only once, but they do not comply with the external inventory restrictions. However, the solution $\frac{1}{2}Z_1^{Ut} + \frac{1}{2}Z_2^{Ut}$ does and it pays the composition change penalty only once.

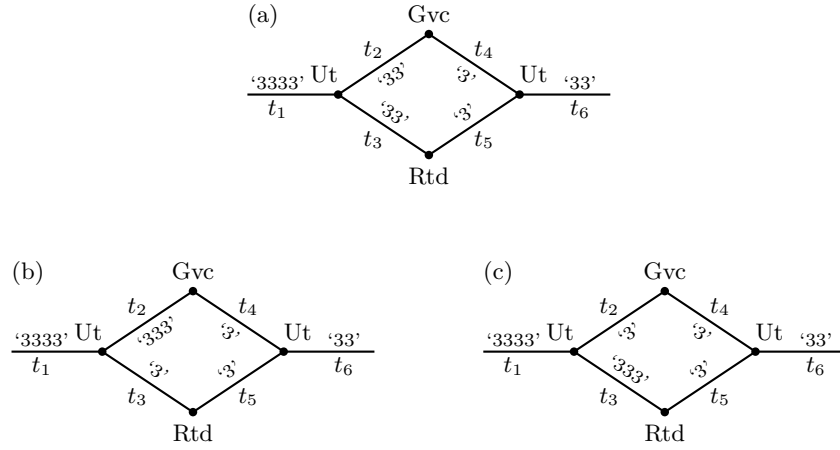


Figure 3.11: Three different ways of splitting and combining in Utrecht.

The following example shows that integer solutions may not exist even if the linear relaxation of the problem has a feasible solution.

Example. Consider the two splitting-combining scenarios in Figure 3.12(a) and Figure 3.12(b). They both give feasible ways to obtain a composition '3443' on trip t_6 from a composition '3333' on trip t_1 . However, taking the convex combination of the corresponding variables Z^{Ut} with coefficients $\frac{1}{2}$ would describe a composition change shown in Figure 3.12(c): uncoupling a unit of length 3 and coupling a unit of length 4 at the same time both in The Hague and in Rotterdam. This is not allowed in our shunting model.

In order to avoid such troubles when relaxing the integrality of the variables Z^{Ut} , we introduce the binary variables U^{Gvc} and U^{Rtd} that express whether or not units

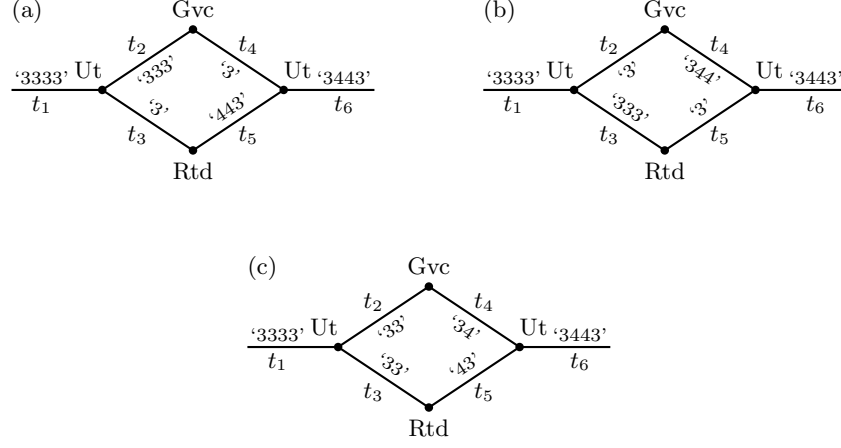


Figure 3.12: Two legal splitting-combining scenarios (a) and (b) and an impossible one (c).

are uncoupled after the arrival of trip t_2 and t_3 . Similarly, we introduce the binary variables C^{Gvc} and C^{Rtd} that express whether or not units are coupled before the departure of trips t_4 and t_5 . These variables can also be written as sums of the variables Z^{Ut} :

$$U^{\text{Gvc}} = \sum_{p_1, \dots, p_6: |p_2| > |p_4|} Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad (3.51)$$

$$U^{\text{Rtd}} = \sum_{p_1, \dots, p_6: |p_3| > |p_5|} Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad (3.52)$$

$$C^{\text{Gvc}} = \sum_{p_1, \dots, p_6: |p_4| > |p_2|} Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad (3.53)$$

$$C^{\text{Rtd}} = \sum_{p_1, \dots, p_6: |p_5| > |p_3|} Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{\text{Ut}} \quad (3.54)$$

The following lemma states that these additional variables in fact suffice to relax the integrality of the variables Z^{Ut} . The proof uses the following specific properties of the Noord-Oost line group:

- There are two rolling stock types: units with 3 carriages and units with 4 carriages.
- Each trip has at most 15 carriages.

- At most two units can be coupled or uncoupled at the same time.
- Every time when a train is split in Utrecht, its left-most part continues to The Hague and its right-most part continues to Rotterdam. Similarly, every time when trains are combined in Utrecht, the left-most part arrives from The Hague and the right-most part arrives from Rotterdam.

These properties may or may not hold for other instances.

Lemma 3.5. *Consider the Composition Model and modify it for a single pair of splitting-combining events as described above. That is, delete the variables X, Y, Z, Z^s and Z^c that are attached to the trips $t_2, \dots, t_5$. Moreover, require $C_{t_i, m}$ and $U_{t_i, m}$ to be integral for $i = 2, \dots, 5$ and for each $m \in \mathcal{M}$. Introduce the binary variables $C^{Gvc}, C^{Rtd}, U^{Gvc}$ and U^{Rtd} , as well as continuous variables Z^{Ut} in the range $[0; 1]$. Finally, add the constraints (3.43) – (3.54) to the model. Then this model has an integral optimal solution whenever it has a feasible solution.*

Proof. Consider an optimal solution of the model. We claim that for each splitting-combining scenario $(p_1, \dots, p_6)$ with $Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{Ut} > 0$ and for each $m \in \mathcal{M}$, we have

$$U_{t_2, m} = \max \{ \nu(p_2)_m - \nu(p_4)_m, 0 \}.$$

That is, in each splitting-combining scenario that has positive value in the solution, $U_{t_2, m}$ units of type m are uncoupled at The Hague after trip t_2 .

Indeed, if $U^{Gvc} = 0$, then the claim follows from (3.47) and (3.51). Suppose now that $U^{Gvc} = 1$. Then each splitting-combining scenario with positive Z^{Ut} value uncouples either one or two units in The Hague after trip t_2 . For each rolling stock type, zero, one or two units are uncoupled in any splitting-combining scenario. Equation (3.47) can be interpreted as the weighted sum of the positive variables Z^{Ut} where the weights $\nu(p_2)_m - \nu(p_4)_m$ are either zero or one or two:

$$U_{t_2, m} = \sum_{\substack{p_1, \dots, p_6: |p_2| > |p_4| \text{ and} \\ Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{Ut} > 0}} (\nu(p_2)_m - \nu(p_4)_m) \cdot Z_{t_1, p_1, p_2, p_3, p_4, p_5, p_6}^{Ut}. \quad (3.55)$$

If no weight zero or two appears in (3.55), then all weights must be equal – just as we claimed – since $U_{t_2, m}$ is an integer number. Thus we may assume that both weights zero and two occur in (3.55).

Let m' denote the other rolling stock type. Then the number of uncoupled units per type can be as follows:

m	m'	Sum of Z^{Ut} values
2	0	α_1
1	0	α_2
1	1	α_3
0	1	α_4
0	2	α_5

The numbers α_i are the sums of the variables Z^{Ut} that correspond to the numbers of uncoupled units per type. These numbers α_i satisfy

$$\alpha_1, \dots, \alpha_5 \geq 0,$$

$$\sum_{i=1}^5 \alpha_i = 1.$$

We also have

$$\alpha_1 > 0,$$

$$\alpha_4 + \alpha_5 > 0$$

since these are the sums of the variables Z^{Ut} that have weight zero and two in (3.55), respectively. Moreover,

$$U_{t_2, m} = 2\alpha_1 + \alpha_2 + \alpha_3,$$

$$U_{t_2, m'} = 2\alpha_5 + \alpha_3 + \alpha_4.$$

Therefore, using the fact that $U_{t_2, m}$ and $U_{t_2, m'}$ are equal to 0, 1 or 2, one easily derives that

$$\alpha_1 = \alpha_5 > 0.$$

That is, there exists a splitting-combining scenario where two units of type m are uncoupled in The Hague and there exists another splitting-combining scenario where two units of type m' are uncoupled. Therefore, the composition on trip t_1 (which is the same in every splitting-combining scenario with positive value Z^{Ut}) must contain both of the substrings mm and $m'm'$. Using units with 3 or 4 carriages and having

an upper bound of 15 carriages on the train lengths, it means that the composition used in trip t_1 must be either $mmm'm'$ or $m'm'mm$.

Suppose that t_1 has composition $mmm'm'$ (the other case being analogous). After splitting, always the left-hand side of the split train goes to The Hague. Thus the splitting-combining scenario which uncouples two units of type m' requires that the entire train of t_1 goes to The Hague, not leaving any unit for trip t_3 . However, every splitting-combining scenario allocates at least unit to each of the trips $t_2, \dots, t_5$.

One can show similarly that all splitting-combining scenarios with positive value Z^{U_t} intend to uncouple $U_{t_3,m}$ units of type m at Rotterdam and to couple $C_{t_4,m}$ and $C_{t_5,m}$ units of type m at The Hague and Rotterdam. That is, these splitting-combining scenarios may be different but their connection to the rest of the model is identical. Then an optimal solution uses only one of them: the one that has the smallest coefficient in the objective function. ■

3.9 Solution Approaches

Having developed the Composition Model for the rolling stock circulation problem, we still have to solve it. In this section, we describe two solution approaches.

Peeters and Kroon (2003) describe the model above for line groups without splitting and combining. Their main idea is to focus on the network flow structure of the problem. Being a single-commodity flow problem in the transition graph, the model admits Dantzig-Wolfe decomposition for solving the linear relaxation quickly: define a variable for each directed path in the graph and write the network flow as the sum of the path flows. The paths in the transition graph only interact in the inventory constraints. Once the linear relaxation is solved, a branch-and-price framework is applied to obtain integer solutions. This approach works very well when implemented for real-life instances of NSR without splitting and combining. Within a couple of seconds, the instances can be solved to optimality.

This solution method extends for splitting and combining only if after splitting a train, the split parts are combined with each other again as shown in Figure 3.10. However, about one third of the splitting-combining events in the Noord-Oost line group behave differently. If a train is split in Amersfoort, the part going to Amsterdam will not meet again the other going to Schiphol. Both parts do turn back, but the part arriving back from Schiphol can only catch a train from Amsterdam that was split in Amersfoort an hour later (see Figure 3.13). This property of the timetable makes the appealing Dantzig-Wolfe decomposition approach impossible.

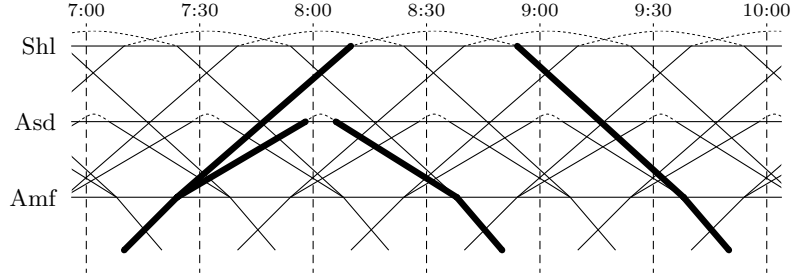


Figure 3.13: Trains between Amersfoort (Amf), Amsterdam (Asd) and Schiphol (Shl). After splitting in Amersfoort, the two parts do not meet any more.

Focusing from the beginning on the Noord-Oost problem, in this thesis we use a commercial MIP solver to obtain solutions. In order to obtain solutions quickly, various techniques need to be applied. We discuss the computational results for the Noord-Oost problem in the next section. Computational results for the Composition Model on other instances of NSR are reported in Sections 3.11.5 and 5.5.

3.10 Computational Results for the Noord-Oost

For our computations, we use the modelling software ILOG OPL Studio 3.7 and the mixed integer programming solver ILOG CPLEX 9.0 on a PC with an Intel Pentium IV 3.0 GHz processor and with 512 Mb internal memory. Our goal is to find feasible solutions of good quality in reasonable time. Mostly, we stop the computations after a couple of hours of CPU time.

We implement the Composition Model for the Noord-Oost line group for a generic Tuesday. Lacking specified initial and final inventories, we look for cyclic solutions. We consider several problem instances with the same timetable, the instances differ in explicit bounds on the objective criteria and in their weight factors in the objective function.

We compare our solutions to a manually created rolling stock circulation. In Table 3.1 we give the values of the three objective criteria in this reference solution. Experiments are carried out with several objective functions. The weight factors for carriage-kilometres (CKM), seat-shortage kilometres (SKM) and the number of composition changes (CCH) are also given in Table 3.1. According to the preferences of NSR, we give a low weight factor to CKM in the objective function but require

	CKM	SKM	CCH
value in practise	317,853	555,215	135
weight in Obj1	1	1	5,000
weight in Obj2	1	1	1,000
weight in Obj3	1	1	100
weight in Obj4	1	10	1,000
weight in Obj5	1	10	100

Table 3.1: The values of the objective criteria in the reference solution and the weight factors used in the experiments.

	Without aggregation		With aggregation	
	MIP	Reduced MIP	MIP	Reduced MIP
# integer var.	16,992	–	15,848	–
# contin. var.	33,901	–	44,769	–
# variables	50,893	34,228	60,617	39,251
# constraints	32,308	17,896	28,149	15,348
# non-zeros	195,184	136,105	287,286	177,622

Table 3.2: Dimensions of the MIP's: number of variables (integer and continuous), constraints and non-zeros in the matrix. Columns under 'With aggregation' are obtained by implementing the variables and constraints described in Section 3.8.7.

the constraint

$$\text{CKM} \leq 318,000 \quad (3.56)$$

expressing that the number of carriage-kilometres is not higher than in the reference solution. In additional experiments we explicitly minimise the carriage-kilometres while giving bounds on seat-shortages and the number of composition changes.

The first two columns of Table 3.2 (under 'Without aggregation') contain the dimensions of the mixed integer program as well as the size of the reduced MIP, created by CPLEX in the pre-processing phase.

Finding feasible solutions of good quality turns out to be quite time-consuming. In order to speed up the solution process, we apply various techniques.

3.10.1 CPLEX Parameters

Fine-tuning the parameters of CPLEX has a large impact on the solution times. We use the barrier method to solve the root node of the branch-and-bound tree. Then

we apply the dual simplex method for any other node. Using the built-in *heuristics* frequently, applying *probing*, *perturbing* the objective function and using *branching priorities* (explained below) turns out to be particularly helpful.

The variables have a hierarchical structure. The variables $N_{t,m}$ describe the number of units of a given type m used for trip t . The variables $Y_{t,b}$ represent decisions one level lower, while the variables $X_{t,p}$ specify the finer details. As we observed, the objective criteria are basically functions of the variables $Y_{t,b}$.

It turns out to be advantageous to branch first on the variables $N_{t,m}$, then on the variables $Y_{t,b}$ and at last on the variables $X_{t,p}$. This can be explained as follows. The values $N_{t,m}$ describe a rough estimate of the solution. Nevertheless, early decisions on train lengths determine a large part of the objective function. Therefore, the branch-and-bound process can work out the exact compositions subject to good lower bounds. We mentioned in Section 3.8.6 that, once the variables $Y_{t,b}$ are integral, the variables $X_{t,p}$ are likely integral, too. So, our branching order often has the effect that we did not need to branch on the variables $X_{t,p}$ at all.

Giving a higher branching priority to the decision variables corresponding to trips in the rush hours also leads to an improvement of the solution time. Seat shortages mostly occur in such trips, thus very good lower bounds on the seat shortages can be computed in the early stage of the algorithm. Then the rolling stock assignment for the rush hours extends to the rest of the day, providing good suboptimal solutions relatively quickly.

3.10.2 Exploiting the Structure of the Instances

When applying the aggregation described in Section 3.8.7, the mixed integer program contains more variables but less constraints. The dimensions of the mixed integer program are given in the two last columns of Table 3.2 (under ‘With aggregation’).

The LP relaxation with aggregation provides slightly better bounds than in the case of no aggregation. Although the difference is small, about 0.1–0.9% of the optimal objective value, even this improvement results in speeding up the solution process. The effect of using the aggregation is presented in Table 3.3. In three test instances, we impose the constraint (3.56) and different bounds (if any) on CCH. We test all instances with and without aggregation, minimising the 5 objective functions in Table 3.1. In each row, we give the optimality gap proved in 2 hours of computation and after how many seconds the best solution was found. A field without numbers means that no feasible solution was found within 2 hours. We can see that the aggregation significantly improves the performance of the branch-and-bound process.

CKM	CCH	Aggr	Obj1	Obj2	Obj3	Obj4	Obj5
318,000	—	Yes	1.69%	0.22%	opt	opt	opt
			3,000	3,200	2,000	1,900	2,700
318,000	—	No	2.56%	0.76%	opt	0.12%	opt
			5,400	5,000	5,000	4,300	4,400
318,000	120	Yes	2.30%	0.50%	1.24%	2.22%	2.32%
			3,500	7,100	3,400	6,500	5,100
318,000	120	No	2.06%	0.67%	—	11.20%	13.16%
			4,800	6,400	—	6,300	5,000
318,000	90	Yes	4.80%	2.67%	—	11.43%	—
			6,000	7,100	—	6,400	—
318,000	90	No	4.90%	16.11%	—	—	—
			5,700	5,300	—	—	—

Table 3.3: For each test instance: *i*) optimality gap after 7200 seconds, *ii*) time elapsed till the best solution was found (in seconds).

We analyse the performance of the model with aggregation in the following sequence of tests: We minimise the seat-shortage kilometres while using the bound of 318,000 on CKM and imposing every possible upper bound between 50 and 155 on CCH. We allow 2 hours of CPU time for each instance. The computational results are summarised in Figure 3.14. The x - and y -axis of the graph correspond to SKM and CCH. The continuous curve indicates the linear programming lower bound, the dashed curve shows the best lower bound obtained after 2 hours of computations. The dots indicate the solutions themselves. It turns out that the linear programming lower bound is hardly improved during the branch-and-bound procedure. We can conclude that performance of the solution process highly depends on the upper bound of CCH: with upper bounds of 115 or higher, almost optimal solutions can be found, while upper bounds up to 95 hardly admit any feasible solutions within 2 hours.

3.10.3 Heuristic Approaches

Search Around the LP Optimum

For some instances, no feasible solution can be found in 2 hours of computation or the quality of the solutions is not satisfactory. This is often the case if bounds on the number of composition changes are too restrictive. Also minimising the carriage-kilometres only while imposing upper bounds on seat-shortages and the number of

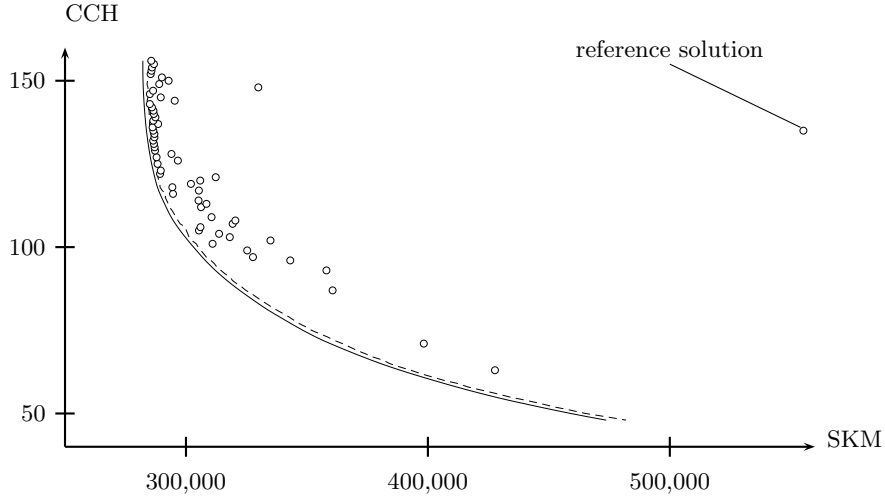


Figure 3.14: Minimising seat-shortage kilometres (SKM) with different bounds on the number of composition changes (CCH) and with at most 318,000 carriage-kilometres.

composition changes often hits the CPU-time limit of a couple of hours without providing solutions with a small optimality gap.

Computational results in the previous section show that the difference between the objective value of the linear relaxation and the best lower bound proved during the branch-and-bound process is small, at most 2%. Moreover, solutions with a small optimality gap indicate that the Composition Model is a quite tight description of the convex hull of the integral solutions. This justifies the following heuristic method.

We first solve a relaxation of the problem, dropping the integrality requirements for some variables (e.g. taking the linear relaxation itself). Then we extract information from the optimal (fractional) solution of the relaxation and based on it, we add extra constraints to the original model. In the simplest form, we require $X_{t,p} = 0$ whenever this variable has value zero in the optimal solution of the relaxed model. As another example, we allow a composition change (that is, extra shunting) after trip t only if the optimal solution of the relaxation has a variable $Z_{t,p,p'} > 0$ with $\nu(p) \neq \nu(p')$.

Reducing the solution space in this way allows us to find feasible solutions within 2 hours for each instance. The optimality gap is, however, still over 10% in some instances. Nevertheless, even these solutions are in all objective criteria better than the reference solution from practise.

When trying several ways to use optimal fractional solutions in order to guide the MIP solution process, we experienced that the improvement in running time strongly depends on the instances themselves. Therefore, we cannot conclude which is the best way of using the relaxed solutions. Nonetheless, after a couple of attempts and a couple of hours of CPU time, we almost always find good integer solutions.

Local Search

Once a feasible solution Σ was found, we can run the model in some “neighbourhood” of this solution. We define the neighbourhood by constraints like saying that for each trip, the composition may be at most one carriage shorter or longer than in Σ . In case of smaller neighbourhoods, the reduced solution space can be enumerated relatively quickly, but often do not contain better solutions. For larger neighbourhoods we allow running times up to 4 hours. After a few local search steps, feasible solutions with optimality gaps of 2–5% can be found.

3.10.4 Summary of the Solutions

In Figure 3.15 we summarise all rolling stock circulations we computed where the number of carriage-kilometres is not higher than in the reference solution. White dots indicate solutions obtained when solving the Composition Model directly by CPLEX; these are also drawn in Figure 3.14. Grey dots represent solutions that we get by using CPLEX in a heuristic way as described in the previous section. As in Figure 3.14, the curve indicates the linear programming lower bound. Solutions that lie far from this line or that are dominated by other solutions that are obtained by additional optimisation criteria: penalising composition changes during peak hours and first class seat-shortages more heavily.

All the solutions in Figure 3.15 obey the constraint $CKM \leq 318,000$. The solutions inside the dashed circle in Figure 3.15 are computed by minimising the number of carriage-kilometres only and imposing bounds on the other objective criteria. These solutions have only 297,000–303,000 carriage-kilometres. Compared to 318,000 carriage-kilometres in the reference solution, it amounts to a reduction in operational costs by nearly 6%. The number of composition changes in these solutions is similar to that in the reference solution, but they have no more than 440,000 seat-shortage kilometres (SKM) while the reference solution has 555,215 seat-shortage kilometres. Note that the linear programming relaxations yield the following lower bounds: Solutions with $SKM \leq 555,215$ have at least 278,982 carriage-kilometres and solutions with $SKM \leq 440,000$ have at least 286,415 carriage-kilometres.

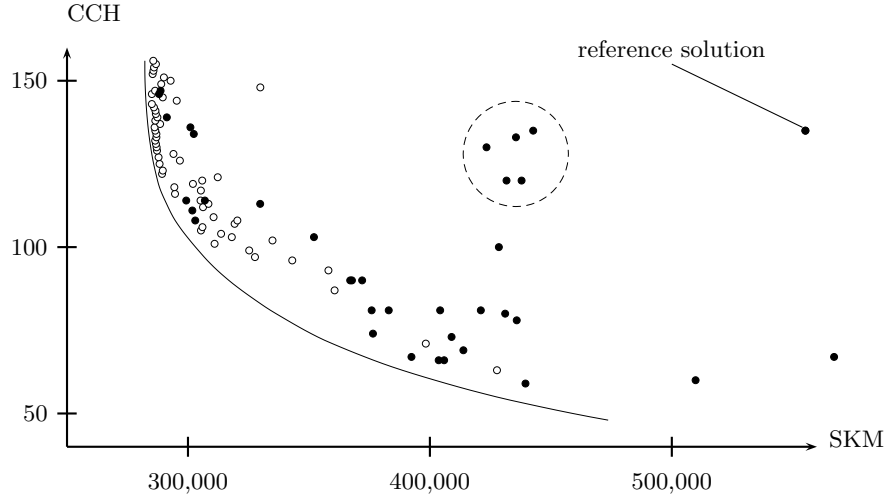


Figure 3.15: Seat-shortage kilometres (SKM) and number of composition changes (CCH) in the solutions with at most 318,000 carriage-kilometres. Solutions inside the dashed circle have 297,000–303,000 carriage-kilometres only.

3.11 The Job Model

In this section we present an alternative model for the tactical rolling stock circulation problem, we call it the ‘Job Model’. It can only be used for problem instances without splitting and combining of trains. Unfortunately, this model provided quite poor computational results on instances of NSR, especially compared to the Composition Model that we described in the previous sections. Nevertheless, we believe that it is interesting to investigate different approaches to solve the same problem and to see their limitations. We note that Lingaya et al. (2002) describe a model for operational rolling stock planning that has some similarities with the Job Model.

3.11.1 Motivation

At the beginning of a day, units are stored at the stations; they are part of the inventories. When carrying out any rolling stock circulation, a particular unit is coupled to a certain departing trip and serves in a number of successive trips until being uncoupled and becoming part of the inventory of a station again. A ‘job’ is such a sequence of successive trips. The duty of a unit consists of jobs: a job starts when the unit is coupled to a departing trip and ends when it is uncoupled from an arriving trip. Coupling and uncoupling may take place at prescribed sides of the

compositions, the problem specification determines the coupling- and uncoupling-sides for each trip. The list of possible jobs can be directly computed from the timetable.

The rolling stock circulation problem amounts to assigning a chain of jobs to each unit. Jobs may not be chosen arbitrarily to be combined to a rolling stock circulation. For instance, if coupling and uncoupling is only possible on the left-hand side, then each train acts like a Last-In-First-Out stack: earlier coupled units must be uncoupled later. That is, if two jobs cover a trip, the earlier starting job must finish later. Thus constraints are needed to avoid such conflicts between jobs.

A job can be assigned several units, too. This means that several units are coupled at the same time, serve in the trips covered by the job and are uncoupled together. These units always stay next to each other, hence their order in the compositions is not relevant.

Jobs route units from the inventory of a station to the inventory of another station. According to the shunting assumptions in Section 3.3, uncoupled units do not appear immediately in the destination inventory but ϱ minutes later where ϱ is the re-allocation time of the last trip in the job.

Similarly to the Composition Model in Section 3.8, we do not compute the rolling stock duties explicitly. Instead, we require that the inventories of the stations are non-negative. This makes sure that the jobs can in fact be composed to duties.

3.11.2 Basic Job Model

In this section we describe the basic form of the Job Model. In the subsequent sections we extend the model and discuss tighter formulations.

A *job* j is a sequence $t_1, \dots, t_k$ of trips with $t_{i+1} = \sigma(t_i)$ for each $i = 1, \dots, k-1$. The set of jobs is denoted by $\mathcal{J}$. A job j is characterised by the set $\mathcal{R}_j := \{t_1, \dots, t_k\}$, by the departure station $s_d(j)$ and departure time $\tau_d(j)$ which are the departure data of trip t_1 and by the arrival station $s_a(j)$ and arrival time $\tau_a(j)$ which are the arrival data of trip t_k . The re-allocation time $\varrho(j)$ is defined as the re-allocation time of trip t_k . We use the symbols $\varphi_j := t_1$ and $\lambda_j := t_k$ to denote the first and last trips in the job. The coupling side right before trip t_1 is denoted by $\gamma_d(j)$, the uncoupling side right after trip t_k is denoted by $\gamma_a(j)$. The coupling and uncoupling sides may be left (L) or right (R).

Let $\hat{\mu}_j$ be an upper bound on number of units that can be assigned to job j . This is determined by the maximally allowed train lengths for trips $t_1, \dots, t_k$ and by (3.2). The total length (in kilometres) of the trips in job j is denoted by d_j . As before, c_m

denotes the number of carriages in a unit of type m , $\kappa_{m,c}$ the seat capacity of a unit for service class c and $\delta_{t,c}$ the passenger demand.

The main decision variables in the Job Model are the following:

$$\begin{aligned} X_j &\in \{0, 1\} && \text{whether job } j \text{ is selected } (X_j = 1) \text{ or not } (X_j = 0). \\ N_{j,m} &\in \mathbb{Z}_+ && \text{the number of units of type } m \text{ used for job } j. \end{aligned}$$

In addition, we use binary variables H_t to express whether a composition change takes place right before trip t ($H_t = 1$) or not ($H_t = 0$). We also use variables $I_{j,m}$ for the inventory of type m at station $s_d(j)$ immediately after the departure of job j . Variables $I_{s,m}^0$ and $I_{s,m}^\infty$ denote the number of units of type m stored at station s at the beginning and at the end of the day. Finally, we use variables $S_{t,c}$ to measure the seat shortages of class c on trip t .

Now, the *Job Model* is formulated as follows.

$$\text{Minimise } f_1 \cdot \sum_{j \in \mathcal{J}} \sum_{m \in \mathcal{M}} d_j c_m N_{j,m} + f_2 \cdot \sum_{t \in \mathcal{T}} \sum_{c \in \mathcal{C}} d_t S_{t,c} + f_3 \cdot \sum_{t \in \mathcal{T}} H_t \quad (3.57)$$

subject to

$$X_j \leq \sum_{m \in \mathcal{M}} N_{j,m} \quad \forall j \in \mathcal{J} \quad (3.58)$$

$$\hat{\mu}_j X_j \geq \sum_{m \in \mathcal{M}} N_{j,m} \quad \forall j \in \mathcal{J} \quad (3.59)$$

$$H_{\varphi_j} \geq X_j \quad \forall j \in \mathcal{J} : \varphi_j \notin \mathcal{T}_0 \quad (3.60)$$

$$H_{\sigma(\lambda_j)} \geq X_j \quad \forall j \in \mathcal{J} : \lambda_j \notin \mathcal{T}_\infty \quad (3.61)$$

$$\mu_t^{\min} \leq \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J} : \mathcal{R}_j \ni t} c_m N_{j,m} \quad \forall t \in \mathcal{T} \quad (3.62)$$

$$\mu_t^{\max} \geq \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J} : \mathcal{R}_j \ni t} c_m N_{j,m} \quad \forall t \in \mathcal{T} \quad (3.63)$$

$$S_{t,c} \geq \delta_{t,c} - \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J} : \mathcal{R}_j \ni t} \kappa_{m,c} N_{j,m} \quad \forall t \in \mathcal{T}, c \in \mathcal{C} \quad (3.64)$$

$$\begin{aligned} I_{j,m} = I_{s_d(j),m}^0 - & \sum_{\substack{j' \in \mathcal{J} : s_d(j') = s_d(j), \\ \tau_d(j') \leq \tau_d(j)}} N_{j',m} \\ & + \sum_{\substack{j' \in \mathcal{J} : s_a(j') = s_d(j), \\ \tau_a(j') \leq \tau_d(j) - \varrho(j')}} N_{j',m} \quad \forall j \in \mathcal{J}, m \in \mathcal{M} \end{aligned} \quad (3.65)$$

$$I_{s,m}^{\infty} = I_{s,m}^0 - \sum_{\substack{j \in \mathcal{J}: \\ s_d(j)=s}} N_{j,m} + \sum_{\substack{j \in \mathcal{J}: \\ s_a(j)=s}} N_{t,m} \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.66)$$

$$I_{s,m}^0 = i_{s,m}^0 \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.67)$$

$$I_{s,m}^{\infty} = i_{s,m}^{\infty} \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.68)$$

$$X_j + X_{j'} \leq 1 \quad \forall j \in \mathcal{J}, j' \in \mathcal{J}: \sigma(\lambda_j) = \varphi_{j'} \quad (3.69)$$

$$X_j + X_{j'} \leq 1 \quad \forall j \in \mathcal{J}, j' \in \mathcal{J}: \mathcal{R}_j \cap \mathcal{R}_{j'} \neq \emptyset, \gamma_a(j) = \gamma_d(j') \text{ and} \\ \tau_d(j) < \tau_d(j') < \tau_a(j) < \tau_a(j') \quad (3.70)$$

$$X_j + X_{j'} \leq 1 \quad \forall j \in \mathcal{J}, j' \in \mathcal{J}: \mathcal{R}_j \cap \mathcal{R}_{j'} \neq \emptyset, \gamma_a(j) \neq \gamma_d(j) \text{ and} \\ \tau_d(j') < \tau_d(j) < \tau_a(j) < \tau_a(j') \quad (3.71)$$

$$X_j \in \{0, 1\} \quad \forall j \in \mathcal{J} \quad (3.72)$$

$$N_{j,m} \in \mathbb{Z}_+ \quad \forall j \in \mathcal{J}, m \in \mathcal{M} \quad (3.73)$$

$$H_t \in \{0, 1\} \quad \forall t \in \mathcal{T} \quad (3.74)$$

$$S_{t,c} \in \mathbb{R}_+ \quad \forall t \in \mathcal{T}, c \in \mathcal{C} \quad (3.75)$$

$$I_{j,m} \in \mathbb{R}_+ \quad \forall j \in \mathcal{J}, m \in \mathcal{M} \quad (3.76)$$

$$I_{s,m}^0, I_{s,m}^{\infty} \in \mathbb{R}_+ \quad \forall s \in \mathcal{S}, m \in \mathcal{M} \quad (3.77)$$

The three terms of the objective function (3.57) count the carriage-kilometres, the seat-shortage kilometres and the number of composition changes. The factors f_1 , f_2 and f_3 express the relative importance of the three objective criteria.

Constraints (3.72) – (3.77) specify the domains of the variables. Constraints (3.58) and (3.59) describe the connection between variables X_j and $N_{j,m}$. Constraints (3.60) and (3.61) identify whether or not a composition change takes place. Constraints (3.62) and (3.63) make sure that for each trip, the train lengths comply with the problem specifications. Constraints (3.64) state that for each trip t and for each class c , the seat-shortage of class c is at least the difference between the passenger demand and the allocated seat capacity. The variables $S_{t,c}$ have positive coefficients in the objective function. Therefore, in each optimal solution we have

$$S_{t,c} = \max \left\{ 0; \delta_{t,c} - \sum_{m \in \mathcal{M}} \sum_{j \in \mathcal{J}: t \in \mathcal{R}_j} \kappa_{m,c} N_{j,m} \right\} \quad \forall t \in \mathcal{T}, c \in \mathcal{C}.$$

Constraints (3.65) and (3.66) compute the inventories of the stations: the initial inventories are decreased or increased by the number of units used on departing and arriving jobs, taking also the re-allocation times into account. Constraints (3.67) and (3.68) set the initial and final inventories. Constraints (3.69) express that coupling and uncoupling cannot both take place between a trip and its successor.

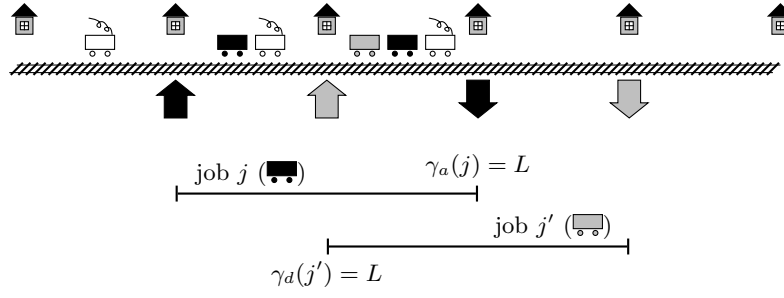
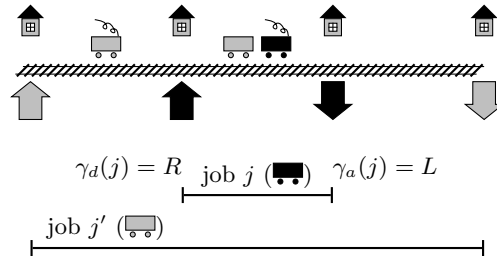
Constraints (3.70) and (3.71) make sure that jobs with $X_j = 1$ can in fact be combined to duties. This can be seen as follows. Whenever a job starts, units are coupled to the left- or right hand side of the departing train, this is always possible. We have to prove that the units can be uncoupled at the departure station of the job. For instance, if a particular unit is to be uncoupled at the left-hand side of the train, all other units to the left of the unit are also to be uncoupled.

We specify the coupling order of jobs that start with a given trip. Let $j_1^L, \dots, j_k^L$ be the jobs that start with trip t and that have $X_{j_i^L} = 1$ and $\gamma_a(j_i^L) = L$; they are ordered by increasing arrival time. Similarly, let $j_1^R, \dots, j_\ell^R$ be the jobs that start with trip t and that have $X_{j_i^R} = 1$ and $\gamma_a(j_i^R) = R$; they are ordered by decreasing arrival time. Units are coupled just before trip t such that their order on trip t is $j_1^L, \dots, j_k^L, j_1^R, \dots, j_\ell^R$. Then units that are coupled before trip t cannot block each other.

We claim that if units are coupled in this order, they can always be uncoupled when their jobs finish. For simplicity we assume that each job j with $X_j = 1$ is assigned one unit only. Suppose that a job j finishes with being uncoupled from the left-hand side of the train (the case of the right-hand side being analogous). There is one reason only why unit u assigned to job j cannot be uncoupled after trip λ_j : There is another unit u' assigned to job j' with $X_{j'} = 1$ such that j' covers trip λ_j , that u' lies to the left of u and that u' is not to be uncoupled yet. Then we have $\mathcal{R}_j \cap \mathcal{R}_{j'} \neq \emptyset$ and $\tau_a(j) < \tau_a(j')$.

If unit u' was coupled later than u was (i.e. if $\tau_d(j) < \tau_d(j')$), then u' could only be coupled at the left-hand side of the train (otherwise u' would not block u). Hence jobs j and j' violate constraint (3.70). An example is shown in Figure 3.16 where the black unit is u and the grey unit is u' . Suppose now that unit u' was coupled earlier than u was (i.e. that $\tau_d(j') < \tau_d(j)$). Since u' lies to the left of u , unit u was coupled at the right-hand side. Then jobs j and j' violate constraint (3.71). We give an example in Figure 3.17 where again, the black unit is u and the grey unit is u' . This completes the proof that the jobs with $X_j = 1$ give rise to a rolling stock circulation.

One immediately verifies that the variables $S_{t,c}$, $I_{s,m}^0$, $I_{s,m}^\infty$ and $I_{j,m}$ have integral values in every feasible solution.

Figure 3.16: Job j is blocked by a job j' with $\tau_d(j) < \tau_d(j')$.Figure 3.17: Job j is blocked by a job j' with $\tau_d(j') < \tau_d(j)$.

3.11.3 Adding Secondary Constraints

The model (3.57) – (3.77) implements all primary constraints of the problem formulation. As in the Composition Model, secondary constraints can be added to the model.

Cyclic rolling stock circulations can be formulated in exactly the same way as in Section 3.8.3 for the Composition Model. For sake of completeness, we repeat it here. One replaces the constraints (3.67) and (3.68) by the constraints

$$n_m = \sum_{s \in \mathcal{S}} I_{s,m}^0 \quad \forall m \in \mathcal{M}, \quad (3.78)$$

$$I_{s,m}^0 = I_{s,m}^\infty \quad \forall s \in \mathcal{S}, m \in \mathcal{M}. \quad (3.79)$$

The following lemma is essentially the same as Lemma 3.2 and states that variables $S_{t,c}$, $I_{s,m}^0$, $I_{s,m}^\infty$ and $I_{j,m}$ still do not need to be defined as integer.

Lemma 3.6. *Consider the model (3.57) – (3.66), (3.69) – (3.77) and (3.78) – (3.79) for the problem of finding a cyclic rolling stock circulation and suppose that the model has a feasible solution. Then it has an integral optimal solution.*

Limitations on storage capacity of stations can be taken into account in the same way as in Section 3.8.3: one inserts the constraints

$$\sum_{m \in \mathcal{M}} \ell_m \cdot I_{j,m} \leq W_s \quad \forall j \in \mathcal{J}: s_d(j) = s$$

and similar constraints for $I_{s,m}^0$ and $I_{s,m}^\infty$. Again, the variables $I_{s,m}^0$ must be integral.

Finally, we add the continuity requirement. Consider a timetable service composed of trips $t_1, \dots, t_k$. Then the constraint

$$\sum_{\substack{j \in \mathcal{J}: \\ t_1, \dots, t_k \in \mathcal{R}_j}} X_j \geq 1$$

says that at least one job will be selected which covers the entire timetable service.

3.11.4 Tighter Formulations

The linear relaxation of the integer program (3.57) – (3.77) does not describe the convex hull of the integer solutions very well. In this section we point out three sets of possibly weak constraints in the model and propose methods to tighten them.

Tightening the Seat-shortage Constraints

Here, we illustrate how the seat-shortage constraints (3.64) can be improved. Consider the second class seat-shortages on a single trip t only. Assume that two types are available: units with 3 and 4 carriages, having 166 and 317 seats, respectively¹. Moreover, let the passenger seat demand be equal 403. The composition on trip t may have 3–15 carriages.

For simplicity, denote the number of units assigned to trip t by N_3 and N_4 for the two types and instead of $S_{t,2}$, we write S . Then N_3 and N_4 are sums of the variables $N_{j,m}$:

$$N_m = \sum_{j \in \mathcal{J}: t \in \mathcal{R}_j} N_{j,m} \quad \forall m \in \{3, 4\}.$$

The maximal train length of 15 carriages allows the following 14 pairs of integer values for (N_3, N_4) : (1, 0), (2, 0), (3, 0), (4, 0), (5, 0), (0, 1), (1, 1), (2, 1), (3, 1), (0, 2), (1, 2), (2, 2), (0, 3), (1, 3). Of these 14 possibilities, only 3 combinations give seat shortages:

N_3	N_4	S
1	0	237
2	0	71
0	1	86

Consider the polyhedron P described by the linear relaxation of the Job Model and project P to the 3-dimensional space of variables N_3 , N_4 and S . Then the projection of P is determined by the following constraints:

$$S \geq 403 - 166N_3 - 317N_4$$

$$15 \geq 3N_3 + 4N_4$$

$$S \geq 0$$

$$N_3 \geq 0$$

$$N_4 \geq 0$$

We illustrate this 3-dimensional polyhedron in Figure 3.18.

¹These numbers do not refer to any rolling stock type of NSR. We chose the values in order to have nicer figures in this section. Units of type “Koploper” have 166 or 224 second class seats.

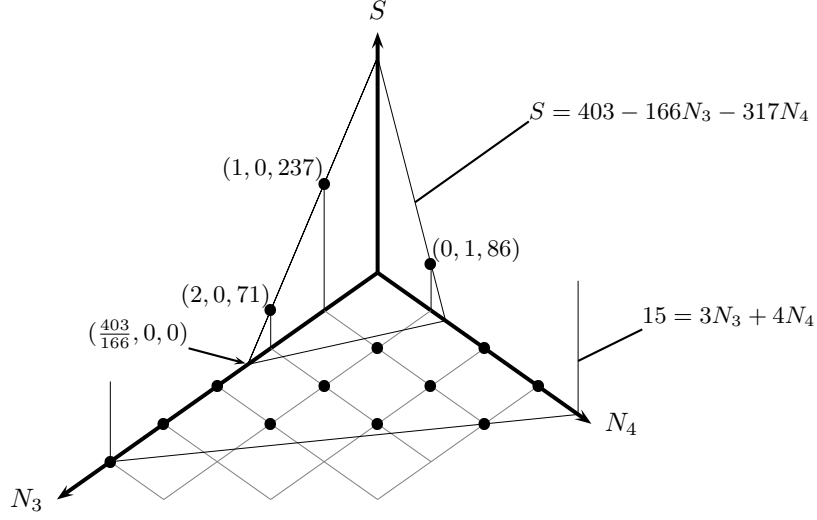


Figure 3.18: Projection of the solution spaces of the linear relaxation onto the (N_3, N_4, S) -space.

In any optimal solution of the mixed integer program, the value of S equals to the actual seat shortage on trip t when using the given numbers N_3 and N_4 of units. That is, each optimal integer solution is projected to one of the fat dots in Figure 3.18. So, no optimal integer solution is cut off if we add the restrictions to the original model that the projection of the solutions must lie in the convex hull of the vectors

$$(N_3, N_4, \max \{0, 403 - 166N_3 - 317N_4\}) \quad (3.80)$$

for appropriate pairs (N_3, N_4) .

For example, the points $(3, 0, 0)$, $(1, 1, 0)$ and $(2, 0, 71)$ determine a facet of the convex hull described by the inequality

$$213 - 71N_3 - 142N_4 \leq S. \quad (3.81)$$

This cutting plane is shown in Figure 3.19. Observe that $(\frac{403}{166}, 0, 0)$ is a vertex of the projected image of the polyhedron P which arises from a vertex of P . This vertex is cut off by the cutting plane (3.81).

The convex hull of the vectors (3.80) can be easily computed in a pre-processing step for each trip and for each service class. We note that similarly defined cutting

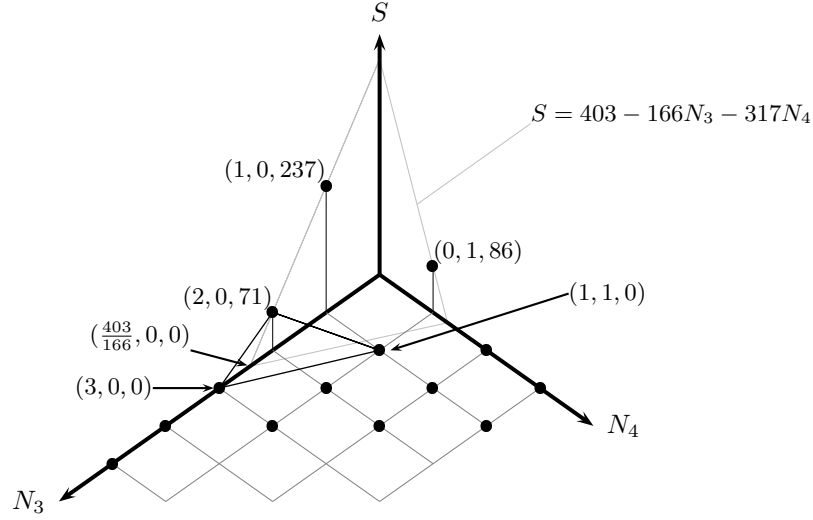


Figure 3.19: A cutting plane through the points $(3, 0, 0)$, $(1, 1, 0)$ and $(2, 0, 71)$ on the (N_3, N_4, S) -space.

planes turned out to be crucial in the rolling stock circulation models of Schrijver (1993), Groot (1996), Van Montfort (1997) and Alfieri et al. (2002) (although these papers do not deal with seat shortages).

Tightening the Job Conflict Constraints

Another point where the basic Job Model can be tightened are the constraints (3.69) – (3.71) that exclude conflicting pairs of jobs. These inequalities can be interpreted as follows. Let $G = (V, E)$ be an undirected graph on the node set $\mathcal{J}$ with the edge set

$$\begin{aligned} & \{(j, j') \in \mathcal{J} \times \mathcal{J} \mid \sigma(\lambda_j) = \varphi_{j'} \\ & \quad \text{or } (\mathcal{R}_j \cap \mathcal{R}_{j'} \neq \emptyset, \gamma_a(j) = \gamma_d(j'), \text{ and} \\ & \quad \quad \tau_d(j) < \tau_d(j') < \tau_a(j) < \tau_a(j')) \\ & \quad \text{or } (\mathcal{R}_j \cap \mathcal{R}_{j'} \neq \emptyset, \gamma_d(j) \neq \gamma_a(j), \text{ and} \\ & \quad \quad \tau_d(j') < \tau_d(j) < \tau_a(j) < \tau_a(j'))\}. \end{aligned}$$

Then the solutions of (3.69) – (3.71) and (3.72) are the incidence vectors of the stable sets in G . (A stable set is a subset of nodes without any edge between them.) The convex hull of these integer solutions is called the *stable set polytope* of the graph G .

When taking the linear relaxation of the model, the stable set polytope gets relaxed to the polytope

$$\{x \in [0, 1]^V \mid x_u + x_v \leq 1 \text{ for each } (u, v) \in E\}.$$

However, this provides a quite poor approximation of the stable set polytope, a tighter formulation may be essential to obtain shorter solution times. Giving a tight linear description of the stable set polytope is a topic on its own. One of the easiest ways to sharpen the constraints is to require

$$\sum_{v \in K} x_v \leq 1 \tag{3.82}$$

for each maximal clique in G , or at least for some of them. A maximal clique is a containment-wise maximal subset K of nodes where each pair of members of K is joined by an edge.

Getting Rid of Big-M Constraints

In the model (3.57) – (3.77), several units may be assigned to each job. Constraints (3.59) are needed to set the link between the binary variables X_j and the non-negative integer variables $N_{j,m}$. These constraints are obtained by the “big-M” method, although the numbers $\hat{\mu}_j$ are not that big in real-life applications: they range from 2–5, the most frequent value is 2. We note here that in many instances of NSR the majority of the values $\hat{\mu}_j$ equals one; there is not much to tighten on the constraints (3.59) then.

Constraints (3.59) concern the number of units only. An alternative way to formulate the connection between variables X_j and $N_{j,m}$ is to count the number of carriages. Then constraints (3.59) are replaced by

$$\mu_j X_j \geq \sum_{m \in \mathcal{M}} c_m N_{j,m} \quad \forall j \in \mathcal{J} \tag{3.83}$$

where μ_j is an upper bound on the number of carriages that can be assigned to job j . These are still “big-M” constraints.

The linear relaxation of “big-M” constraints is often very weak. By introducing new decision variables as follows, constraints (3.59) or (3.83) can be eliminated. We

apply the same idea as in Section 3.8.5. For each job j , we define the set $\mathcal{B}_j \subseteq \mathbb{Z}_+^{\mathcal{M}}$. Each vector $b \in \mathcal{B}_j$ represents the choice to assign b_m units of type m to job j . Then we define the binary decision variables

$$Y_{j,b} \in \{0, 1\} \quad \forall j \in \mathcal{J}, b \in \mathcal{B}_j$$

and we replace constraints (3.58) – (3.59) by the following restrictions:

$$N_{j,m} = \sum_{b \in \mathcal{B}_j} b_m Y_{j,b} \quad \forall j \in \mathcal{J}, m \in \mathcal{M}, \quad (3.84)$$

$$X_j = \sum_{b \in \mathcal{B}_j} Y_{j,b} \quad \forall j \in \mathcal{J}. \quad (3.85)$$

Although the number of decision variables increases, the branch-and-bound process might become faster as the linear relaxation provides better lower bounds.

3.11.5 Computational Results for the Job Model

In order to investigate the performance of the Job Model and to compare it with the Composition Model, we consider two instances and solve them using both models by the commercial MIP software CPLEX. The instance **SPR-1** is a circulation for a single day with 401 trips. The instance **SPR-2** concerns two consecutive days with 803 trips. The timetable is based on the rolling stock type “Sprinter” of NSR. To obtain multiple rolling stock types, we divide the available units artificially into two types. Although the instance is not an instance of NSR, its structure is very similar to real-life rolling stock circulation problems. The dimensions of the mixed integer programs are given in Table 3.4 for both models and for both test instances.

Similarly to the computational test described in Section 3.10, we minimise a linear combination of the objective criteria CKM, SKM and CCH with five different settings (Obj1, Obj2, Obj3, Obj4, Obj5); the weight factors are given in Table 3.1.

In Section 3.11.4 we describe three ways to obtain a tighter linear relaxation of the Job Model. The computational results below arise from an implementation where only the tighter seat-shortage constraints are incorporated. Adding a large number of clique inequalities of the form (3.82) did not have any effect on the solution times and qualities. In this research we did not consider other kinds of inequalities for the stable set polytope. Also, replacing constraints (3.59) by (3.84) – (3.85) did not have any positive effect on the solution process.

		MIP		Reduced MIP		
		# columns	# rows	# columns	# rows	# non-zeros
CM	SPR-1	36,387	16,261	33,738	15,431	167,086
JM	SPR-1	10,845	78,480	9,064	49,107	502,805
CM	SPR-2	71,202	31,406	69,546	31,292	480,773
JM	SPR-2	21,656	156,927	18,124	98,335	1,363,031

Table 3.4: Dimensions of the Composition Model (CM) and Job Model (JM) for instances SPR-1 and SPR-2.

Each instance is given 1,800 seconds of CPU time. The best performance is achieved by using for both models the CPLEX settings described in Section 3.10.1 with the exception that the linear relaxation of the Job Model is solved much faster by the dual simplex method than by the barrier method.

We summarise the numerical computational results in Table 3.5. Each instance of the Composition Model is solved to optimality within a couple of minutes. Observe that the objective value of the linear relaxation hardly differs from the value of the optimal integral solution. The optimum value of the linear relaxation of the basic Job Model (i.e. without constraint tightening) lies 15–20% under the integer optimum and no feasible solution is found within 1,800 seconds. Tightening the seat-shortage constraints significantly improves the linear relaxation. Then CPLEX is able to generate additional cuts, thereby increasing the lower bounds close to the integer optimum: one instance has a difference of 3.11%, the lower bound in the other instances differs from the integer optimum by less than 1.40%.

The SPR-1 instance is well-tractable by the Job Model: feasible solutions with optimality gaps of 0–7% are found within 30 minutes. Here the optimality gap is taken between the best integral solution and the best lower bound. The instance SPR-2, however, turns out to be too difficult for the CPU-time limit of 30 minutes: only one of the five objective functions admitted feasible solutions.

The Composition Model clearly outperforms the Job Model in every test instance. Yet, it may be possible to improve the performance of the Job Model, for instance by using a more sophisticated description of the stable set polytope. Nevertheless, we believe that the outstanding solution times of the Composition Model are not easy to achieve.

In addition, the Job Model strongly relies on the fact that splitting and combining do not occur. Splitting and combining of trains makes it very difficult to describe the jobs and to determine the conflicts between jobs. Therefore, the Job Model can hardly be applied to instances of NSR where trains are split and combined.

Instance	CM LP	CM IP	CM time	basic JM LP	JM LP	JM bound	JM IP	JM gap	JM time
SPR-1 Obj1	574,487	574,487	20	499,303	541,903	574,487	574,487	0	380
SPR-1 Obj2	549,847	549,847	37	474,970	526,992	548,458	550,618	0.36%	1,800
SPR-1 Obj3	534,383	534,587	220	456,171	511,021	532,276	544,787	2.26%	1,800
SPR-1 Obj4	1,227,855	1,227,855	22	1,029,365	1,127,453	1,217,416	1,243,529	2.14%	1,800
SPR-1 Obj5	1,187,445	1,187,345	27	998,748	1,094,507	1,173,352	1,257,956	7.16%	1,800
SPR-2 Obj1	1,223,161	1,227,524	889	1,067,141	1,167,506	1,218,381	1,233,367	1.23%	1,800
SPR-2 Obj2	1,179,577	1,180,360	550	1,012,483	1,132,731	1,171,294	—	—	1,800
SPR-2 Obj3	1,143,959	1,144,595	397	973,983	1,098,878	1,139,045	—	—	1,800
SPR-2 Obj4	2,550,280	2,550,440	217	2,145,382	2,357,540	2,473,283	—	—	1,800
SPR-2 Obj5	2,466,645	2,467,967	81	2,086,728	2,291,781	2,433,899	—	—	1,800

‘Basic JM’ stays for the Job Model without constraint tightening; ‘JM’ includes tightening the seat-shortage constraint. ‘LP’ refers to the objective value of the linear relaxation, ‘IP’ to the best solution found, ‘bound’ to the lower bound proved by the branch-and-bound method within 1,800 seconds. Solution time is measured in seconds.

Table 3.5: Numerical results for the instances SPR-1 and SPR-2 solved by the Com-position Model (CM) and the Job Model (JM).

3.12 Conclusions

In this chapter, we considered the tactical rolling stock circulation problem of NSR and we described two ways of modelling it. The models can express various technical and market requirements. In computational experiments, the Composition Model provided good solutions with reasonable running times, the Job Model performed much worse.

We tested the Composition Model on fairly large instances of NSR. The model turned out to be robust enough to cope with very different weights on the objective criteria. When solved by a commercial MIP solver, it provided rolling stock circulations with different objective characteristics in a couple of hours of computation time. The decision makers can choose a solution that matches the practical requirements best. Planners at NSR agreed that the solutions of the Composition Model are in any respect better than the manually created plans.

The computer-generated rolling stock circulations must still be checked and sometimes adjusted by human planners. Nevertheless, the implemented model turned out to be a very useful planning tool at the Logistics Department of NSR. From 2004 on, the basic shape of the tactical rolling stock circulations at NSR are partly computed by solving this model. These computer-generated circulations have a significantly lower number of carriage-kilometres than the plan of the previous years while providing the same service quality. This allowed NSR to reduce its annual operational costs by a couple of millions of euros.

Chapter 4

Maintenance Routing

Short-term planning at NSR includes a number of planning steps that are currently carried out without computer-aided planning tools. Of these problems of railway practise, we only study the maintenance routing problem of NSR in this thesis. Maintenance routing makes sure that rolling stock units that need a regular preventive maintenance check visit the maintenance facility in time.

We present two integer programming models for maintenance routing. The ‘Interchange Model’ is designed to take as many details of reality into account as possible. However, the large amount of input data needed for the Interchange Model is difficult to obtain. This motivates the ‘Transition Model’ which is a much simpler formulation of the problem. Implementations of real-life test data give promising results for both models: we find good solutions in an acceptable amount of time.

This chapter is structured as follows. First we describe the maintenance routing process of NSR in detail and formulate the maintenance routing problem. Thereafter we give a brief literature overview. Subsequently we describe the Interchange Model and the Transition Model. For both models, we discuss theoretical complexity issues and computational results on test instances of NSR. At the end of the chapter, we draw some conclusions.

4.1 Maintenance Routing in Practise

The operational rolling stock plan only concerns duties for anonymous units. In short-term planning in practise, at most 3 days before executing the operational plan the physical units are assigned to the duties. This gives the *regular plan* which describes

what the individual units have to do during the forthcoming couple of days. The regular plan is composed of a rolling stock circulation and a corresponding shunting plan. Details of the shunting plans are only known to local planners. From the central planners' point of view, the regular plan is a rolling stock circulation that admits a shunting plan.

A *task* is a smallest indivisible train movement in the regular plan to be carried out by a single unit. So, if a trip is assigned a composition with more than one unit, it corresponds to several tasks. A pair (t, t') of tasks is a *regular transition* if t' is the successor of task t . That is, t and t' are carried out consecutively by the same unit in the regular plan. A *regular duty* is the sequence of all tasks of a single unit during the forthcoming days.

Units require a preventive maintenance check after a certain number of kilometres. This limit depends on the rolling stock type; if a unit travelled that distance since its last maintenance check, then it may not carry passengers any more before undergoing a maintenance check. Units visit the maintenance facility roughly once a month. Note that units must also undergo smaller safety inspections once every two days. These can take place at many stations and are handled on an ad hoc basis by local planners. These small safety inspections are not considered in this chapter at all.

Maintenance of rolling stock is carried out by NedTrain while NSR is responsible for sending units to the maintenance facilities in time. The checks take about a day and can only take place at specialised stations. In tactical and operational planning, maintenance is taken into account in a limited way. The timetable contains *maintenance tasks*; if a unit is assigned to such a task, it undergoes a maintenance check. However, this assignment is done provisionally, without caring about which unit will be urgent and when. The actual maintenance routing is handled in the short-term phase by central planners at the Materieelregelcentrum (MRC). Maintenance routing is carried out every day with a rolling planning horizon, always considering the forthcoming 1–3 days.

Every day, maintenance routing planners receive the list of the *urgent* units: those units that need a maintenance check within the planning horizon. Usually, these are the units that made the highest number of kilometres since their last maintenance check. These units may have different urgencies, expressed by a deadline (for instance, saying that the maintenance check should take place within 2 days). Maintenance routing has to make sure that the urgent units carry out a maintenance task within the planning horizon. To that end, the part of the regular plan that lies within the planning horizon must be adjusted.

Maintenance routing planners modify the regular plan by interchanging units: They consider a small number of regular transitions (t_i, t'_i) (that is, pairs of consecutive tasks of single units) such that the arrival station of the tasks t_i is the same. Then, they replace these regular transitions simultaneously by a collection of new transitions between the tasks t_i and t'_i . The maintenance routing planners look for combinations of such interchanges that lead each urgent unit to a maintenance task within its deadline.

Example. Figure 4.1 shows an example for a single urgent unit. The bold lines are the tasks and the dotted lines are the regular transitions. The arrows show the interchanges that route the urgent unit to the maintenance task.

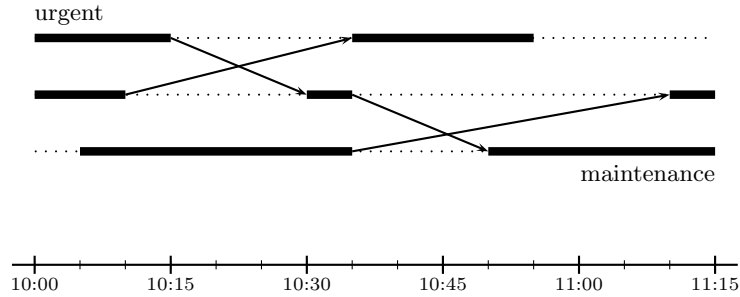


Figure 4.1: An instance of maintenance routing with one urgent unit.

The modifications of the regular plan may require adjusted shunting plans at the stations. Generating shunting plans is a difficult problem in itself. So, maintenance routing planners themselves cannot decide whether or not their modifications can be implemented in practise. Therefore they check with local planners if the preferred modifications can be carried out. Since this communication with local planners takes time, maintenance routing planners cannot test too many possibilities. They are usually satisfied with the first solution found.

If maintenance routing planners do not find a solution, they may decide that an urgent unit goes from one station to another as an empty train (and that a non-urgent unit goes back to restore the rolling stock balance). This is quite expensive and may conflict with the schedules of the train drivers and with the capacity of infrastructure.

4.2 Maintenance Strategy

We described above the maintenance routing process at NSR; our research addresses this problem. However, other maintenance strategies are also possible and they would lead to routing problems of other type.

One might wonder whether the entire maintenance routing process could be eliminated at all by taking maintenance considerations in earlier phases of the rolling stock planning into account. That is, whether it would make sense to create a rolling stock circulation for a month such that each unit visits the maintenance facility exactly once. This is the usual approach in the airline industry.

The answer is that it would not work at NSR. First, it would mean that one has to consider individual units in tactical and operational circulations. Due to the size and complexity of the problem, the computation times at these planning phases would probably become huge. Second, even if we had time for all these long computations, disruptions and delays in the operations would make it unlikely that the rolling stock schedule can be carried out exactly as planned in a period longer than a couple of days. So, we need short planning horizons in maintenance routing.

Fortunately, the Dutch railway network contains a large number of frequently operated and relatively short train lines. This provides many interchange possibilities for the units. Therefore, as experience shows, a couple of days is usually enough to route an urgent unit to a maintenance facility. This explains why the maintenance routing planners consider a planning horizon of 1 to 3 days and only a small number of urgent units to be routed.

In the current maintenance practise, NSR and NedTrain agree on the number of maintenance tasks in an early phase of the planning process. In short-term planning, all these maintenance tasks must be filled with a unit even if no unit is close to the kilometre-limit. Studies at NSR showed that more maintenance is carried out currently than strictly necessary.

Alternative maintenance strategies may help to reduce maintenance costs. A natural idea is to send a unit to a maintenance task only if it really has reached the kilometre-limit (or if it is broken). Central coordination is still needed for the case when too many units would require a maintenance check on the same day. A key element is to make routing decisions as late as possible. To that end, one might even accept a higher number of empty trains. The long-time benefit of such an alternative maintenance strategy is currently being investigated at NSR and NedTrain. Changing the maintenance strategy requires significant adjustments in tactical, operational

and short-term rolling stock planning. Nonetheless, we believe that the modelling approaches of this chapter can be carried over for altered problem specifications.

4.3 Problem Formulation

In this section, we give a (slightly vague) formulation of the maintenance routing problem. It is rather a list of wishes that must be taken into account in models for the problem.

The maintenance routing problem amounts to modifying the regular rolling stock duties such that the following requirements are satisfied.

1. The new duty of each urgent unit contains a maintenance task within the deadline.
2. Empty trains are to be avoided as much as possible.
3. Passenger traffic may not suffer from maintenance routing. In particular, *each task in the timetable must be covered by a unit*.
4. The rolling stock circulations should not be affected too much. To ensure that, maintenance routing planners only interchange units of the same type. In other words, *maintenance routing handles the different rolling stock types separately*.
5. In order to agree quickly with local planners on the modified shunting plans, maintenance routing should not require too many changes in the shunting plans. We emphasise again that maintenance routing planners do not have direct information about the shunting process. Based on their experience, however, they can mostly guess quite well whether or not interchanging a couple of units at a given station is *likely* to be approved by local planners.

The fifth requirement above is not well-defined yet. It is not clear at all how one should measure the deviation from the regular shunting plan without even knowing it in detail. The models we are going to describe in this chapter show two radically different ways to do it.

The requirements become less and less binding as time approaches the point when the trains have to run. In real-life maintenance routing at NSR, the third and fourth requirements may actually be violated if no other solution is found. Timetable services may be operated with a smaller number of units than planned, trains may even be cancelled. Also, units of different types are interchanged. Both kinds of violations are strongly undesirable and used only as a last option.

The objective criteria of tactical planning (see Section 3.6) are not really important any more. Instead, the main goal is to find quickly a solution that can be smoothly implemented in practise. There is no natural quantity to measure how good a solution is. Different maintenance planners may judge the same solution differently. Therefore, there is quite a choice for an objective function in mathematical models. Important is that the objective function implements the main goals listed above and that most maintenance planners find the solutions reasonably good.

Using any method in short-term planning, there is not enough time to plan the rolling stock from scratch mainly because the modified shunting plans cannot be checked easily. But we may use the fact that we have the regular plan for which feasible shunting plans exists. Applying a small number of modifications, one may hope that the shunting plans can in fact be adjusted.

Each solution of maintenance routing needs approval of the local planners. Therefore, the final goal of this research is to develop an *interactive* decision support system that proposes candidate solutions to maintenance routing planners: they can accept the proposal or run the system again with modified specifications. Results on single instances can also be used to refine the models. In real-life maintenance planning, local planners must be contacted several times until a solution is approved. So, it is essential to produce candidate solutions quickly. Waiting 15 minutes or more would slow down the planning process. Moreover, methods that provide reasonably good solutions quickly are much more likely to be accepted both by decision makers and by maintenance routing planners themselves.

Finally, reality is always much more complex than what mathematical models can describe effectively. In case of emergency, almost every rule may be violated. Having this in mind, we focused in our models and test implementations on the *rules* rather than on the *exceptions*. Then the models may become slightly rigid. Nonetheless, if computer models can help in solving the easy but time-consuming maintenance routing instances, then the planners have more time to deal with instances that really require creativity and ad hoc ideas.

The input for the maintenance routing problem contains the set of tasks. A task t is characterised by the departure station $s_d(t)$, arrival station $s_a(t)$, departure time $\tau_d(t)$ and arrival time $\tau_a(t)$ (so $\tau_d(t) \leq \tau_a(t)$). The successor task of task t is $\sigma(t)$. The pair $(t, \sigma(t))$ is a regular transition; regular transitions satisfy $s_a(t) = s_d(\sigma(t))$ and $\tau_a(t) < \tau_d(\sigma(t))$. We assume that each task lies entirely within the planning horizon. If a task in a railway application departs earlier than the beginning of the planning horizon, we set its departure time to the beginning of the planning period. Similarly, we cut off the parts of tasks after the end of the planning horizon.

The input also specifies the set of urgent units and the set of maintenance tasks. For each urgent unit u , the set of those maintenance tasks is given that can be assigned to u ; these sets may express different urgencies. Finally, we assume that the number of urgent units equals the number of maintenance tasks.

4.4 Literature Overview

Ziarati et al. (1997) set up a large-scale integer programming model for locomotive assignment where maintenance routing plays an important role. Lingaya et al. (2002) describe a model for supporting the operational management of locomotive-hauled railway cars. They seek for a maximum expected profit schedule that satisfies various constraints, among them also maintenance requirements.

Other papers focus on aircraft maintenance routing. For example, Barnhart et al. (1998a), Clarke et al. (1997), Feo and Bard (1989), Gopalan and Talluri (1998) and Talluri (1998) deal with this problem. Andereg et al. (2003) describe models for railway applications that are similar to the aircraft routing models.

These models consider maintenance routing as a part of the medium- and long-term vehicle scheduling problem. Small safety inspections (that are arranged at NSR by the local planners on an ad hoc basis) form an important part of the problem specification and also the larger-scale maintenance checks are taken into account. These models compute the vehicle circulation for the forthcoming month such that each vehicle is scheduled for maintenance exactly once. Also, the vehicle schedules are created from scratch, without taking shunting into account. Therefore, in a railway application there is just a small probability that the output of the models in the literature can be carried out in practise.

4.5 The Interchange Model

The basic decision of maintenance routing planners is to consider a number of tasks and to re-arrange the successors of these tasks. The set of all such simultaneous interchanges can be very large. In fact, each modified plan can be interpreted as a large simultaneous interchange. In order to keep the size of the model tractable, we consider only a relatively small set of elementary modifications called ‘interchanges’. An essential part of the model in this section is that interchanges can be combined, that is, interchanges can be carried out after each other. Then solutions with complex patterns can be obtained by using the interchanges as building blocks.

4.5.1 Input and the Output of the Interchange Model

The input of the model contains the list of allowed modifications of the regular plan. We assume that these allowed modifications have a special form, we call them ‘interchanges’. An interchange describes which unit should take over the role of which other unit and it specifies exactly when this must take place. We make several assumptions about interchanges.

An *interchange* is specified by two time moments $m_1 < m_2$ in the planning period, by $k \geq 2$ regular transitions $(t_i, \sigma(t_i))$ and by a permutation $I: \{1, \dots, k\} \rightarrow \{1, \dots, k\}$. For each $i = 1, \dots, k$, we require

$$\begin{aligned}\tau_a(t_i) &\leq m_1, \\ m_2 &\leq \tau_d(\sigma(t_i)).\end{aligned}$$

In other words, for each $i = 1, \dots, k$, the time interval $[m_1, m_2]$ is contained in the time interval between the tasks t_i and $\sigma(t_i)$.

A single rule $i \rightarrow I(i)$ of an interchange is interpreted as follows. The unit that arrived as task t_i keeps its planned position on the shunting yard till moment m_1 . Then it is repositioned to the place where another unit is stored according to the shunting plan: this other unit was planned to carry out task $\sigma(t_{I(i)})$. The repositioning is ready at the moment m_2 : from that moment on, the original shunting plan is executed again. When all rules of an interchange are applied, each task is covered by a unit.

The time difference $m_2 - m_1$ must be enough to carry out the necessary shunting operations, so it may not be shorter than a certain buffer time ϱ . Assuming that ϱ minutes are enough for the extra shunting, we only allow interchanges where the difference $m_2 - m_1$ equals ϱ . Based on discussion with planners, we set ϱ to 10 minutes.

We also assume that for each single interchange, the shunting plans can in fact be modified so that the interchange is carried out. That is, that the individual interchanges are approved by local planners or at least by maintenance routing planners. However, we do not know which combinations of interchanges can be carried out together.

Example. Figure 4.2 indicates three units that arrive according to the regular plan at a station and depart from there later. In this example the buffer time equals to 10 minutes. The interchange is described by the given tasks and by

$$\begin{aligned} m_1 &= 10:00, & m_2 &= 10:10, \\ 1 &\rightarrow I(1) = 2, & 2 &\rightarrow I(2) = 3 \quad \text{and} \quad 3 \rightarrow I(3) = 1. \end{aligned} \quad (4.1)$$

As one can see in Figure 4.2, the same interchange structure between the same three units may also take place 10 minutes later. In the input, it is represented by another interchange with $m_1 = 10:10$, $m_2 = 10:20$ and with the same interchange rules as in (4.1).

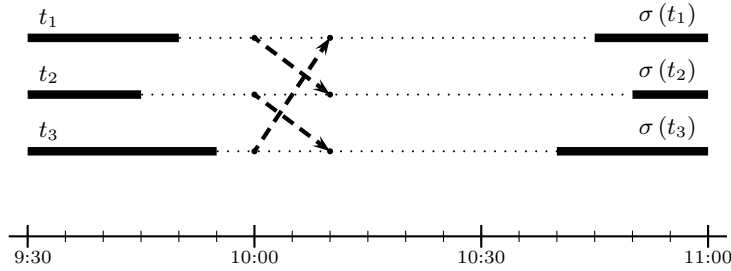


Figure 4.2: An example of an interchange.

This is a basic notion of interchanges. In our implementations, we used a slightly extended version of interchanges that is described in Section 4.5.4. Here we restricted ourselves to this basic notion for the sake of a simpler, more intuitive notation.

Applying one interchange only provides very limited possibilities. So, we wish to combine interchanges in order to model shunting in a more flexible way.

Definition 4.1. Two interchanges given by $(t_1, \dots, t_k, I, m_1, m_2)$ and $(t'_1, \dots, t'_{k'}, I', m'_1, m'_2)$ are independent if the sets of tasks $\{t_1, \dots, t_k\}$ and $\{t'_1, \dots, t'_{k'}\}$ are disjoint or if the time intervals $[m_1; m_2]$ and $[m'_1; m'_2]$ are disjoint.

In this model of the shunting process, only pair-wise independent interchanges can be combined. This can be motivated as follows. From the maintenance routing planners' perspective, it is difficult to check whether several interchanges together can be implemented in practise. By using independent interchanges, we localise their effects on the regular plan, making it more likely that the entire new plan can be carried out. It is possible that a combination of certain non-independent interchanges

can still be carried out, it may even decrease the number of shunting movements. As the information about local shunting possibilities is limited, we do not allow such non-independent interchanges to stay on the safe side.

It is interesting to compare this shunting model to that in Section 3.3 (also used in Chapter 5). In tactical (and also operational planning), the order of the units on the shunting yards is not considered at all. In contrary, we assume that every composition change is possible. In short-term planning, however, we keep track of the position of every individual unit.

The input of the Interchange Model contains a penalty value for each interchange, expressing how much additional shunting effort it causes. These penalties also reflect the likelihood that the interchange can be carried out successfully: more risky interchanges get higher penalties. In Section 4.5.5 we will discuss how we can obtain a list of interchanges and penalty values.

The output of the model is a collection of independent interchanges such that applying them, the urgent units reach an appropriate maintenance task. We can interpret this output as a collection of new duties (i.e. chains of tasks) for the units plus suggestions to local planners how the new rolling stock plan can be implemented in practise.

4.5.2 Graph Representation

We give a flow-type model for the maintenance routing problem, thus we first need a graph $G = (V, A)$. The units are labelled by $d = 1, \dots, D$. Index the integer minutes in the planning period by $0, \dots, H$. Create a node for every pair (d, m) with $d \in \{1, \dots, D\}$ and $m \in \{0, \dots, H\}$ such that m does not lie strictly between the departure time and the arrival time of any task which is carried out by unit d . For a node $v = (d, m)$, let $\text{time}(v) := m$. *First* nodes are the nodes with $\text{time}(v) = 0$, *last* nodes are those with $\text{time}(v) = H$. Let V_0 denote the set of first nodes and V_∞ the set of last nodes.

For any task t which is carried out by unit d , insert a *task arc* from $(d, \tau_d(t))$ to $(d, \tau_a(t))$. Moreover, for each $d \in \{1, \dots, D\}$ and for each $m \in \{0, \dots, H\}$, join node (d, m) to $(d, m + 1)$ if these nodes exist and if they are not yet joined by a task arc. The arcs defined so far are called *regular arcs*. They form a collection of disjoint directed paths, each path representing the regular duty of one unit.

Each node (d, m) of the graph refers to a station as follows. Unit d has a latest arrival event as task t before m or it has an earliest departure event as task t' after

m . Then node $v = (d, m)$ refers to $s_a(t)$ or $s_d(t')$, whichever of them exists; this station is denoted by $\text{stat}(v)$. Note that if both t and t' exist, then $s_a(t) = s_d(t')$.

Let $\mathcal{I}$ denote the set of interchanges. Consider an interchange $C = (t_1, \dots, t_k, I, m_1, m_2) \in \mathcal{I}$. For each $i = 1, \dots, k$, let $v_i = (d_i, m_1)$ and $w_i = (d_i, m_2)$ where d_i is the unit that carries out tasks t_i and $\sigma(t_i)$ in the regular plan. Create for each $i = 1, \dots, k$ an *interchange arc* $(v_i, w_{I(i)})$ and identify C with the set of arcs $\{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$. If a particular pair (v, w) is an interchange arc in several interchanges, we have parallel arcs from v to w .

Independence of interchanges is defined in Definition 4.1. In this graph representation of the problem, the definition can be stated as follows.

Definition 4.2. Let $C = \{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$ and $C' = \{(v'_i, w'_{I'(i)}) \mid i = 1, \dots, k'\}$ be interchanges. Let P be the union of the arc sets of those regular arc paths that connect the nodes v_i to the nodes w_i . Let P' be defined similarly for the interchange C' . Then C and C' are said to be independent if the arc sets P and P' are disjoint.

The arcs of the graph have weights to estimate the shunting effort of the interchanges. For each interchange arc a , set $\text{weight}(a)$ to the penalty of the complete interchange containing arc a . The weight of the regular arcs is zero.

Example. Figure 4.3 shows the graph for the six tasks in Figure 4.2 and the interchange in (4.1). Horizontal arcs are regular, the six task arcs are denoted by t_1, t_2, t_3 and $\sigma(t_1), \sigma(t_2), \sigma(t_3)$. The interchange is represented in the graph by the arcs e, f, g .

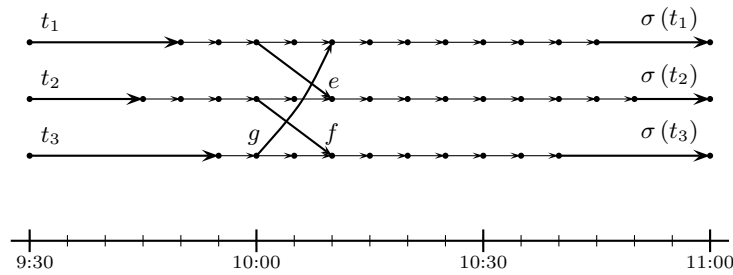


Figure 4.3: The graph representation of the interchange in Figure 4.2.

Let U be the set of nodes $(u, 0)$ where $u \in \{1, \dots, D\}$ is the index of an urgent unit. We identify the urgent units with the nodes in U . For each maintenance task that is carried out in the regular plan by unit d and that starts at time moment

m , we mark node (d, m) as a *maintenance node*. The set of maintenance nodes is denoted by M . For urgent unit u , M_u denotes the set of those maintenance nodes that can be assigned to unit u .

This completes the definition of the graph G . Each arc points from an earlier time moment to a later one, therefore the graph does not contain directed circuits.

According to the regular plan, the D units follow D node-disjoint paths consisting of the regular arcs. Executing an interchange $\{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$ corresponds to the following operation. We replace the regular arc paths that connect the nodes v_i to the nodes w_i by the arcs $(v_i, w_{I(i)})$ of the interchange. Starting from the original D node-disjoint regular arc paths and executing a collection of pairwise independent interchanges, we obtain D new node-disjoint paths that cover all the task arcs. We consider these paths as a modified rolling stock plan.

Having built up the graph from the input, the Interchange Model for the maintenance routing problem is stated as follows:

Definition 4.3. Interchange Model: Select a collection of pairwise independent interchanges such that, when executing the selected interchanges, the urgent units are routed to a node representing an appropriate maintenance task. Minimise the total weight of the solution, defined as the sum of the individual weights of those arcs that are used by the urgent units until they reach a maintenance task.

Example. Consider again the maintenance routing problem shown in Figure 4.1. Assuming that the input contains 6 interchanges, Figure 4.4(a) shows the graph representation of this problem: each interchange corresponds to a pair of diagonal arcs. The starting position of the urgent unit is node u . This unit must be routed to the maintenance node μ . Figure 4.4(b) indicates a feasible solution where two interchanges are selected. In both figures, task arcs are bold.

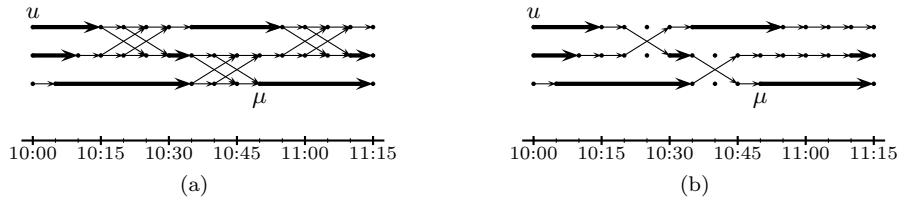


Figure 4.4: (a) The graph defined for the instance in Figure 4.1. (b) A feasible solution for this instance.

Note that, by executing pairwise independent interchanges, each task arc remains covered by a unit. Therefore, we do not need explicit constraints for this requirement.

The objective function is intended to estimate the total shunting effort required to carry out a solution. There are several ways to do it. A natural objective function would be to take the sum of the penalties of the selected interchanges. The objective function in the problem formulation, however, counts the penalty for an interchange multiple times if more than one urgent unit uses that interchange. This implements the idea that an interchange being responsible for routing several urgent units is much more critical than an interchange with only one urgent unit. We also prefer our choice because it enables us to compute good lower bounds for measuring the performance of heuristic solution methods. Nevertheless, the ultimate justification of an objective function is whether or not the planners in practise are satisfied by the solutions that the model provides.

4.5.3 Integer Programming Formulation

We formulate the problem stated above for the graph $G = (V, A)$ as a binary linear program. The set of arcs entering node v is denoted by $\delta^{\text{in}}(v)$ and the set of arcs leaving v by $\delta^{\text{out}}(v)$.

The binary decision variables are the following:

$$\begin{aligned} y &: \mathcal{J} \rightarrow \{0, 1\}, \\ z &: A \rightarrow \{0, 1\}, \\ x_u &: A \rightarrow \{0, 1\} \quad \forall u \in U. \end{aligned}$$

The decision variables y_C are related to the interchanges: $y_C = 1$ if and only if interchange $C \in \mathcal{J}$ is selected in the solution. The functions x_u are network flows, indicating the paths of the urgent units to the maintenance tasks, while function z is a network flow that represents the collection of paths of all units during the entire planning period. The model is formulated as follows.

$$\text{Minimise } \sum_{a \in A} \left(\text{weight}(a) \cdot \sum_{u \in U} x_u(a) \right) \quad (4.2)$$

subject to

$$\sum_{a \in \delta^{\text{in}}(v)} z(a) = \sum_{a \in \delta^{\text{out}}(v)} z(a) \quad \forall v \in V \setminus (V_0 \cup V_\infty) \quad (4.3)$$

$$\sum_{a \in \delta^{\text{out}}(v)} z(a) = 1 \quad \forall v \in V_0 \quad (4.4)$$

$$\sum_{a \in \delta^{\text{in}}(v)} z(a) \leq 1 \quad \forall v \in V \quad (4.5)$$

$$\sum_{a \in \delta^{\text{out}}(v)} z(a) \leq 1 \quad \forall v \in V \quad (4.6)$$

$$\sum_{a \in \delta^{\text{in}}(v)} x_u(a) = \sum_{a \in \delta^{\text{out}}(v)} x_u(a) \quad \forall u \in U, v \in V \setminus (\{u\} \cup M_u) \quad (4.7)$$

$$\sum_{a \in \delta^{\text{out}}(v)} x_u(a) = 1 \quad \forall u \in U \quad (4.8)$$

$$\sum_{u \in U} \sum_{a \in \delta^{\text{out}}(v)} x_u(a) = 0 \quad \forall v \in M \quad (4.9)$$

$$\sum_{u \in U: v \notin M_u} \sum_{a \in \delta^{\text{in}}(v)} x_u(a) = 0 \quad \forall v \in M \quad (4.10)$$

$$\sum_{u \in U} x_u(a) \leq z(a) \quad \forall a \in A \quad (4.11)$$

$$z(a) = y_C \quad \forall C \in \mathcal{I}, a \in C \quad (4.12)$$

$$z(a) \in \{0, 1\} \quad \forall a \in A \quad (4.13)$$

$$x_u(a) \in \{0, 1\} \quad \forall u \in U, a \in A \quad (4.14)$$

$$y_C \in \{0, 1\} \quad \forall C \in \mathcal{I} \quad (4.15)$$

Constraints (4.3) and (4.4) state that the function z is a network flow where the first nodes are sources of value 1 and where the last nodes are the sinks. Constraint (4.5) and (4.6) make sure that every node has a throughput of at most 1 in this flow. Then the arcs with $z(a) = 1$ form a system of node-disjoint paths, connecting the first nodes to the last nodes.

Similarly, constraints (4.7) and (4.8) express that the functions x_u are network flows such that u is the only source node and that the nodes in M_u are the sinks. Constraints (4.9) make sure that no flow leaves the maintenance nodes in any of the network flows x_u . Moreover, (4.10) states that a maintenance node v may have incoming flow only in a flow x_u if v is an allowed maintenance node for urgent unit u . According to constraints (4.11), the paths indicated by the flows x_u form a subsystem of the path system indicated by the network flow z . Constraints (4.12) express that either all arcs of an interchange are selected for the network flow z or none of them. Finally, constraints (4.13) – (4.15) set the domains of the variables.

The binary integer program does not have any explicit constraints saying that the selected interchanges should be independent. The following lemma states that such constraints are not necessary.

Lemma 4.4. *Consider any feasible solution of the model (4.2) – (4.15). Then for any interchange $C = \{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$ with $y_C = 1$ and for $i = 1, \dots, k$, the regular arc path between nodes v_i and w_i does not contain a node $v \notin \{v_i, w_i\}$ such that v is traversed by the flow z .*

Proof. Suppose there exists an interchange C with $y_C = 1$ and a node $v = (d, m)$ lying between nodes v_i and w_i that is traversed by z (see Figure 4.5). Choose such a pair (C, v) so that m is as small as possible. Then the nodes strictly between v_i and v have no throughput in z . Let e be the arc entering node v with $z(e) = 1$. This arc e cannot be a regular arc, so $e \in C'$ for some interchange $C' = \{(v'_j, w'_{I'(j)}) \mid j = 1, \dots, k'\}$. This interchange C' must contain an arc f with $z(f) = 1$ that leaves a node (d, m') with $m' < m$. The tail of arc f is traversed by z , therefore $\text{time}(\text{tail}(f)) \leq \text{time}(v_i)$. (Here, $\text{tail}(f)$ denotes the tail of arc f .) Moreover, $v_i \neq \text{tail}(f)$, since otherwise node v_i would have an outflow in z larger than 1. Thus we have $\text{time}(\text{tail}(f)) < \text{time}(v_i)$.

Then interchange C' has $y_{C'} = 1$, node v_i is traversed by z and lies between v'_j and w'_j for an index j that satisfies $v = w'_j$. This contradicts the choice of interchange C and node v . ■

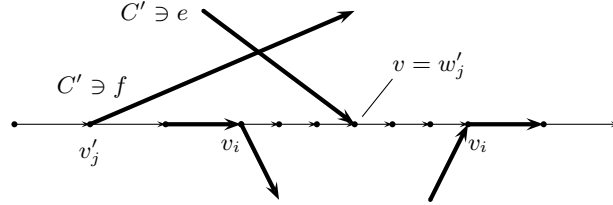


Figure 4.5: Illustration for the proof of Lemma 4.4. Bold arcs have z -value 1.

Lemma 4.4 immediately implies

Corollary 4.5. *Let (x, y, z) be a feasible solution of the binary integer program (4.2) – (4.15) and suppose that $y_{C_1} = y_{C_2} = 1$ for interchanges $C_1, C_2 \in \mathcal{I}$. Then C_1 and C_2 are independent.*

Thus, in every feasible solution of the binary linear program, the set of arcs with $z(a) = 1$ arises from the regular transitions by executing the interchanges with $y_C = 1$ and these interchanges are pairwise independent. Therefore, every feasible solution

of the binary linear program admits a feasible solution of the Interchange Model and these solutions have the same objective value. One can easily prove the other direction, too: every solution of the Interchange Model corresponds to a solution of the binary linear program.

4.5.4 Extending the Notion of Interchanges

The notion of interchanges does not cover some modifications of the regular plan that are in fact possible and useful in practise.

Example. Consider the example in Figure 4.6. A train consisting of the white and the grey unit arrives at a station and after a couple of minutes, it departs again without changing the direction. Suppose furthermore that at that time, a third (black) unit is stored at the station.

It might be possible to place the black unit to the arrival platform, then to couple the arriving train to it and to uncouple the white unit (see Figure 4.7). This modification is shown schematically in Figure 4.8. In fact, such modifications of the regular plan occur quite often in maintenance routing.

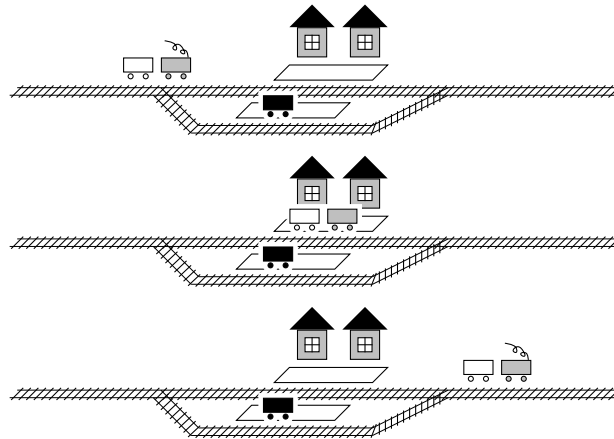


Figure 4.6: A train arrives at a station and a couple of minutes later, it departs as another timetable service.

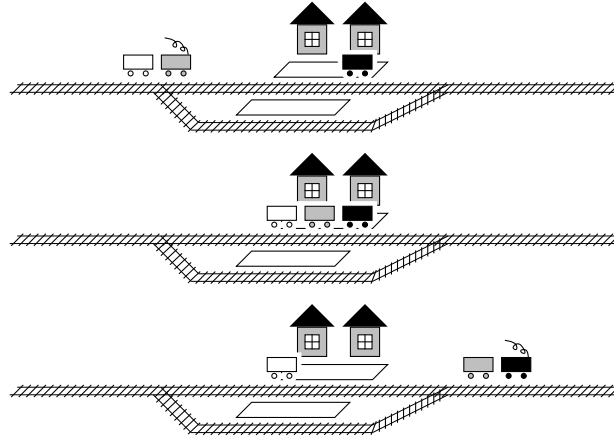


Figure 4.7: Coupling the black unit to the arriving train and uncoupling the white unit from it.

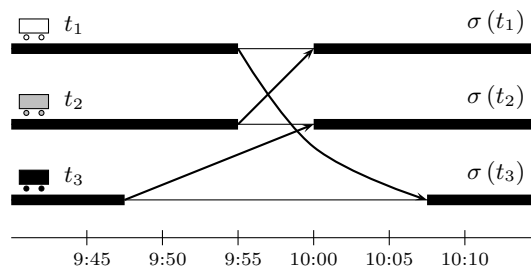


Figure 4.8: The modification of the regular plan in Figure 4.7 schematically.

The modification in the example above does not comply with the definition of interchanges if the time difference between the arrival and departure is less than the buffer time ϱ . This motivates to extend the notion of interchanges.

For tasks $t_1, \dots, t_k$, let

$$\alpha := \max\{\tau_a(t_i) : i = 1, \dots, k\},$$

$$\beta := \min\{\tau_d(\sigma(t_i)) : i = 1, \dots, k\}.$$

So far, we allowed interchanges $(t_1, \dots, t_k, I, m_1, m_2)$ with $m_2 = m_1 + \varrho$ if $\beta - \alpha \geq \varrho$. We now introduce interchanges for the case $\alpha < \beta < \alpha + \varrho$. That is, the time intervals $[\tau_a(t_i); \tau_d(\sigma(t_i))]$ intersect but the intersection is shorter than ϱ minutes. We require for an extended interchange between tasks t_i and $\sigma(t_i)$ for $i = 1, \dots, k$ the following:

$$\begin{aligned} \tau_a(t_i) = \alpha \quad \text{or} \quad \tau_a(t_i) \leq \beta - \varrho & \quad \forall i = 1, \dots, k, \\ \tau_d(\sigma(t_i)) = \beta \quad \text{or} \quad \tau_d(\sigma(t_i)) \geq \alpha + \varrho & \quad \forall i = 1, \dots, k. \end{aligned}$$

That is, the arrival time of a task is either exactly the latest arrival time or it is smaller than the earliest departure time minus the buffer time. A similar restriction is stated for the departure times. The interchanges defined in Section 4.5.1 satisfy this definition. Note that although not part of the definition, non-basic interchanges are only intended to model the case where all tasks with arrival time α belong the one trip, and also all tasks with departure time β belong the one trip. Such a case is shown in Figures 4.7 and 4.8.

Just like in Section 4.5.1, the substantial content of an extended interchange is described by the permutation $I: \{1, \dots, k\} \rightarrow \{1, \dots, k\}$ which indicates how the modified plan matches tasks $t_1, \dots, t_k$ to the tasks $\sigma(t_1), \dots, \sigma(t_k)$.

Extended interchanges can easily be added to the graph defined in Section 4.5.2. Suppose that tasks $t_1, \dots, t_k$ form an extended interchange with $\alpha > \beta - \varrho$ (i.e. they cannot form a basic interchange) and suppose that for each $i = 1, \dots, k$, task t_i is carried out by unit d_i in the regular plan. Then we represent the extended interchange with the following set of arcs:

$$\text{from node } (d_i, \max\{\alpha; \beta - \varrho\}) \text{ to node } (d_{I(i)}, \min\{\beta; \alpha + \varrho\}).$$

Again, we identify the extended interchange with this set of arcs in the graph. The graph representation of the extended interchange considered above is given in Figure 4.9.

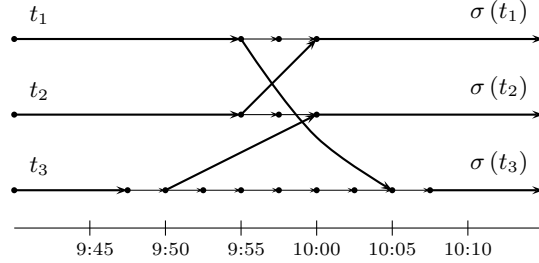


Figure 4.9: The modification of the regular plan indicated in Figure 4.7 is represented in the graph.

Using extended interchanges, the formulation of the Interchange Model and the binary linear program (4.2) – (4.15) remain unchanged; independence of interchanges is defined by Definition 4.2. Then each claim about the binary program, in particular Corollary 4.5, can be proved without any change.

We emphasise two important properties of extended interchanges in the graph representation of the problem.

1. For an extended interchange $\{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$, the time intervals $[\text{time}(v_i); \text{time}(w_i)]$ intersect.
2. If an extended interchange contains two arcs a_1 and a_2 such that the tail of a_1 refers to an earlier time moment than the tail of a_2 does (formally: if $\text{time}(\text{tail}(a_1)) < \text{time}(\text{tail}(a_2))$) then the tail of a_2 is the head of a task arc. Similarly, if $\text{time}(\text{head}(a_1)) < \text{time}(\text{head}(a_2))$ then the head of a_1 is the tail node of a task arc.

4.5.5 Obtaining the Input Data

The Interchange Model requires a large amount of information about the shunting process. Collecting all these details is not easy. Another problem is to define appropriate numerical penalty values: Different local planners may judge the difficulty of certain interchanges quite differently.

For our test implementation, we create the interchanges ourselves using the extended notion of interchanges. First we specify the buffer time. As mentioned, discussions with the planners lead to a value of 10 minutes. Then we look for pairs

of tasks t_1 and t_2 with

$$\begin{aligned} s_a(t_1) &= s_a(t_2), \\ \tau_a(t_1) &\leq \tau_d(\sigma(t_2)) - \varrho, \\ \tau_a(t_2) &\leq \tau_d(\sigma(t_1)) - \varrho. \end{aligned}$$

Such pairs provide candidates for interchanges. We consider the (at most) four trips that tasks t_1 , t_2 , $\sigma(t_1)$ and $\sigma(t_2)$ belong to. Interchanges between units in these trips are defined by applying simple rules like exchanging the leftmost units or exchanging the whole compositions if they have the same length.

We also insert empty trains of the following form. If two units are stored at different stations but have no tasks in the regular plan during a period of, say, 2 hours, we define an interchange between these units during this time interval. Such empty trains are only allowed at night and only to bring units directly to the maintenance facility.

We use the following factors to define the penalty values of the interchanges:

1. The number of extra shunting movements and extra storage tracks that are needed for carrying out the interchange; these parameters depend on the positions of the units in the compositions as well as on the infrastructure of the stations.
2. The length of the time interval between the arrival and departure events: the shorter this interval, the higher the penalty value.
3. Stations with heavy traffic provide less shunting possibilities, therefore the penalty there is higher.
4. Interchanges during the morning or evening peak hours are difficult; interchanges at night and during off-peak hours are easier.

Maintenance routing planners may prefer modifications in the regular plans as early as possible. Such considerations can easily be incorporated in the penalty values.

When applying the Interchange Model in practise, the most reliable source for obtaining the interchanges and their penalties would be the local planners. However, the interchanges and the penalties highly depend on the regular plan. Whenever the regular plan is adjusted, the interchanges might also need to be updated. Therefore, successful application of the Interchange Model requires the automation of generating and evaluating the interchanges.

4.5.6 NP-Completeness of the Interchange Model

In this section we show that the feasibility problem of the Interchange Model is NP-complete. The size of the graph is determined by the number of tasks and by the size of the list $\mathcal{J}$ of interchanges. Note that the number of interchanges itself depends on the length of the planning horizon. The construction below shows that the problem is NP-complete if $|\mathcal{J}| = O(d)$ where d denotes the number of units and if the longest path in the acyclic graph has length 3.

Theorem 4.6. *It is NP-complete to decide whether or not the Interchange Model has a feasible solution.*

Proof. We show that 3SAT can be polynomially reduced to the feasibility of the Interchange Model. Consider a conjunctive normal form

$$\varphi = \bigwedge_{i=1}^n (x_{a_i}^{\varepsilon_{a_i}} \vee x_{b_i}^{\varepsilon_{b_i}} \vee x_{c_i}^{\varepsilon_{c_i}})$$

in Boolean variables $x_1, \dots, x_\ell$ where $\varepsilon_j \in \{+1, -1\}$ and

$$x_j^{\varepsilon_j} = \begin{cases} x_j & \text{if } \varepsilon_j = +1, \\ \neg x_j & \text{if } \varepsilon_j = -1. \end{cases}$$

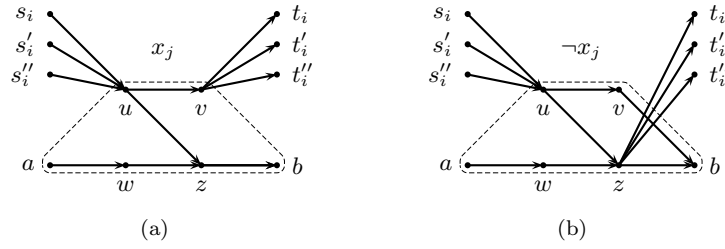
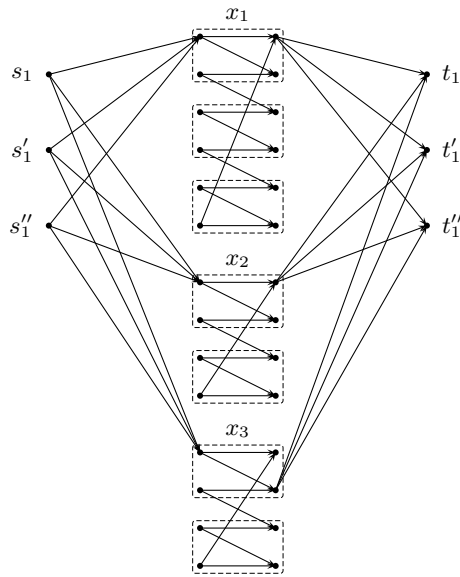
It is well known that the problem of deciding whether or not such a conjunctive normal form φ can be satisfied is NP-complete (see Cook (1971)).

For any conjunctive normal form φ , we build up an instance of the Interchange Model such that the required maintenance paths in the graph exist if and only if φ can be satisfied. The construction is polynomial in the size of φ .

Let $d = 6n$. We start from an empty graph and an empty list $\mathcal{J}$. For each clause $x_{a_i}^{\varepsilon_{a_i}} \vee x_{b_i}^{\varepsilon_{b_i}} \vee x_{c_i}^{\varepsilon_{c_i}}$, we create the nodes s_i, s'_i, s''_i and t_i, t'_i, t''_i .

For every occurrence of a literal x_j or $\neg x_j$ in clause i , create a *box* consisting of 6 new nodes u, v, w, z, a, b and draw the arcs $uv, wz, uz, aw, s_i u, s'_i u$ and $s''_i u$. If the non-negated literal x_j appears in clause i , insert the arcs vt_i, vt'_i, vt''_i, zb . If $\neg x_j$ appears in clause i , insert the arcs zt_i, zt'_i, zt''_i, vb (see Figure 4.10). Mark the arcs uv, wz, aw, vb and zb as horizontal.

We join the boxes corresponding to the same Boolean variable by fixing any cyclic ordering on them and drawing an arc from the w -node of any box to the v -node of the next box in the ordering. Then we obtain a graph as shown in Figure 4.11. In this figure, the dashed rectangles contain the u, v, w , and z -nodes of the boxes, the a and b -nodes are not drawn.

Figure 4.10: Boxes corresponding to literals x_j and $\neg x_j$.Figure 4.11: The graph for $\varphi = (x_1 \vee x_2 \vee \neg x_3) \wedge \dots$.

For each variable x_j , we create an interchange $C \in \mathcal{J}$ consisting of the arcs of type uz inside and the arcs of type wv between the x_j -boxes.

For every clause i , consider the nodes s_i, s'_i, s''_i and the 3 u -nodes they are connected to. These 6 nodes span a complete bipartite subgraph. Decompose these 9 arcs into 3 disjoint perfect matchings, mark the arcs in one of these matchings as horizontal and add the other two matchings as interchanges with 3 arcs to $\mathcal{J}$. Then do the same at the other side: at the nodes t_i, t'_i, t''_i and their neighbours.

This gives an instance of the Interchange Model. Each node represents a task with a length of zero minutes. The horizontal arcs are the regular transitions, the nodes s_i are the first tasks of the urgent units, t_i is the unique maintenance task assigned to s_i and $\mathcal{J}$ is the list of interchanges. Note that the longest path in this graph contains 3 arcs and that $\mathcal{J}$ has $\ell + 4n$ elements.

We claim that the Interchange Model for this graph has a feasible solution if and only if φ can be satisfied.

Suppose there exists an assignment of the Boolean variables making φ true. If $x_j = 1$, select the horizontal arcs inside all the x_j -boxes, otherwise select the uz and wv arcs inside or between the x_j -boxes. For every clause i , choose a literal making it true and consider the box corresponding to this occurrence of the variable. If the arc e from s_i to the u -node of this box is not horizontal, select e as well as the two other arcs that appear in the same member of $\mathcal{J}$ as e does. If e is horizontal, select the horizontal arcs incident to s_i, s'_i, s''_i . Select also the arc leading from the box to t_i together with the other two arcs that were defined similarly.

Finally, select for every box the arc leaving the a -node of the box and the arc entering the b -node of the box. Then the selected arcs form the required path system: we only use horizontal (i.e. regular) arcs and all arcs of some of the interchanges.

Conversely, suppose there exists such a path system. Assign the value true to a variable x_j if and only if the horizontal arcs are used in the x_j -boxes. Then for any i , the $s_i - t_i$ path shows that the i th clause of φ is satisfied. ■

4.5.7 The Case of One Urgent Unit

In this section we show that the case of one urgent unit is easy to solve. Each feasible solution of the Interchange Model contains a path from a first node that corresponds to an urgent unit to an allowed maintenance node. In case of a single urgent unit, it takes only very mild extra assumptions to make the converse, that each such path can be extended to a solution of the Interchange Model, also true. These extra assumptions are satisfied in each realistic railway application. With

these assumptions, the case of a single urgent unit can be reduced to a shortest path problem. This simple algorithm will serve as the basic step of a heuristic solution approach in Section 4.5.8 for routing several urgent units.

The key idea of solving the case of a single urgent unit is to ensure that if a directed path in the graph representation of the problem contains two interchange arcs then the two corresponding interchanges are independent. Clearly, this is true when using basic interchanges. Indeed, the tail nodes of all arcs in a basic interchange correspond to the time moment m_1 and the head nodes correspond to the time moment $m_2 = m_1 + \varrho$. So, interchange arcs in a directed path correspond to disjoint time intervals. Therefore, the corresponding interchanges are pair-wise independent.

However, using extended interchanges, one can easily construct instances of the Interchange Model where a directed path contains arcs from two non-independent interchanges. The following lemma states that this cannot happen if the empty trains satisfy some requirements and if the buffer time ϱ is not too large compared to a certain parameter of the timetable.

Empty trains appear in the model as interchanges $\{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$ where the nodes v_i and w_i do not refer to the same station. We require the following three properties of such interchanges.

$$\text{time}(v_i) = \text{time}(v_j) \quad \forall 1 \leq i, j \leq k, \quad (4.16)$$

$$\text{time}(w_i) = \text{time}(w_j) \quad \forall 1 \leq i, j \leq k, \quad (4.17)$$

$$\begin{aligned} &\text{The units involved in the interchange do not have any task } \varrho \text{ minutes} \\ &\text{before and after the interchange.} \end{aligned} \quad (4.18)$$

A *round-trip* is a sequence $t_1, \dots, t_k$ of tasks with $s_a(t_i) = s_d(t_{i+1})$ and $\tau_a(t_i) < \tau_d(t_{i+1})$ for $i < k$ and $s_d(t_1) = s_a(t_k)$. In other words, it is a sequence of tasks that could be carried out by a unit consecutively such that the starting and ending station of the sequence coincide. The time duration of a round-trip is $\tau_a(t_k) - \tau_d(t_1)$.

Lemma 4.7. *Suppose that every round-trip has a time duration of at least 2ϱ . Moreover, suppose that every interchange that describes empty trains satisfies (4.16) – (4.18). Then for any directed path P and any pair of interchange arcs $e, e' \in P$, the interchanges containing the arcs e and e' are independent.*

Proof. We may assume that arc e precedes arc e' in P . Let $C = \{(v_i, w_{I(i)}) \mid i = 1, \dots, k\}$ and $C' = \{(v'_i, w'_{I'(i)}) \mid i = 1, \dots, k'\}$ be the interchanges with $e = (v_1, w_{I(1)})$ and $e' = (v'_1, w'_{I'(1)})$. Suppose that C and C' are not independent. That is, with appropriate indices, the regular arc path from v_2 to w_2 intersects the regular

arc path from v'_2 to w'_2 . In particular, the time intervals $[\text{time}(v_2); \text{time}(w_2)]$ and $[\text{time}(v'_2); \text{time}(w'_2)]$ intersect. The properties of extended interchanges imply that

$$\text{time}(v_2) < \text{time}(w_{I(1)}), \quad (4.19)$$

$$\text{time}(v'_1) < \text{time}(w'_2). \quad (4.20)$$

Clearly, we have

$$\text{time}(w_{I(1)}) \leq \text{time}(v'_1) \quad (4.21)$$

implying that

$$\text{time}(v_2) < \text{time}(w'_2). \quad (4.22)$$

Therefore C and C' can only be non-independent if

$$\text{time}(v'_2) < \text{time}(w_2). \quad (4.23)$$

Figure 4.12 shows an example. The waved line together with arcs e and e' form the path P . The interchange arcs f and f' belong to C and C' , respectively.

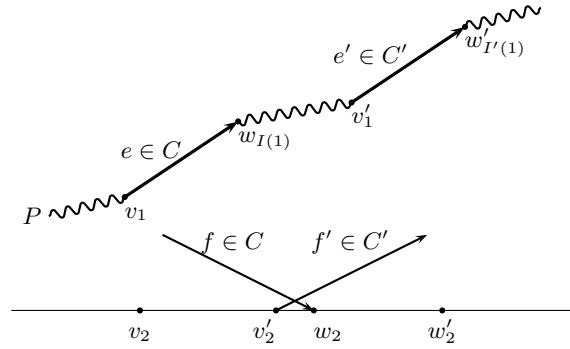


Figure 4.12: The path P having arcs from non-independent interchanges.

Observe that there is no task arc on the regular arc path that connects v'_2 to w_2 . Therefore, these two nodes refer to the same station s :

$$\text{stat}(w_2) = \text{stat}(v'_2) = s. \quad (4.24)$$

It follows from (4.19) – (4.23) that $\text{time}(w_{I(1)}) < \text{time}(w_2)$ or $\text{time}(v'_2) < \text{time}(v'_1)$. We may assume that the first holds (the other case being analogous). Then by (4.17), C cannot contain empty trains. In particular, $\text{stat}(w_{I(1)}) = s$. Moreover, using the properties of extended interchanges again, a task arc starts at $w_{I(1)}$ and we have

$$\text{time}(w_2) < \text{time}(w_{I(1)}) + \varrho, \quad (4.25)$$

$$\text{time}(v'_2) > \text{time}(v'_1) - \varrho. \quad (4.26)$$

Therefore

$$\begin{aligned} \text{time}(v'_1) - \text{time}(w_{I(1)}) &< (\text{time}(v'_2) + \varrho) + (\varrho - \text{time}(w_2)) \\ &= 2\varrho - \text{time}(w_2) + \text{time}(v'_2) < 2\varrho. \end{aligned} \quad (4.27)$$

If $\text{stat}(v'_1) = s$, then the tasks that are passed by the $w_{I(1)} - v'_1$ -segment of P form a round-trip with a time duration smaller than 2ϱ , a contradiction.

If, however, $\text{stat}(v'_1) \neq s$, then C' contains empty trains. Since there is a task arc on the segment of P between $w_{I(1)}$ and v'_1 , (4.18) implies that

$$\text{time}(v'_2) = \text{time}(v'_1) \geq \text{time}(w_{I(1)}) + \varrho,$$

which contradicts (4.23) and (4.25). ■

Corollary 4.8. *Suppose that the time duration of every round-trip is at least 2ϱ and that the interchanges describing empty trains satisfy (4.16) – (4.18). Then the Interchange Model with one urgent unit can be solved in $O(|A|)$ time.*

Proof. Lemma 4.7 implies that the Interchange Model for one urgent unit is equivalent to the shortest path problem in G . Since G does not contain directed cycles (the arcs are directed according to time), the shortest path algorithm can be implemented very efficiently, the running time is proportional to the number of arcs. ■

In practise, the additional assumptions mean hardly any restriction. We set $\varrho = 10$ minutes; real-life instances are very unlikely to have a round-trip shorter than 20 minutes. Moreover, empty trains are mostly used in the late evening or at night in practise, when constraints (4.16) – (4.18) are satisfied.

4.5.8 Heuristic Algorithm

We have seen in Section 4.5.7 that the case of a single urgent unit is easy to solve. The following *Iterated Shortest Path Heuristic* (ISPH) is a natural approach to find solutions for more than one urgent unit. The basic idea is that the urgent units are routed one-by-one to an appropriate maintenance task.

We fix an order of the set U of urgent units. We iterate on U , applying the following steps for the actual $u \in U$:

1. Look for a shortest path P from u to one of the maintenance nodes that are allowed for u . We stop with an error if there is no path from u to M_u . Let $e_1, \dots, e_\ell$ be the interchange arcs in P and let $C_1, \dots, C_\ell$ be the interchanges with $e_j \in C_j$.
2. Delete each interchange C' from the graph that is not independent from any of the interchanges $C_1, \dots, C_\ell$.
3. For each interchange C_j , represented here as $\{(v_i, w_{I(i)}) \mid i = 1, \dots, k_j\}$, delete all the nodes that lie on the regular arc paths connecting the nodes v_i to w_i strictly between v_i and w_i . (That is, the nodes v_i and w_i are not to be deleted.)
4. Delete each arc that is incident to any of the nodes in P except for the arcs of P .
5. Having updated the graph, we continue iterating on U .

Theorem 4.8 implies that, in each iteration of ISPH, the path for the currently considered urgent unit can be obtained by applying independent interchanges $C_1, \dots, C_\ell$. Afterwards, we explicitly forbid for next iterations all interchanges that are in conflict with the already selected interchanges and we make sure that the later iterations do not touch the path of the currently routed urgent unit. Therefore ISPH provides a feasible solution of the Interchange Model if it terminates without an error.

The running time of ISPH is $O(|U| \cdot |A|^2)$. Indeed, there are at most $|U|$ iterations. In each iteration, the shortest path is computed in an acyclic graph with at most $|A|$ arcs. Thereafter the graph is updated in $O(|A|^2)$ steps: one has to find and delete at most $|A|$ arcs for each interchange arc in P .

The order in which the urgent units are to be processed by ISPH is not specified yet. Due to the short running time, we can simply run ISPH with several orders and select the best solution; these orders may depend on the instances themselves.

4.5.9 Lower Bounds

In the previous section, we described a heuristic solution approach. In order to evaluate its performance without solving the model exactly, we need easy-to-compute lower bounds. A simple way for this is to determine the length ℓ_u of a shortest path from urgent node u to the set of allowed maintenance nodes M_u . Then

$$\sum_{u \in U} \ell_u$$

is a lower bound on the objective value of any feasible solution. Recall that in the objective function (4.2), the weight of an interchange arc is the entire penalty of the interchange.

This simple lower bound can slightly be sharpened as follows. Partition the set U of urgent units into subsets $U_1, \dots, U_h$ such that members of a subset U_i have identical maintenance requirements. That is, if u and u' belong to a set U_i , then $M_u = M_{u'}$; let $M^i := M_u$ for any $u \in U_i$. Consider the graph representation of the Interchange Model. For each $i = 1, \dots, h$, compute a minimum cost node-capacitated network flow with source set U_i , with sinks set M^i and with node capacities 1. Let $\text{MF}(i)$ denote the minimum cost of such a flow. Then

$$\text{FlowBound} := \sum_{i=1}^h \text{MF}(i) \quad (4.28)$$

is a lower bound on the objective value of any feasible solution. Indeed, dropping constraint (4.12) from the Interchange Model, the remaining problem is a multi-commodity flow problem instance where each set of urgent units with identical urgencies is a commodity. Then, by relaxing the constraint that the commodities have to share the node capacities and the maintenance nodes themselves, we obtain the lower bound (4.28).

We mentioned in Section 4.5.2 that several other natural objective functions could be used for the Interchange Model, for instance the sum of the penalties of the interchanges selected in the solution. However, it is not easy at all to find sharp but yet quickly computable lower bounds for those alternative objective functions.

	Planning horizon			
	2 days	3 days	4 days	5 days
Number of nodes	8,998	13,165	17,177	21,358
Number of arcs	38,090	55,179	71,975	88,351

Table 4.1: Number of nodes and arcs in the graph representation

4.6 Computations for the Interchange Model

4.6.1 Test Case

We implement the model for the unit type “Sprinter”. There are 47 units serving about 800 tasks per day. The typical length of the compositions is 1 or 2 units. There are only a few exceptional trains with 3 units. This makes the weight estimates for the interchanges easier. The regular plan contains two maintenance tasks per day. They take place at the maintenance facility in Leidschendam and start in the early morning. Concerning the number of tasks and interchanging possibilities, this unit type forms a medium-size case at NSR. All computations are been carried out on a PC with an Intel P4 3.0 GHz processor and 512 Mb internal memory.

We collected and evaluated the interchanges by applying some simple rules to the regular plan. Thus we computed about 5,700 interchanges per day. Table 4.1 contains the dimensions of the graph model. The penalty values of the interchanges are chosen between 0 and 1000. An arc with a weight higher than 100 corresponds to an interchange which is considered to be quite difficult but still possible in practise. The penalty for empty trains is set to 1000. By allowing empty trains, the model has a feasible solution for any choice of the urgent units. We also ran the model without empty trains; then some instances turned out to be infeasible.

We consider planning horizons of $h = 2, 3, 4$ and 5 days. We always start on Monday morning. So, depending on h , the planning horizon lasts till the late evening of Tuesday, Wednesday, Thursday or Friday. There are $2h$ urgent units: 2 units assigned to the maintenance tasks starting early morning on Tuesday, 2 starting early morning on Wednesday and so on. In each of the instances, every urgent unit is assigned to a unique day within the planning horizon when it has to undergo a maintenance check. The most frequently used planning horizons in the maintenance planning process at NSR are $h = 2$ and $h = 3$.

4.6.2 Experiments

We generate two sets of instances: **RAND** and **DIFF**. Each instance contains the list of urgent units and their assignment to the maintenance days. In the instance set **RAND**, the urgent units are selected randomly; **RAND** has 1000 instances for each possible length of the planning horizon. The set **DIFF** consists of instances where the heuristic algorithm is likely to have difficulties since the paths of the urgent units disturb each other. The set **DIFF** contains 500 instances for each of the planning horizons. For each instance, two stations are chosen and then randomly selected units that start at these two stations are set urgent, the deadlines are also selected randomly.

We apply two solution methods. First, we use the heuristic algorithm implemented in C++. As mentioned in Section 4.5.8, we run **ISPH** with many orders on the urgent units. For each of the planning horizons $h = 2, 3, 4$ and 5 , we define a set of orders of the $2h$ urgent units. These sets have 24, 28, 36 and 52 orders for $h = 2, 3, 4$ and 5 ; they include all possible orders of the 4 urgent units with a deadline of one or two days as well as the possibility to swap two units with identical deadlines. Note that increasing the size of these sets did not lead to significantly better solutions. Below, ‘**HEUR**’ refers to the heuristic solution method and also to its solution value. The flow bound (4.28) is denoted referred as **FlowBound**.

Second, all instances are solved as binary linear programs using the modelling software **ILOG OPL Studio 3.7** and the integer programming solver **ILOG CPLEX 9.0**. The optimal solution values are denoted by ‘**IP**’.

When solving the integer program, we restrict the solving time to 3 hours and use the following **CPLEX** settings. The linear relaxation is solved by the barrier method which outperforms any built-in simplex algorithm by far. Then the branch-and-bound procedure uses the dual simplex method. It turns out to be very useful to perturb the objective function and to apply Best Estimate Search as node selection strategy. We also use all the MIP cuts that **CPLEX** provides, but these have little effect on the solution times. Despite extensive computational experiments, we did not find any other features of **CPLEX** that improved the performance of the branch-and-bound method significantly.

4.6.3 Performance of the Algorithms

The heuristic algorithm **HEUR** gives a feasible solution for every test instance and for every planning horizon of 2, 3, 4 and 5 days.

Test set	h	# inst.	Relative gap		Absolute gap	
			average.	maximum	average	maximum
RAND	2	1,000	1.9%	71%	1.1	39
RAND	3	1,000	5.6%	89%	4.9	54
RAND	4	1,000	8.2%	85%	8.9	70
RAND	5	1,000	14.3%	82%	19.1	103
DIFF	2	500	2.6%	71%	9.1	45
DIFF	3	500	7.3%	134%	25.9	93
DIFF	4	500	7.0%	108%	31.5	114
DIFF	5	500	9.3%	81%	44.7	134

Table 4.2: Comparing HEUR to FlowBound

First we compare the flow bounds to the heuristic solutions. The absolute gap is defined by $\text{HEUR} - \text{FlowBound}$, while the relative gap is defined by $\frac{\text{HEUR} - \text{FlowBound}}{\text{FlowBound}}$. The average and maximum value of the relative gaps and of the absolute gaps are presented in Table 4.2. As we could expect, the gap increases as the length of the planning period and thereby the number of commodities grows.

The absolute gap between the lower bound and the heuristic solution is significantly larger in the set DIFF than in the set RAND. On the other hand, instances in DIFF have typically much higher solution values (and lower bounds), leading to smaller relative gaps. In all instance sets, the relative gap can be quite large. However, the largest absolute gap is never higher than the weight of just one considerably expensive arc, although we computed paths for up to 10 urgent units. We also note that in our computations on further 5-day instances, the highest absolute gap we experienced is 189 which is still about the penalty of one moderately difficult interchange.

None of the solutions we found uses an interchange with a penalty higher than 109 (apart from the very expensive empty trains which are sometimes unavoidable). This indicates that the heuristic solutions tend to use a larger number of interchanges, each of them having a moderately high penalty. More discussions with planner are needed to reveal whether this is advantageous in practise.

We report now our experiments with solving the binary integer program by CPLEX. The dimensions of the integer program are shown in Table 4.3. We also give there the size of the reduced problem instances obtained after the CPLEX pre-processing phase.

The behaviour of the two sets of instances is different. All test instances in RAND can be solved to optimality, despite the large size of the binary linear programs. The

	Planning horizon			
	$h = 2$	$h = 3$	$h = 4$	$h = 5$
Number of variables	265,053	353,404	441,755	530,106
Number of constraints	74,961	125,850	183,213	250,976
Columns in reduced IP	$\sim 35,000$	$\sim 80,000$	$\sim 130,000$	$\sim 200,000$
Rows in reduced IP	$\sim 30,000$	$\sim 55,000$	$\sim 80,000$	$\sim 110,000$
Non-zeros in reduced IP	$\sim 140,000$	$\sim 300,000$	$\sim 520,000$	$\sim 770,000$

Table 4.3: Size of the binary linear programs

Test set	h	# inst.	Relative gap		Absolute gap	
			average	maximum	average	maximum
RAND	2	1,000	0.8%	42%	0.5	27
RAND	3	1,000	2.5%	57%	2.4	37
RAND	4	1,000	3.8%	53%	4.4	44
RAND	5	1,000	6.0%	48%	8.7	54
DIFF	2	500	0.5%	20%	1.6	32
DIFF	3	500	1.5%	45%	6.3	51
DIFF	4	500	1.8%	38%	9.2	53
DIFF	5	500	2.7%	31%	15.5	68

Table 4.4: Comparing HEUR to IP.

solution process mostly requires very few nodes in the branch-and-bound tree, quite often the linear programming relaxation has an integral optimal solution. Some instances require, however, a longer solution process taking up to 5,000 seconds for proving optimality. Even in those cases, an almost optimal solution is found after a couple of branchings.

The instances in DIFF turn out to be harder. All 2-day instances and most of the instances with a longer planning horizon can be solved to optimality before hitting the time limit of 3 hours per instance. In case of 3, 4 and 5-day instances, 9, 15 and 32 instances are not solved to optimality. However, a much longer branch-and-bound procedure is needed to reach optimality. In many cases, CPLEX finds a feasible solution quite close to a lower bound, but closing the gap requires hours of branching. There are a few test instances where the heuristic approach provides better solutions than the best solution found by CPLEX within 3 hours. Finally we mention that CPLEX is not able to find any feasible solution for a couple of particularly hard instances.

For comparing the quality of the heuristic and the optimal solutions, we use the absolute gap $\max\{\text{HEUR} - \text{IP}, 0\}$ and the relative gap $\frac{\max\{\text{HEUR} - \text{IP}, 0\}}{\text{IP}}$. Table 4.4

Test set	h	# inst.	Relative gap		Absolute gap	
			average	maximum	average	maximum
RAND	2	1,000	0.8%	30%	0.6	39
RAND	3	1,000	2.4%	40%	2.6	39
RAND	4	1,000	3.5%	36%	4.5	44
RAND	5	1,000	6.6%	36%	10.4	55
DIFF	2	500	1.8%	40%	7.5	43
DIFF	3	500	4.2%	54%	19.6	80
DIFF	4	500	4.0%	49%	22.3	80
DIFF	5	500	5.2%	39%	29.2	83

Table 4.5: Comparing FlowBound to IP.

Test set and algorithm		$h = 2$	$h = 3$	$h = 4$	$h = 5$
RAND, DIFF	FlowBound	1–4	1–5	1–5	2–6
RAND, DIFF	HEUR	2–4	5–6	7–10	15–18
RAND	IP	16–107	58–1,969	222–1,389	734–5006
RAND	IP (average)	20.4	87.4	297.0	1,054.6
DIFF	IP	22–616	83–10,800	278–10,800	845–10,800
DIFF	IP (average)	42.9	352.1	1,170.4	2,507.4

Table 4.6: Running times (in seconds).

shows the average and maximum values of these performance indicators. The quality of the flow bounds is examined in Table 4.5, the measures are the absolute gap $IP - \text{FlowBound}$ and the relative gap $\frac{IP - \text{FlowBound}}{IP}$. It turns out that the average and maximum gaps between IP and FlowBound are larger than between HEUR and IP. This shows that the quality of the heuristic solutions is significantly better than what the easy-to-compute FlowBound indicates.

Table 4.6 shows the running times for computing the flow bounds, for the heuristic algorithm and for solving the binary linear programs by CPLEX. We also give the average running times for the CPLEX computations. We can see that instances in DIFF require 2 to 4 times longer computation times than instances in RAND.

4.6.4 Conclusions for the Interchange Model

The heuristic solution method solves all test instances, the objective difference between the heuristic solution and the optimal integral solution is not large in the weight structure of the instances. This can be explained by the structure of the problem. The number of urgent units is small compared to the total number of units

(especially in the case of a shorter planning horizon, such as 2 or 3 days), so the urgent units do not have too much interaction with each other. Also, the timetable allows enough interchanging possibilities to provide several almost optimal paths for the urgent units.

In the solutions of the Interchange Model, the only highly penalised interchanges are empty trains on the first day of the planning horizon for units with a deadline of 24 hours. This indicates that, when solving a sequence of problems every day and using each day's output as input for the next day, the system would very rarely use expensive arcs. However, disturbances and delays may prevent the local shunting crew to carry out the desired interchanges. This may lead in practise to using more expensive interchanges, including empty trains.

Despite the large size of the binary integer program, the average solution times of CPLEX are acceptable, especially for the practically important planning horizons of 2 and 3 days. For longer periods (but also for some 2 and 3 days instances) the running times are still a bit longer than convenient in an interactive decision support application.

The heuristic algorithm we propose is able to solve all test instances. However, it might not find a feasible solution with other data. In such a case, we may give up the heuristic approach and just run CPLEX. As an alternative, HEUR (or any other heuristic algorithm) can be used in an interactive manner. When the heuristic displays a solution for a subset of urgent units, the maintenance routing planner can see the conflicts between this partial solution and some candidate paths for urgent units that have not been routed yet. Then the planner can forbid or enforce some interchanges and run the heuristic algorithm again. Since the heuristic is fast, the planner can examine several alternatives within minutes. By storing the planner's preferences and using these in later runs, the model may 'learn' from the planners.

4.7 The Transition Model

In the previous section we described the Interchange Model for maintenance routing. Despite promising computational results, the model also has a drawback: It requires too many details about the shunting possibilities; more than is likely to be available in practise. Therefore, we present in this section the alternative 'Transition Model' which needs less input data. While the Interchange Model tries to reflect the shunting process as accurately as possible, the Transition Model takes very few details explicitly into account.

The Transition Model incorporates no other information on shunting possibilities than a description of which new transitions are allowed. The sole criterion for that is whether there is enough time to carry out the shunting operations involved with the particular transition without causing delays. We assign a weight to every allowed new transition and define the weight of a solution as a simple function of the weights of the new transitions in the solution. Since the Transition Model is less involved than the Interchange Model, it may produce less acceptable solutions. On the other hand, its input data are a lot easier to obtain and in the setting of an interactive decision support system the weight function can be tuned based on rejections or approvals of local shunting crew. In any case, the real strength or weakness of the model can only be evaluated by implementing it for real-life instances.

The Transition Model itself is a variant of the multi-depot vehicle routing problem, studied in detail by Desrosiers et al. (1995) and Toth and Vigo (2002). Our results and methods are related to those by Carraraesi and Gallo (1984) and by Bertossi et al. (1987).

4.7.1 Maintenance Routing Graphs

A regular transition is a pair of tasks that are planned to be carried out by a unit consecutively in the regular plan. Regular transitions $(t, \sigma(t))$ satisfy

$$\begin{aligned} s_a(t) &= s_d(\sigma(t)), \\ \tau_a(t) &< \tau_d(\sigma(t)). \end{aligned}$$

A *candidate transition* is a pair of tasks (t, t') such that it is not a regular transition and such that

$$\begin{aligned} s_a(t) &= s_d(t'), \\ \tau_a(t) &\leq \tau_d(t') - \varrho. \end{aligned}$$

A candidate transition can be interpreted as a pair of an arriving and a departing event at the same station with a time difference of at least ϱ minutes. It represents the possibility to join the tasks to duties in another way than the regular plan. The buffer time ϱ reserves time for possibly needed extra shunting operations, moreover, it protects against spreading of delays. We assume that the first tasks in the regular duties have no in-coming candidate transition and the last tasks of the regular duties have no out-going candidate transition. Regular and candidate transitions together are called *transitions*.

The set of tasks together with the set of transitions forms an acyclic graph. Any directed graph that can be obtained in this way is called a *maintenance routing graph* (or *MR-graph*). Note that MR-graphs are always acyclic.

In a directed graph, a *sink* is a node without in-coming arcs and a *source* is a node without out-going arcs. The set of sources is denoted by V_0 and the set of sinks by V_∞ . In an MR-graph, the first tasks in the regular duties are the sources, while the last tasks in the regular duties are the sinks.

Let M be the set of *maintenance nodes*. The *urgencies* in an MR-graph $G = (V, A)$ are given by a set U of sources in G and by sets $M_u \subseteq M$ for each $u \in U$. Members of U are *urgent nodes*. In a maintenance routing application, maintenance nodes are the maintenance tasks. Furthermore, a node is urgent if it is the earliest task of one of the units that must reach a maintenance facility in the planning period. For $u \in U$, M_u is the set of maintenance tasks that can be assigned to u in order to satisfy the maintenance requirement of the corresponding unit.

In real-world maintenance routing, we may allow the buffer time to vary from arc to arc, but here, an MR-graph always has a uniform buffer time. It also happens in real world maintenance routing that some practical instances can only be solved when allowing empty trains. If desired, we can model such movements by transitions between tasks that arrive and depart at different stations. However, such more general transitions are not part of the definition of MR-graphs.

4.7.2 Model Formulation

From the timetable and the buffer time ϱ , we construct the corresponding MR-graph. The sequences of tasks that can be assigned to a single unit over a whole planning period are precisely the directed paths in the corresponding MR-graph that connect a source to a sink. A *path cover* in an acyclic directed graph $G = (V, A)$ is a set of arcs that forms a collection of node disjoint paths such that the paths cover V and that each path connects a source of G with a sink of G . The regular transitions in a maintenance routing graph form a path cover. So, the maintenance routing problem amounts to finding a path cover such that the path in the path cover that starts at urgent node u contains a node in M_u .

In order to evaluate shunting efforts, each transition a has a weight, denoted by $c(a)$; this weight estimates how difficult it is to use for a unit. In Section 4.7.7, we discuss briefly how we obtain these weights. We measure the overall shunting difficulty of a solution by the sum of the individual transition weights.

With these data, the maintenance routing problem becomes the following integer multi-commodity flow model; we define it for general acyclic graphs for later reference.

Transition Model: *Given an acyclic graph $G = (V, A)$, a set U of sources and the urgencies $(M_u : u \in U)$, as well as a weight function $c : A \rightarrow \mathbb{R}_+$, find a path cover Z minimising*

$$\sum_{a \in Z} c(a)$$

such that for each $u \in U$, the path starting at u contains a member of M_u .

We formulate this as a binary linear program. The variables are:

$$\begin{aligned} z : A &\rightarrow \{0, 1\}, \\ x_u : A &\rightarrow \{0, 1\} \quad u \in U. \end{aligned}$$

The function z is a network flow where the arcs with $z(a) = 1$ form a path cover, that is, the paths of all units during the entire planning period. The functions x_u are also network flows where the arcs with $x_u(a) = 1$ form the path of urgent node u to a maintenance node. An urgent node $u \in U$ is a source while members of M_u are sinks with respect to the network flow x_u .

The integer program reads as follows.

$$\text{Minimise } \sum_{a \in A} c(a) \cdot z(a) \tag{4.29}$$

subject to

$$\sum_{a \in \delta^{\text{in}}(v)} z(a) = 1 \quad \forall v \in V \setminus V_0 \tag{4.30}$$

$$\sum_{a \in \delta^{\text{out}}(v)} z(a) = 1 \quad \forall v \in V \setminus V_\infty \tag{4.31}$$

$$\sum_{a \in \delta^{\text{in}}(v)} x_u(a) = \sum_{a \in \delta^{\text{out}}(v)} x_u(a) \quad \forall u \in U, v \in V \setminus (\{u\} \cup M_u) \tag{4.32}$$

$$\sum_{a \in \delta^{\text{out}}(u)} x_u(a) = 1 \quad \forall u \in U \tag{4.33}$$

$$\sum_{u \in U} \sum_{a \in \delta^{\text{out}}(v)} x_u(a) = 0 \quad \forall v \in M \tag{4.34}$$

$$\sum_{\substack{u \in U: \\ v \notin M(u)}} \sum_{a \in \delta^{\text{in}}(v)} x_u(a) = 0 \quad \forall v \in M \quad (4.35)$$

$$\sum_{u \in U} x_u(a) \leq z(a) \quad \forall a \in A \quad (4.36)$$

$$z \in \{0, 1\}^A \quad (4.37)$$

$$x_u \in \{0, 1\}^A \quad \forall u \in U \quad (4.38)$$

Constraints (4.30) – (4.31) state that z is a network flow, having a unit throughput through each node. Constraints (4.32) describe flow conservation for each flow x_u . Constraints (4.33) set the sources of the flows x_u . Constraints (4.34) – (4.35) express that a maintenance node has no out-going flow in any of the flows x_u and that it only accepts in-going flow from an urgent node u if it can be assigned to u . Constraints (4.36) make sure that the paths indicated by x_u are parts of the path cover z . Finally, constraints (4.37) – (4.38) specify the domain of the variables.

When comparing the Transition Model to the Interchange Model, the most important difference between both approaches is the collection of constraints in (4.12): they describe the interaction between units. In reality (and also in the Interchange Model), a candidate transition may only be cheap because some other candidate transitions are carried out at the same time. The Transition Model cannot see any connection between the transitions, therefore it may underestimate the real shunting difficulty. It will be interesting to see whether or not the Transition Model provides unrealistic solutions.

4.7.3 Objective Function

As mentioned in Section 4.3, the objective criteria in the maintenance routing problem are quite vaguely formulated. Therefore, several different objective functions could be used. We choose (4.29) because it is probably the most simple and intuitive objective function. However, one can also use

$$\text{Minimise } \sum_{a \in A} c(a) \cdot \left(\sum_{u \in U} x_u(a) \right). \quad (4.39)$$

In that case, only the urgent units contribute to the total weight of the solution. In order to make the objective function (4.39) realistic, the weight of a candidate transition a should reflect the shunting difficulty of those candidate transitions which arise in the path cover as consequences of a . It is, however, hopeless to incorporate the

consequences of *several* candidate transitions selected for urgent units. Therefore, the alternative objective (4.39) may seriously underestimate the shunting difficulty of the overall solution. The only clear advantage of (4.39) is that it allows direct comparison between the results of the Interchange Model and of the Transition Model. We discuss such a comparison in Section 4.9.

4.7.4 Reducing the Problem Size

An arc in an acyclic graph G is said to be *covered* if it is in at least one path cover of G . Many transitions in a maintenance routing problem have a very long time span. Such transitions will typically not be covered in the corresponding MR-graph, so they make the model larger than necessary. We call an acyclic graph *reduced* if each arc is covered. Finding a path cover and determining which arcs are covered are network flow problems, so we can reduce a graph in polynomial time. Actually when a path cover is given, like in the case of MR-graphs, one can determine the covered arcs in linear time. This is done by standard network flow techniques. For details of network flow theory we refer to Schrijver (2003).

Given an acyclic graph $G = (V, A)$ with a path cover R , construct an auxiliary graph $\hat{G} = (\hat{V}, \hat{A})$ as follows. For each $v \in V$ there are two distinct nodes v_{out} and v_{in} in $\hat{V}$. For each candidate transition $vw \in A$, there is an arc $v_{\text{out}}w_{\text{in}}$ in $\hat{A}$ and for each regular transition $vw \in A$, there is an arc $w_{\text{in}}v_{\text{out}}$ in $\hat{A}$. A small example is given in Figure 4.13. (See also Appendix B.)

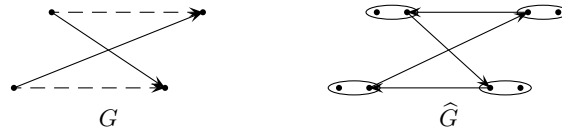


Figure 4.13: An acyclic graph with a path cover (dashed arcs) and the auxiliary graph.

Now an arc in G is covered if and only if it is either in R or the corresponding arc in $\hat{G}$ is in a directed circuit. Therefore, identifying the covered arcs amounts to finding the strongly connected components of $\hat{G}$ which can be done in linear time, indeed (Karzanov (1970)).

4.7.5 Complexity Results

In this section we derive complexity results for the Transition Model. We prove that the feasibility problem in an arbitrary acyclic graph is NP-complete even when

there is only a single urgent node. When the number of sources is fixed, however, an optimal solution of the Transition Model can be found in polynomial time with dynamic programming irrespective the number of urgent nodes. For MR-graphs with one urgent node, the feasibility version is polynomially solvable, but the optimisation problem remains NP-hard. The following table summarises these results of this section.

Complexity of the Transition Model	Feasibility	Optimising
Arbitrary acyclic graphs, $ U = 1$	NP-hard	NP-hard
Arbitrary acyclic graphs, fixed number of sources	P	P
MR-graphs, $ U = 1$	P	NP-hard

NP-Hardness in Arbitrary Acyclic Graphs

Theorem 4.9. *It is NP-complete to decide if an acyclic graph has a path cover such that one of its paths connects a given source with a given sink.*

Proof. We reduce the satisfiability problem to the feasibility of the Transition Model. Let $x_1, \dots, x_n$ be Boolean variables, let $\varepsilon \in \{+1, -1\}$ and denote

$$x_j^\varepsilon = \begin{cases} x_j & \text{if } \varepsilon = +1, \\ \neg x_j & \text{if } \varepsilon = -1. \end{cases}$$

Consider the conjunctive normal form

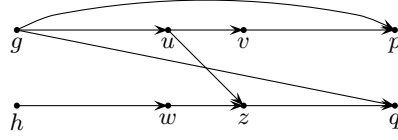
$$\varphi = \bigwedge_{i=1}^k \left(x_{i_1}^{\varepsilon_{i_1}} \vee x_{i_2}^{\varepsilon_{i_2}} \vee \dots \vee x_{i_{m(i)}}^{\varepsilon_{i_{m(i)}}} \right).$$

It is well-known that it is NP-complete to decide whether or not we can assign values to the variables that make φ true (Cook, 1971).

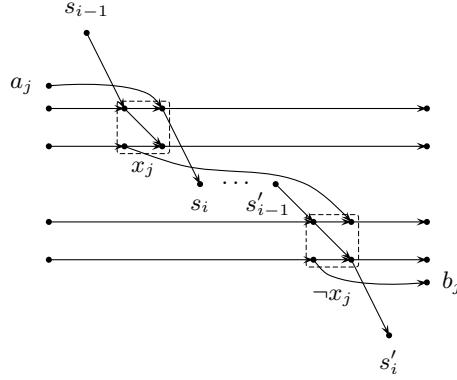
We start with an empty graph. Insert two nodes a_j, b_j for each variable x_j and draw an arc $a_j b_j$. For each $i = 0, \dots, k$, create three nodes s_i, c_i, d_i and insert the arcs $c_i s_i, s_i d_i$ and $c_i d_i$. Moreover, insert the arc $c_k d_0$. Let $s = c_0$ and $t = d_k$.

For the occurrence of a variable x_j (either as x_j or as $\neg x_j$) in clause i , create a ‘box’ as shown in Figure 4.14. Create 8 new nodes u, v, w, z, g, h, p, q and draw the arcs $uv, uz, wz, gu, hw, vp, zq, gp$ and gq . Insert the arc $s_{i-1}u$. Moreover, draw the arc vs_i if the non-negated x_j occurs in clause i and draw the arc zs_i if $\neg x_j$ occurs in clause i .

Order the occurrences of x_j by increasing clause index. Draw an arc from a_j to the v -node of the first x_j -box. Insert an arc from the w -node of any x_j -box to the

Figure 4.14: An x_j -box.

v -node of the next x_j -box. Finally, join the w -node of the last x_j -box to b_j . Then we obtain a picture as in Figure 4.15. Note that the arcs $a_j b_j$, $c_i s_i$, $s_i d_i$, gu , uv , vp , hw , wz , zq form a path cover.

Figure 4.15: Joining the x_j -boxes.

We claim that a path cover connecting s with t exists in this graph if and only if φ is satisfiable.

Suppose there exists such a path cover. Observe that for any pair of occurrences of x_j , the paths use in both boxes either the uv -arc or the uz -arc. This follows easily for two consecutive occurrences (i.e. for those that are joint by a wv -arc) using the fact that the v -nodes and the z -nodes have exactly two entering arcs while the u -nodes and the w -nodes have exactly two leaving arcs. Set x_j true if the uv -arcs are used in the x_j -boxes, otherwise set it false. Observe that every $s - t$ path in the graph contains all the nodes s_i . Then the existence of the $s - t$ -path proves that every clause is satisfied.

Conversely, suppose that the variables have an assignment satisfying φ . For any clause i , label any of the literals that makes it true.

If x_j has value true, select the arc $a_j b_j$, as well as the uv , wz , hw and zq arcs of all the x_j -boxes. If an occurrence of x_j is labelled for clause i , select the $s_{i-1}u$, vs_i and gp -arcs of this box, otherwise select the gu and vp -arcs.

If x_j has value false, select the uz , hw and vp -arcs of the x_j -boxes, the wv -arcs between the x_j -boxes as well as the arcs connecting the nodes a_j and b_j to the x_j -boxes. If the occurrence is labelled for clause i , select the arcs $s_{i-1}u$, zs_i and gq in this x_j -box, otherwise select the arcs gu and zq .

Finally, select the arcs $c_i d_i$ for $i = 1, \dots, k-1$ and the arc $c_k d_0$. Then the selected arcs form a path cover connecting s with t . ■

Bounded Number of Units

In railway applications, the number of sources equals the number of available rolling stock units. Maintenance routing instances often share this number, the length of the planning horizon (and thereby the number of tasks) is the varying parameter. Then the question arises what can be said about the complexity of the Transition Model if the number of source nodes in the acyclic graph is constant. The following theorem implies that the Transition Model with a constant number of source nodes is solvable in polynomial time.

Theorem 4.10. *For any acyclic graph $G = (V, A)$, the optimisation problem (4.29) – (4.38) can be solved in $O((|V||A|)^{O(d)})$ time where d is the number of sources.*

Proof. Let $n = |V|$, $m = |A|$ and let $k = |U|$. We assume that the sources are $S = \{1, \dots, d\}$ and that $U = \{1, \dots, k\}$. The optimisation problem can be reduced to a shortest path problem in an acyclic graph $G' = (V', A')$ with at most $(2mn)^d$ nodes and at most $(2mn)^{2d}$ arcs. The sketch of the construction is as follows.

We colour the nodes of G red. By subdividing arcs, we can transform G into a graph where the node set is partitioned into disjoint subsets $W_0, \dots, W_\ell$ such that each arc connects W_i to W_{i+1} for some index i . We may also assume that the arcs between W_0 and W_1 form a matching of size d . The enlarged graph has at most mn nodes and at most mn arcs. We colour the newly created nodes black. Then a path cover in the original graph corresponds to a system of paths in the enlarged graph covering all the red nodes.

Now we construct $G' = (V', A')$. Create a member of V' for every $(d+k)$ -tuple $v' = (a_1, \dots, a_d, \varepsilon_1, \dots, \varepsilon_k)$ where $a_1, \dots, a_d$ are arcs connecting subsets W_{t-1} and W_t for some index t such that these arcs cover all the red nodes in $W_{t-1} \cup W_t$, and where $\varepsilon_j \in \{0, 1\}$ for each $j = 1, \dots, k$. The arcs a_i indicate the routes of the units through

the network, while ε_j encodes whether urgent unit j has undergone maintenance so far.

Let s' be the node in V' with arcs a_i between W_0 and W_1 and with $\varepsilon_1 = \dots = \varepsilon_k = 0$. Let T' contain all nodes $v' \in V'$ such that the arcs a_i in v' lie between the subsets $W_{\ell-1}$ and W_ℓ and such that $\varepsilon_j = 1$ for each j .

Let $v' = (a_1, \dots, a_d, \varepsilon_1, \dots, \varepsilon_k)$ and $w' = (b_1, \dots, b_d, \zeta_1, \dots, \zeta_k)$ nodes of G' . Then (v', w') is an arc in G' if

- for every i , the head of arc a_i is the tail of arc b_i ;
- the paths (a_i, b_i) for $i = 1, \dots, d$ are node disjoint;
- for every $j = 1, \dots, k$, $\zeta_j = 1$ if and only if $\varepsilon_j = 1$ or if the head of arc b_j is an allowed maintenance node for urgent unit j .

We define the weight of the arc (v', w') as $\sum_{i=1}^d c(b_i)$.

Then a system of d disjoint paths in G covering all the red nodes and satisfying all the maintenance requirements corresponds to an $s' - T'$ -path in G' with the same weight and vice versa. ■

4.7.6 MR-Graphs with One Urgent Unit

We have seen in the previous section that the feasibility problem with a single urgent node is NP-complete in general acyclic graphs. It is, however, polynomially solvable in MR-graphs.

Theorem 4.11. *Let $G = (V, A)$ be an MR-graph and let A_0 be the set of covered arcs. Moreover, let $s, t \in V$ and suppose that P is an $s - t$ -path in $G_0 = (V, A_0)$ with inclusion-wise minimal node set. Then there exists a path cover containing P .*

Proof. Let R denote the set of regular transitions. Consider the auxiliary graph $\widehat{G}$ of G as defined in Section 4.7.2. Let W be the set of nodes v_{out} with v in P and $v \neq t$. Let W_N be the set of nodes $v_{\text{out}} \in W$ with $vw \in P \setminus R$.

We show that no pair of different nodes $v_{\text{out}} \in W_N$ and $v'_{\text{out}} \in W$ lie in the same strong component of $\widehat{G}$.

Suppose the contrary. Let vw and $v'w'$ be arcs in the path P . We may assume that v' lies on P between v and t , the other case being analogous. Let K be the strong component of $\widehat{G}$ that contains v_{out} and v'_{out} . Then $w_{\text{in}} \in K$ since $v_{\text{out}}w_{\text{in}}$ belongs to a directed circuit in $\widehat{G}$. Moreover, if $v'w'$ is a regular transition, then every $v_{\text{out}} - v'_{\text{out}}$ -path contains the node w'_{in} since $w'_{\text{in}}v'_{\text{out}}$ is the only arc that enters

v'_{out} . If, however, $v'w'$ is a candidate transition, then $v'_{\text{out}}w'_{\text{in}}$ is contained in a circuit of $\widehat{G}$. In both cases we have $w'_{\text{in}} \in K$. In particular, $\widehat{G}$ contains a $w'_{\text{in}} - v_{\text{out}}$ -path $\widehat{P}$.

For every arc $x_{\text{out}}y_{\text{in}} \in \widehat{A}$, we have $s_a(x) = s_d(y)$ and for every arc $x_{\text{in}}y_{\text{out}} \in \widehat{A}$, we have $s_d(x) = s_a(y)$. Therefore $s_a(v) = s_d(w')$. Moreover,

$$\tau_a(v) \leq \tau_d(w) - \varrho \leq \tau_d(w') - \varrho.$$

Thus vw' is either a regular transition (and thus it is covered) or a candidate transition. In the latter case, $\widehat{P} \cup \{v_{\text{out}}w'_{\text{in}}\}$ is a circuit in $\widehat{G}$. Therefore vw' is covered. This contradicts the minimality of the path P . Therefore v_{out} and v'_{out} cannot lie in the same strong component of $\widehat{G}$.

For every arc $vw \in P \setminus R$, fix a directed circuit $C_{v,w}$ in $\widehat{G}$ containing the arc $v_{\text{out}}w_{\text{in}}$. Let C be the set of those edges in G that correspond in $\widehat{G}$ to an edge in $\bigcup \{C_{v,w} : vw \in P \setminus R\}$. Then the circuits $C_{v,w}$ are node disjoint, therefore $(R \setminus C) \cup (C \setminus R)$ is a path cover of G . Moreover, no arc in $P \cap R$ belongs to C , therefore $(R \setminus C) \cup (C \setminus R)$ contains P . ■

This implies

Corollary 4.12. *The feasibility problem of the Transition Model in an MR-graph with one urgent node can be solved in polynomial time.*

The NP-hardness of the feasibility problem in arbitrary acyclic graphs is partly caused by sparsity of some acyclic graphs that are not MR-graphs. The proof above is based on the property of MR-graphs that the existence of a certain pair of arcs vw and $v'w'$ implies the existence of the arc vw' .

Theorem 4.11 requires that the node set of path P is inclusion-wise minimal. One may ask whether each directed $s - t$ -path in an MR-graph extends to a path cover. Figure 4.16 shows an example that this is not so: the path formed by the bold arcs cannot be extended to a path cover.

The algorithm for the feasibility problem does not extend to the optimisation version of the single urgent unit case. Reason is that the weight of a path serving the urgent unit does not reflect the weight of a corresponding solution, as also non-urgent units will use new transitions and these do contribute to the weight of the solution as well. Actually, finding an optimal solution of the Transition Model with one urgent node is also NP-hard in MR-graphs.

Theorem 4.13. *It is NP-hard to find an optimal solution of the transition model on MR-graphs with one urgent unit, one maintenance task and $\{0, 1\}$ -valued weight function.*

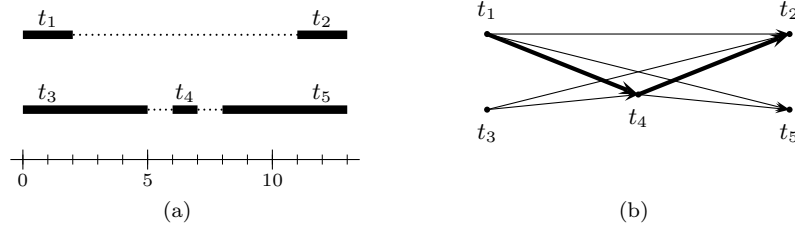


Figure 4.16: (a) A maintenance routing instance with $\rho = 4$. (b) The corresponding MR-graph. The bold arcs form a path that cannot be extended to a path cover.

Proof. First we show that each acyclic directed graph G with a path cover R can be extended to an MR-graph G' . Indeed, there is only one station and the arcs in R are the regular transitions. We define departure and arrival times as follows. Consider a topological order of G , that is, an order on $V = \{v_1, \dots, v_n\}$ such that $v_i v_j \in A$ implies $i < j$. Let $\tau_d(v_i) = 2i - 1$, $\tau_a(v_i) = 2i$ and let $\rho = 0$. Then every arc $vw \in A$ is a regular or candidate transition. We insert an arc for all the other pairs (v, w) that form a candidate transition according to the definition in Section 4.7.1. The obtained graph G' is an MR-graph.

We apply this to the graph G constructed in the proof of Theorem 4.9. We set the weight 0 for the original arcs and we set the weight 1 for the newly inserted arcs. Then deciding whether or not the optimisation problem in the extended graph G' has optimum value zero amounts to solving the feasibility problem in G . ■

4.7.7 Transition Weights

The success of the model in practical applications highly depends on the choice of the weight function. We assume that the regular plan has a corresponding shunting plan that can be carried out in practise. The weight measure how much the regular plan has to be modified in order to carry out the given transitions.

The weight of a transition can be derived directly from the timetable data, by taking the following factors into account:

- the positions of the units in the compositions;
- the time difference between the tasks;
- information about the infrastructure like the physical distance between the arrival and departure tracks;

Planning period	2 days	3 days	4 days	5 days
# of tasks	1,196	1,762	2,326	2,709
# of transitions	98,035	198,614	332,031	508,853
# of covered transitions	11,394	17,067	22,830	28,570

Table 4.7: Dimensions of the graph model.

- the station where the transition takes place; stations with heavy traffic provide less changing possibilities;
- the time that the transition is carried out; changes during rush hours are hard, while changes at night are typically easy.

In our implementations, we followed another method: We used the Interchange Model as a detailed model of the shunting possibilities to compute the weights. We did so in order to be able to compare the output of both models on the same input. We shall explain more details about this in Section 4.9. Note again that whichever method is used to determine the transition weights, the values can be updated when the model is applied for solving real-life problems.

4.8 Computations for the Transition Model

As for the Interchange Model, we use the units of the type “Sprinter” as test example: we consider the same set of tasks. Recall that there are two maintenance tasks on every day. For simplicity, we set the buffer time ϱ uniformly to 10 minutes based on discussions with the planners.

For the computations we used the modelling software ILOG OPL Studio 3.7 and the integer program solver ILOG CPLEX 9.0 on a PC with an Intel P4 3.0 GHz processor and 512 Mb internal memory.

When generating the candidate transitions, the graph turns out to be quite large (see Table 4.7). However, the reduction step described in Section 4.7.4 results in decreasing the number of arcs dramatically. Without this reduction, the binary linear programs would become far too large for acceptable computation times.

We consider planning horizons of 2 up to 5 days. Since the maintenance tasks start at the same time at the same location on every day, the integer program requires only one flow x_u for every possible deadline. The dimensions of the integer programs are given in Table 4.8. We solve the sets RAND and DIFF of instances that we described in Section 4.6.2.

Planning period	2 days	3 days	4 days	5 days
# urgent units	4	6	8	10
# variables in IP	28,453	51,193	79,786	114,096
# constraints in IP	15,337	23,816	32,947	42,672

Table 4.8: Dimensions of the MIP's.

Test set	# of inst.	Horizon	Avg. time	Max. time
RAND	1,000	2 days	4.31	6
RAND	1,000	3 days	8.92	12
RAND	1,000	4 days	19.69	138
RAND	1,000	5 days	43.20	486
DIFF	500	2 days	4.30	11
DIFF	500	3 days	8.39	12
DIFF	500	4 days	20.13	138
DIFF	500	5 days	49.03	1,681

Table 4.9: Average and maximum solution times of the MIP's.

Despite the large sizes of the integer linear programs, optimal solutions could be found in reasonably short time as shown in Table 4.9. It indicates that, although the optimisation problem may be difficult for arbitrary weight structures, real-life instances are much easier to deal with. For most instances, the largest part of the computation is spent on solving the LP relaxation of the model. Then, after a small number of branchings, CPLEX is able to construct an optimal integer solution.

When comparing different CPLEX settings, the barrier method turns out to be the fastest LP solver for the Transition Model. The other LP algorithms of CPLEX require more time for each instance. For example, depending on the length of the planning period, the network simplex method leads to 3–10 times higher average computation times. To obtain the best solution times, we also use the standard technique of perturbing the objective function.

We have seen in Section 4.6 that solving the Interchange Model by CPLEX, instances in the set DIFF are significantly more difficult than those in RAND. Due to its much simpler structure, the Transition Model does not show sharp differences in the solving times of both sets of test instances. The average solving times of both test sets hardly differ, but the maximum solving time for DIFF is much higher on some peculiar instances. Nevertheless, even those maximum solving times could be acceptable for practical applications.

Test set	# of inst.	Horizon	Avg. time	Max. time
RAND	1,000	2 days	3.30	5
RAND	1,000	3 days	6.46	12
RAND	1,000	4 days	16.89	39
RAND	1,000	5 days	39.09	94
DIFF	500	2 days	4.84	8
DIFF	500	3 days	9.46	21
DIFF	500	4 days	20.83	62
DIFF	500	5 days	39.64	171

Table 4.10: Average and maximum solution times of the MIP's when using the alternative objective function (4.39).

All results above concern the Transition Model as described in Section 4.7.2, in particular, using the objective function (4.29). We also solved all test instances with the alternative objective function (4.39). The main goal with these latter computations is to compare the results directly to the results of the Interchange Model. Using the same CPLEX settings as before, the model can be solved to optimality for each test instance. The solutions times, given in Table 4.10, turn out to be slightly less than with the original objective function.

4.9 Comparing the Two Models

In the previous sections, we presented two different models for the maintenance routing problem. In what follows, we show that with appropriate objective functions, the Transition Model becomes a relaxation of the Interchange Model. Moreover, we compare the output of the Interchange Model and the Transition Model.

4.9.1 Deriving the Transition Weights

In this comparison, we take in the Interchange Model the objective function (4.2) and in the Transition Model the objective function (4.39). In both models, we consider the paths that connect the starting nodes of the urgent units to the maintenance nodes and sum up the arc weights in these paths.

Each feasible solution of the Interchange Model naturally corresponds to a feasible solution of the Transition Model as follows. The Interchange Model specifies paths for the units, also implying new duties for the units: sequences of tasks to be carried out by a single unit. Two consecutive tasks in a new duty form a regular or candidate

transition in the Transition Model; these transitions together give a feasible solution of the Transition Model.

The only difficulty is to relate the objective functions to each other. The Interchange Model works with fine details of the shunting possibilities, therefore it is a natural idea to derive the transition weights from the Interchange Model. Consider a transition (t, t') . Then tasks t and t' are represented in the Interchange Model as task arcs (v, w) and (v', w') . Recall that the arc weights in the graph representation of the Interchange Model are denoted by $\text{weight}(a)$. Now we define the weight of transition (t, t') , denoted by $\tilde{c}(t, t')$, as the length of the shortest path from w to v' with respect to the arc lengths $\text{weight}(a)$. With these transition weights, each feasible solution of the Interchange Model with total weight q corresponds to a feasible solution of the Transition Model with total weight of at most q . Therefore we have

Theorem 4.14. *The Transition Model with the objective function (4.39) and transition weights $\tilde{c}$ is a relaxation of the Interchange Model.*

We point out that deriving the transition weights from the Interchange Model may not be possible in a real-life application. The main motivation for the Transition Model is that the vast amount of input data needed for the Interchange Model may likely be inaccessible.

4.9.2 Numerical Results

Let IM denote the optimal objective value of the Interchange Model and TM the optimal objective value of the Transition Model. Then the absolute gap is $\text{IM} - \text{TM}$ and the relative gap is $\frac{\text{IM} - \text{TM}}{\text{TM}}$. The average and maximum values of these gaps, separately computed for the test sets RAND and DIFF, are given in Table 4.11.

It turns out that the objective values of the Transition Model are surprisingly close to the optimum values of the Interchange Model. In fact, the majority of the instances in RAND had the same objective value for both models. Instances in the set DIFF have, as expected, larger differences between the models. However, the largest absolute difference is again smaller than the weight of a single moderately expensive arc. The small relative gaps for instances in DIFF are caused by the fact that the objective values in DIFF are much higher on average than in RAND.

4.10 Conclusions

In this chapter we described the maintenance routing problem of NSR. We presented two mathematical models for this problem: the Interchange Model and Transition

Test set	h	# inst.	Relative gap		Absolute gap	
			average	maximum	average	maximum
RAND	2 days	1,000	0.14%	43%	0.09	16
RAND	3 days	1,000	0.64%	32%	0.55	20
RAND	4 days	1,000	0.92%	26%	0.96	20
RAND	5 days	1,000	1.51%	21%	1.96	26
DIFF	2 days	500	0.68%	7%	7.43	47
DIFF	3 days	500	0.91%	8%	12.26	69
DIFF	4 days	500	1.08%	6%	13.96	70
DIFF	5 days	500	1.42%	5%	17.01	80

Table 4.11: Comparing IM to TM.

Model. We analysed computational complexity issues of both models, proposed solution methods for them and carried out computational tests on real-life instances of NSR.

When solving the problems by commercial mixed integer programming software, the Transition Model turned out to be easy to handle. However, the Interchange Model resulted in a much larger integer program, requiring inconveniently long solving times on some instances with a planning horizon of 4 or 5 days. We also developed a simple but yet powerful heuristic approach for solving the Interchange Model. This algorithm was able to solve all test instances with an acceptable (absolute) optimality gap.

Although the Interchange Model takes much more details about the shunting possibilities into account than the Transition Model, their numerical optimal solution values did not differ much. To some extent, this also holds for the optimal solutions themselves.

The real practical value of the solutions we found is not easy to evaluate. We discussed a number of instances with maintenance routing planners: they found the solutions satisfactory. However, this does not mean that the local planners would accept all of them. In this respect, the solutions we obtained are comparable to the solutions generated by the maintenance routing planners themselves.

Based on our computational results and on the discussions with the maintenance routing planners, we can conclude that both the Interchange Model and the Transition Model are good candidates for the core of a decision support system at NSR.

Chapter 5

Operational Rolling Stock Planning

In this chapter we consider the operational rolling stock planning problem of NSR; the term ‘operational planning’ refers to a planning horizon of 3 days to 2 months. We describe the operational rolling stock circulation problem and the two-phase method the planners at NSR use to handle it. In this thesis we only study the second phase in detail; investigation of the first phase is a subject of further research. We analyse the complexity of the second phase and propose solution methods for it. We illustrate the power and the limitations of the methods on instances of NSR.

5.1 Operational Rolling Stock Circulations

In operational planning the tactical rolling stock schedules are adjusted to the particular weeks. Operational rolling stock planning consists of two tasks: The *operational rolling stock circulations* have to be created by the central planners and the corresponding *operational shunting plans* have to be set up by the local planners. Here we only deal with the rolling stock circulations.

A general overview of operational planning is given in Section 2.2.4. We mentioned there that most effort in operational rolling stock planning is spent on large-scale adjustments of the tactical plans; this is often necessary when parts of the railway infrastructure become temporarily unavailable. We focus in this chapter on problem instances of such character.

Concerning the input, the constraints and the assumptions, the operational rolling stock circulation problem is similar to its tactical variant. The same assumptions are made on the shunting process as described in Sections 3.3 and 3.4 for tactical planning. The available rolling stock has to be assigned to the trips such that the same constraints are satisfied as in tactical planning. These constraints are listed in Section 3.5.

The operational plan is not a stand-alone rolling stock plan: It should fit to the rolling stock circulations before and after the considered planning period. In particular, the initial inventories (i.e. those at the beginning of the planning period) must be equal to the number of units that are located at the stations at the beginning of the planning period. These initial inventories are specified by the rolling stock schedules that are to be carried out before the planning period; these schedules are determined in tactical planning or in earlier stages of operational planning. Similar constraints hold for the final inventories.

It is not always possible to realise all desired inventories. Then the rolling stock balance must be maintained by sending units from one station to another as empty trains. Although this is undesirable, it often happens in practise.

Example. *Figure 5.1(a) shows the tactical rolling stock circulation for a small railway network consisting of 8 trips between stations A, B and C. The grey and black units are of different types. The desired initial and final inventories are also indicated; the tactical plan complies with these desired inventories.*

Cancelling two trips (indicated by dashed lines) enforces one to modify the tactical plans. A possible operational rolling stock circulation is shown in Figure 5.1(b). This operational rolling stock circulation realises the desired final inventories but there is an off-balance in the initial inventories: A grey unit is available at station A but the operational rolling stock circulation requires a black unit. Similarly, a black unit is available at station B but the operational rolling stock circulation requires a grey unit.

5.1.1 Differences between Tactical and Operational Planning

The major differences between tactical and operational rolling stock planning are the planning horizon and the objective criteria. In tactical planning, the rolling stock is scheduled for the forthcoming two months (at least), the emphasis is on a good balance of efficiency, service quality and robustness. It takes several weeks until the

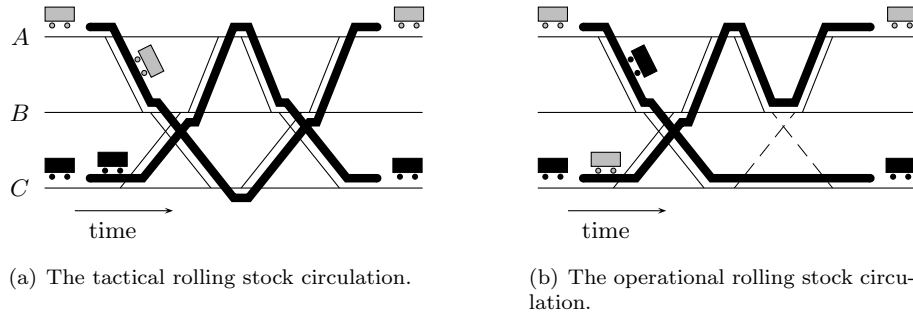


Figure 5.1: An instance of operational rolling stock planning. The desired initial and final inventories are indicated at the beginning and the end of the time-lines of the stations.

tactical rolling stock schedule has been worked out in every detail and is accepted by central planners as well as by local planners.

The operational rolling stock schedules are determined at most 2 months before they are carried out. Time is too short then to work in the same way as in tactical planning. Instead, the main goal is to create the operational rolling stock schedules – including the shunting plans – as quickly as possible, even at the cost of efficiency and service quality. Since infrastructure maintenance works usually take one or two days, potentially higher carriage-kilometres and seat-shortage kilometres on these days do not change much the annual performance of the railway operator.

Another important issue is the influence of the rolling stock schedules on the crew schedules. While a trip needs only one driver, the number of conductors depends on the length of the train. For example, a trip with 6 carriages can be served by one conductor, while 7 carriages require two conductors. If the operational plan needs more conductors for a trip than the tactical crew schedule provides, the planners have to modify the crew schedule or they have to decide that a unit is to be closed for passengers. As neither of these is appealing, the number of such trips should be kept low in the operational rolling stock circulation.

A further difference between tactical and operational rolling stock planning is the problem size. Instances of tactical rolling stock planning typically concern a small number of train lines on a single day. Instances of operational planning may be substantially larger as they contain all trips that are directly or indirectly affected by the unavailable tracks and as the planning period may take several days.

We note that in practise, rules of operational planning are somewhat less strict than rules of tactical planning. Getting closer to executing the rolling stock schedules, there is more space for ad hoc measures to keep the system run smoothly.

5.1.2 Two-phase Approach

Planners at NSR handle the operational rolling stock circulation problem by the following two-phase method. In the first phase, they try to set up a simple rolling stock circulation that satisfies as much of the specifications as possible. To that end, they disregard the constraints on initial inventories and create an *intermediate plan* that satisfies all other constraints. This intermediate plan is intended to have the smallest possible number of composition changes between the trips and their successors. In practise, the planners often start with specifying the compositions of the trips at the end of the planning period. They do it in such a way that the numbers of units per type that end up at the stations are equal to the desired final inventories. Then they proceed by assigning to any trip the same composition as to its successor. Roughly speaking, the trains go up and down without any composition change. In the intermediate plan seat shortages are not minimised explicitly. Nevertheless, the planners make sure that each trip receives a certain minimal seat capacity which is often given in the specifications. This may force the planners to allow a small number of composition changes in the intermediate plan.

The intermediate plan may have a ‘surplus’ or a ‘deficit’ of units at certain stations at the beginning or the end of the planning period. A station has a *deficit* in the initial inventory if the desired initial inventory is smaller than the number of units that start at that station according to the intermediate plan. A station has a *deficit* in the final inventory if the number of units that end up at that station according to the intermediate plan is smaller than the desired final inventory. Surpluses are defined analogously: A station has a *surplus* in the initial inventory if the desired initial inventory is larger than the number of units that start at that station according to the intermediate plan. A station has a *surplus* in the final inventory if the number of units that end up at that station according to the intermediate plan is larger than the desired final inventory. Deficits and surpluses are also called *off-balances*.

Example. Consider again the example in Figure 5.1. The operational rolling stock circulation has no surplus or deficit in the final inventories. Station A has a surplus of one grey unit and a deficit of one black unit, while station C has a surplus of one black unit and a deficit of one grey unit.

In the second phase, the off-balances are resolved, we call this phase *rebalancing*. It amounts to finding a rolling stock circulation where the sum of the surpluses is as small as possible (zero ideally). Compared to the intermediate plan, the output rolling stock circulation may not have new surpluses and deficits at any location and at any time.

Rebalancing is done in practise by simple modifications of the intermediate plan: (i) adjust the train lengths by coupling or uncoupling units; (ii) assign units of another type to some trips (and all their successors and predecessors); (iii) insert extra empty trains. The total number of shunting movements is kept low by applying only a small number of such modifications. This increases the chance that the shunting planners can create the shunting plans quickly.

Planners prefer operational rolling stock circulations that differ little from the intermediate plan in the sense that only few trips are modified in the rebalancing phase. The reason for this is the fact that the modifications are done manually. Obviously, this criterion is obsolete in a fully computer-based decision support system.

Note that in most instances, changing the rolling stock types (i.e. option (ii) above) is undesirable in practise; it is considered to be an ad hoc measure. The models in this chapter do not include this option.

Both in theory and in practise, an important special case of rebalancing is to resolve the *off-balance of just one unit*: a rolling stock circulation has to be modified so that the inventory at station s_1 at time t_1 decreases by one, while the inventory at station s_2 at time t_2 increases by one. Methods for this can be useful also in short-term planning to correct off-balances caused by disruptions.

5.1.3 Modelling Approaches

The successful application of the Composition Model in tactical planning (Chapter 4) suggests that it can also be used in operational planning. Although operational rolling stock planning is still being done manually at NSR, the Composition Model is used time to time to set up the intermediate plan. Creating the intermediate plan manually is very time-consuming for complex instances such as the Noord-Oost line group; using the Composition Model for this speeds up the planning process. Note that the model is solved separately for each day of the planning horizon. The intermediate plan produced by the Composition Model mostly contains some off-balances. In the rebalancing phase, the planners apply ad hoc methods to resolve them.

Most rolling stock circulation instances of NSR have a simpler structure than the Noord-Oost line group: splitting and combining of trains does not occur. As reported in Section 3.11.5, the Composition Model turns out to be solvable on such instances of tactical planning within a couple of minutes. Yet, it has still to be investigated whether solving the Composition Model by a commercial MIP solver can cope with real-life operational planning problems as they may have a much larger size than tactical planning problems.

In this chapter we look for an alternative approach for solving the operational rolling stock circulation problem: We try to automate the two-phase solution method described above, thereby supporting the planning process without changing it substantially. The emphasis is on finding fast yet reliable heuristic methods. Here we only consider instances without splitting and combining of trains. Furthermore, discussions with planners revealed that creating the intermediate plans – even manually – is pretty straightforward for such instances. Therefore we only study the rebalancing problem.

5.1.4 Operational Rolling Stock Planning in the Literature

Since operational rolling stock planning is highly reminiscent of tactical planning, most of the publications listed in Section 3.1 are also relevant to this chapter. We mention in particular the papers of Ben-Khedher et al. (1998), Lingaya et al. (2002) and Ziarati et al. (1997): they concern problems whose planning horizon and formulation are closest to those in this chapter. These three papers also deal with maintenance of the rolling stock but at NSR this is considered only in short-term planning.

5.2 The Rebalancing Problem

The rebalancing problem is formulated as follows:

Definition 5.1. *Rebalancing problem:* *The input is the timetable and the intermediate plan; we assume that trains are not split nor combined. The shunting assumptions are as described in Sections 3.3 and 3.4. The intermediate plan is a rolling stock circulation satisfying all constraints in Section 3.5 except those on the initial and final inventories.*

The objective is to find a rolling stock circulation that resolves off-balances without creating new ones such that the weighted sum of the following three quantities

is minimised: the amount of unresolved off-balances; the number of trips that are modified during rebalancing; and the number of trips whose train length in the output exceeds a certain limit (depending on the trip).

In railway practise, minimising the amount of unresolved off-balances is usually more important than the other two criteria. Note that the third optimisation criterion aims at minimising the number of trips that require more conductors than in the tactical plan.

It is worth mentioning that in the case of a single rolling stock type, the rebalancing problem is a single-commodity flow problem with a non-convex objective function: it may cost the same to assign two or three units to a trip that is served by one unit in the intermediate plan. Here we only consider instances with multiple types.

In what follows, we prove that the feasibility version of the rebalancing problem is NP-complete, even when resolving an off-balance of just one unit. Then we look for solution methods.

5.3 NP-Completeness of the Rebalancing Problem

In this section we prove that it is NP-complete to decide whether the rebalancing problem has a feasible solution even if, with a given integer number k , a single station has a surplus of k units in the initial inventory and another station has a deficit of k units in the final inventory. Subsequently, we extend the construction in the proof and show that the problem remains NP-complete in the case of an off-balance of a single unit. Before that, we introduce some notations and assumptions and describe a ‘gadget’ that plays an important role in the NP-completeness proofs.

The instances of the rebalancing problem in this section contain stations with left-hand shunting side, indicated by [L], and stations with right-hand shunting side, indicated by [R]. The uncoupling side of each arriving trip is the shunting side of its arrival station and the coupling side of each departing trip is the shunting side of its departure station. There are two rolling stock types P and Q . Each trip must receive one or two units.

In the figures throughout the whole section, the railway networks are drawn in time-space diagrams: Each station is represented by a horizontal time-line, the time increases to the right. The trips correspond to diagonal lines between the time-lines. Train stops are indicated by dots. Dotted arcs between arrival and departure events connect the trips to their successors.

In the figures, some trips have no departure or arrival station. Consider these stations to be anonymous. All anonymous stations are different. Trips to or from anonymous stations always have a single unit of a certain type in the intermediate plan. Anonymous stations with a departing trip always have initial inventory 1 for this type and 0 for the other type; the final inventory is 0 for both types. The analogous condition holds for anonymous stations with an arriving trip: the initial inventories are 0, the final inventory is 1 for the type of the arriving unit and 0 for the other type.

In the intermediate plan, trips are operated by a single unit of type P (in the figures below represented by thick solid lines), by a single unit of type Q (thick dotted lines) or by a two-unit composition PP (thick dashed lines).

5.3.1 Building Blocks for the Proofs: the Gadgets

A *gadget* is a part of the railway network shown in Figure 5.2. It contains trips between 8 stations $\alpha_1, \alpha_2, \beta, \gamma, \delta, \varepsilon, \omega_1$ and ω_2 . Trip s_1 from α_1 to γ , trip s_2 from α_2 to β , trip t_1 from ε to ω_1 and trip t_2 from δ to ω_2 (as well as their predecessors and successors) are each operated in the intermediate plan by a single unit of type P . The trips from β to γ , from γ to δ and from δ to ε (as well as their predecessors and successors) are each operated by a single unit of type Q . Stations β and δ allow shunting on the left hand side only, while stations $\alpha_1, \alpha_2, \gamma, \varepsilon, \omega_1$ and ω_2 allow shunting on the right hand side only.

The initial and final inventories of stations β, γ, δ and ε are 0. Stations $\alpha_1, \alpha_2, \omega_1$ and ω_2 have undefined initial and final inventories in type P and they have zero initial and final inventory in type Q .

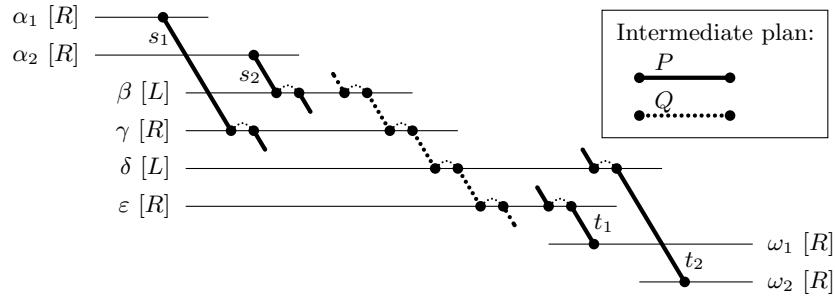


Figure 5.2: A gadget.

Lemma 5.2. *Consider a rolling stock circulation for a gadget that satisfies the given inventory, shunting and train length constraints. Then the following holds:*

- (i) *Trip s_1 has composition PP if and only if trip t_1 has composition PP .*
- (ii) *Trip s_2 has composition PP if and only if trip t_2 has composition PP .*
- (iii) *At most one of the trips t_1 and t_2 can have composition PP .*

Proof. (i) If trip s_1 has composition PP , then a unit can be uncoupled from it at station γ . This unit can be coupled to the right-hand side of the unit that travels from γ towards ε . Then the unit of type P can only be uncoupled at station ε . Actually, this is the only possibility to lead the uncoupled unit to either ω_1 or ω_2 . Moreover, this is the only way to get composition PP for trip t_1 .

(ii) Similar to (i).

(iii) Two extra units of type P can reach stations ω_1 and ω_2 only if the trip between γ and δ has composition PQP . However, this would violate the upper bound on the train length. ■

We use the simplified symbol in Figure 5.3 for a gadget. The main purpose of a gadget is to bring an additional unit either from α_1 to ω_1 , or from α_2 to ω_2 , but not both.

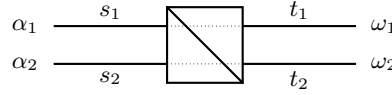


Figure 5.3: A simple symbol for a gadget.

5.3.2 Resolving an Off-balance of k Units

Consider an undirected graph $G = (V, E)$ with $V = \{1, \dots, n\}$ and let k be a positive integer with $k \leq n$. We build an instance of the rebalancing problem that is feasible if and only if G contains a stable set of size k . A *stable set* is a subset of nodes such that no pair of them is joined by an edge. It is well known that deciding whether the graph has a stable set with k nodes is NP-complete (Karp (1972)).

Create two stations A and Z . For every node $v \in V$ with d_v neighbours, we create $d_v + 1$ stations $S_1^v, \dots, S_{d_v+1}^v$. The shunting side of all these stations is the right-hand side.

For each $v \in V$, insert a trip from station A to station S_1^v and insert a trip from station $S_{d_v+1}^v$ to station Z . For each trip from A to a station S_1^v , create a predecessor trip from an anonymous station to A . For each trip arriving at Z , insert a successor from Z to an anonymous station. All trips so far are operated with a single unit of type P in the intermediate plan.

For each node $v \in V$ with neighbours $u_1, \dots, u_{d_v}$, assign stations $S_1^v, \dots, S_{d_v}^v$ to the edges $u_1v, \dots, u_{d_v}v$, bijectively in an arbitrary way. For each edge $uv \in E$ with $u < v$, add a gadget as follows. Let S_i^u and S_j^v be the stations assigned to edge uv . Create four new stations β, γ, δ and ε , set $\alpha_1 = S_i^u$, $\alpha_2 = S_j^v$, $\omega_1 = S_{i+1}^u$, $\omega_2 = S_{j+1}^v$ and insert all the trips described in the definition of a gadget. A station S_j^v with $1 < j < d_v + 1$ belongs to exactly two gadgets and has one arriving and one departing trip. The departing trip is the successor of the arriving trip.

This completes the railway network. Its size is polynomial in n : it contains $O(n^2)$ trips between $O(n^2)$ stations. The network for a small graph is shown schematically in Figure 5.4.

The intermediate plan satisfies the following inventory constraints. The initial and final inventories for type Q are 0 (except for some anonymous stations inside the gadgets). For type P , the initial and final inventories of stations S_j^v are 0. Station A has initial and final inventory k , while station Z has initial and final inventory 0. The initial and final inventories of β -, γ -, δ - and ε -stations of the gadgets are all zero.

The *desired* inventories differ from these at two points. The desired final inventory of station A in type P is 0, the desired final inventory of station Z in type P is k . In some sense, rebalancing means that k units of type P must be routed from A to Z .

Note that the inventory and train length constraints do not leave much choice for feasible rolling stock circulations. Each trip has either the same composition as in the intermediate plan or it receives the original composition extended by a single unit of type P .

Theorem 5.3. *Graph $G = (V, E)$ contains a stable set of size k if and only if the instance of the rebalancing problem constructed above has a feasible solution.*

Proof. Suppose that G contains the stable set $\{v_1, \dots, v_k\}$. A solution of the rebalancing problem can be obtained as follows. Couple the k units of type P at station A to the k trips that depart towards stations $S_1^{v_1}, \dots, S_1^{v_k}$.

Consider any gadget that connects stations $S_j^{v_i}$ and $S_{j+1}^{v_i}$ for some indices i and j . We adjust the intermediate plan inside the gadget as follows. The trips of this gadget that are incident to stations $S_j^{v_i}$ and $S_{j+1}^{v_i}$ get composition PP ; we also make all the necessary modifications to route the additional unit through the gadget from $S_j^{v_i}$ to

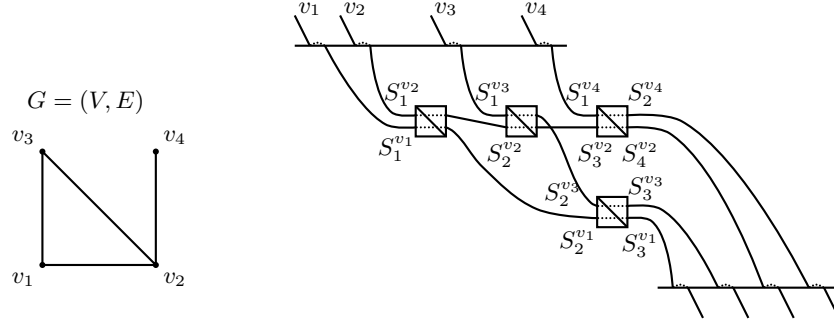


Figure 5.4: An example of the construction of the network.

$S_{j+1}^{v_i}$. The adjustment of the gadgets can be done simultaneously because there is no edge between the nodes $v_1, \dots, v_k$. Then all the k excess units reach station Z where they can be uncoupled. Therefore, the rebalancing problem is feasible.

Conversely, consider a solution of the rebalancing problem. At station A , k units of type P are coupled to trips towards stations, say, $S_1^{v_1}, \dots, S_1^{v_k}$. These units pass through all gadgets that are related to the nodes $v_1, \dots, v_k$ and end up at station Z . Then the nodes $v_1, \dots, v_k$ form a stable set in G as otherwise Lemma 5.2 (iii) would be violated. ■

Corollary 5.4. *The feasibility version of the rebalancing problem is NP-complete in the case of an off-balance of k units.*

5.3.3 Resolving an Off-balance of One Unit

Here we extend the construction described in the previous section. Thereby we prove that the maximum stable set problem can be reduced to the rebalancing problem with an off-balance of one unit.

Let $G = (V, E)$ be an undirected graph with $|V| = n$ and let k be a positive integer with $k \leq n$. Consider the railway network constructed in the previous section. It is represented in Figure 5.5 by stations A and Z and the grey box between them.

Create $k + 1$ new stations $\alpha_1, \dots, \alpha_k$ and ω . Insert $4k$ trips as follows (see Figure 5.5). Create a trip from α_i to α_{i+1} for each $i = 1, \dots, k$ (where $\alpha_{k+1} = \omega$) and insert their predecessors and successors from and to anonymous stations. Also insert a trip that departs from station α_i and returns to the same station and has no predecessor or successor. All these new trips are operated by a single unit of type P in the intermediate plan.

Insert k extra gadgets $g_1, \dots, g_k$. Let $s_1^{(i)}$, $s_2^{(i)}$, $t_1^{(i)}$ and $t_2^{(i)}$ denote the s_1 -, s_2 -, t_1 - and t_2 -trips of gadget g_i .

For each $i = 1, \dots, k$, trip $s_1^{(i)}$ departs from station Z and has a predecessor from an anonymous station. Trip $t_1^{(i)}$ arrives at α_i and has a successor to an anonymous station. Trips $s_1^{(i)}$, its predecessor, $t_1^{(i)}$ and its successor have composition P in the intermediate plan.

For each $i = 1, \dots, k$, trip $s_2^{(i)}$ departs from station A and has a predecessor from an anonymous station. Trip $t_2^{(i)}$ arrives at α_i and has a successor to an anonymous station. Trips $s_2^{(i)}$ and $t_2^{(i)}$ have composition PP , while the predecessor of $s_2^{(i)}$ and the successor of $t_2^{(i)}$ have composition P in the intermediate plan.

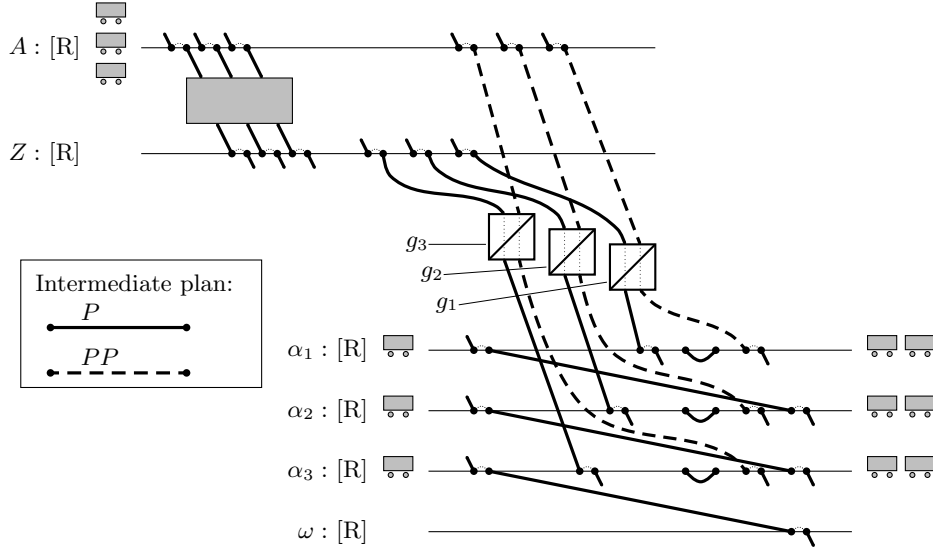


Figure 5.5: One unit to be routed ($k = 3$). At the beginning and the end of the timelines, we give the initial and final inventories of type P realised by the intermediate plan.

The railway network we constructed has polynomial size in n since it contains $O(n^2)$ trips and $O(n^2)$ stations.

The initial and final inventories for units of type Q are 0 except for some anonymous stations inside the gadgets. For type P , the initial inventory of station A is k , at stations $\alpha_1, \dots, \alpha_k$ it is 1 and at stations Z and ω it is 0. The final inventory of stations A , Z and ω is 0, while at stations $\alpha_1, \dots, \alpha_k$ it is 2. Anonymous stations have initial and final inventory zero or one. All other stations (i.e. the β -, γ -, ε - and δ -stations of the gadgets) have zero initial and final inventories.

The goal is to decrease the final inventory of station α_1 in type P by one and to increase the final inventory of station ω in type P by one.

Note that, as in the previous section, the shunting, inventory and train length constraints restrict the possible rolling stock circulations a lot. Each trip must be assigned the same composition as in the intermediate plan, eventually with a unit of type P coupled or uncoupled at the appropriate side. In particular, the circulation of the unit of type Q does not change at all.

Theorem 5.5. *Graph G has a stable set of size k if and only if the intermediate plan can be modified to decrease the final inventory of α_1 in type P by one and to increase the final inventory of ω in type P by one.*

Proof. To increase the final inventory at ω by one, the trip from α_k to ω must get a composition PP . Then the trip from α_k returning to α_k has no unit to serve unless an extra unit arrives earlier from gadget g_k . That is, trip $s_1^{(k)}$ from Z to gadget g_k and trip $t_1^{(k)}$ from gadget g_k to α_k must get composition PP , too. Then trip $s_2^{(k)}$ from A to gadget g_k and trip $t_2^{(k)}$ from gadget g_k to α_k must get composition P only. To correct the final inventory at α_k , the trip from α_{k-1} to α_k must get composition PP . Repeating the argument, it follows that the rebalancing problem can be solved if and only if all the k units that start at A can reach station Z . Invoking Theorem 5.3 completes the proof. ■

Corollary 5.6. *The feasibility version of the rebalancing problem is NP-complete in the case of an off-balance of a single unit.*

5.4 Heuristic Approach for Rebalancing

In this section we describe a heuristic approach for the rebalancing problem. Our goal is to find a conceptually simple heuristic algorithm that can reliably cope with instances in practise such that it admits quick solution times.

First we deal with the special case of resolving an off-balance of one unit. Then we propose an iterative algorithm that resolves an off-balance of one unit in each iteration, making use of the method described for the special case.

5.4.1 Off-balance of One Unit Heuristically

The input is the last updated intermediate plan, we call it simply the *input plan*. The input plan has off-balances in a given type m : There are stations and time moments with a surplus of units of type m and there are stations and time moments with a

deficit of units of type m . The output is a rolling stock circulation with one less surplus at a certain point and with one less deficit of type m at another point while all other initial and final inventories remain the same.

When used in an iterative framework, the input plan is updated according to the output of the algorithm. This updated plan serves as the input plan of the next iteration.

Taking all possible modifications of the input plan into account easily leads to a fairly complex integer programming model like the Composition Model. As the goal of this section is to provide a simple heuristic approach, we restrict the solution space by requiring that the circulation of any type differing from m remains unchanged. That is, in the output circulation each trip gets the same pattern of types differing from m as in the input plan.

Example. Let $\{P, Q, R\}$ be the set of available types with $m = P$ and suppose that a trip has the 3-unit composition PQR in the input plan. Then this trip may get composition QR , PQR , QPR , $QPPR$, etc. in the output plan, however, it cannot get composition PQ or RQP .

When resolving an off-balance of one unit of type m , one may expect that the composition of a trip undergoes only some minor modification like becoming one unit of type m longer or shorter. This motivates the following restriction. If r' is the successor trip of a trip r and units of any type are uncoupled in the input plan right after trip r , then the output plan may not couple units immediately before trip r' . A similar restriction holds if units are coupled to trip r' in the input plan. Note that in an iterative setting, it can happen that early iterations cancel all uncouplings from trip r , so coupling units to trip r' may become possible in later iterations.

Graph Representation

We build a single-commodity flow-type model in an appropriate graph $G = (V, E)$ to describe the circulation of the units of type m . The graph can be seen as an extension of the simplified shunting model in Section 3.3. Below, we use the notations introduced in Section 3.5.

First we define the relevant time moments at a station i . A time moment j is *relevant* at station i if a trip departs at j from station i or if a trip r arrives at $j - \varrho(r)$ at station i where $\varrho(r)$ is the re-allocation time. In addition, the beginning and the end of the planning period are also *relevant*.

Create a *station node* for each pair (i, j) of station i and a relevant time moment j at station i . Moreover, for two consecutive relevant time moments j, j' at station i , draw a *station arc* from the node associated with (i, j) to the node associated with (i, j') .

Consider a trip r and suppose the input plan assigns to it a composition

$$\underbrace{m \dots m}_{k_1^{(r)}} m_1 \underbrace{m \dots m}_{k_2^{(r)}} m_2 \dots m_{\ell_r-1} \underbrace{m \dots m}_{k_{\ell_r}^{(r)}}$$

where $m_1, \dots, m_{\ell_r-1}$ denote types different from m . That is, units of type m appear in ℓ_r (possibly empty) groups, the groups being separated by units of other types. Since the circulation of these other types may not be changed, the only decision of the one-off-balance heuristic algorithm is to change the size of these groups. That is, the algorithms may only change the values $k_1^{(r)}, \dots, k_{\ell_r}^{(r)}$.

Create nodes $u_1^{(r)}, \dots, u_{\ell_r}^{(r)}$ that correspond to the departure of trip r and nodes $v_1^{(r)}, \dots, v_{\ell_r}^{(r)}$ that correspond to the arrival of trip r . Insert a *trip arc* from $u_i^{(r)}$ to $v_i^{(r)}$ for each $i = 1, \dots, \ell_r$.

If trip r has no successor, then draw an arc from each node $v_i^{(r)}$ to the station node corresponding to station $s_a(r)$ and time moment $\tau_a(r) + \varrho(r)$. If a trip has no predecessor, then draw an arc from the station node corresponding to station $s_d(r)$ and time moment $\tau_d(r)$ to each node $u_i^{(r)}$. Then we obtain a graph like in Figure 5.6.

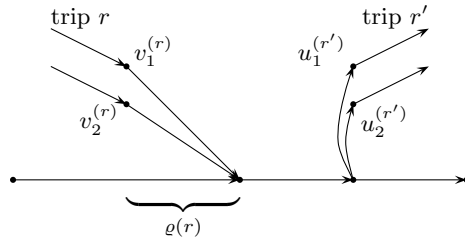


Figure 5.6: The graph representation of trip without a successor or without a predecessor.

Now we turn to modelling the composition changes between a trip r and its successor trip r' . We distinguish three cases, depending on how the input plan changes the composition between trips r and r' . Recall that an allowed composition

change is either coupling or uncoupling some units at the appropriate side of the arriving trip, but not both.

First assume that units are uncoupled from the, say, right-hand side of the arriving trip r . Then the graph representation does not contain the possibility of coupling any unit to trip r' . In this case we have $\ell_r \geq \ell_{r'}$. Physically, the train is split into two parts at a point that lies in the $\ell_{r'}^{\text{th}}$ group of the arriving composition. Then the first (i.e. left-most) $\ell_{r'} - 1$ groups go over unchanged to become the first $\ell_{r'} - 1$ groups of trip r' . The last (i.e. right-most) $\ell_r - \ell_{r'}$ groups (if any) are uncoupled. Units in the $\ell_{r'}^{\text{th}}$ group of trip r can go over to the $\ell_{r'}^{\text{th}}$ group of trip r' or they can be uncoupled. These possibilities are expressed by the arcs shown in Figure 5.7(a) for the case $\ell_r = 4$ and $\ell_{r'} = 2$. The construction can be easily adjusted if uncoupling takes place at the left-hand side of the arriving trip.

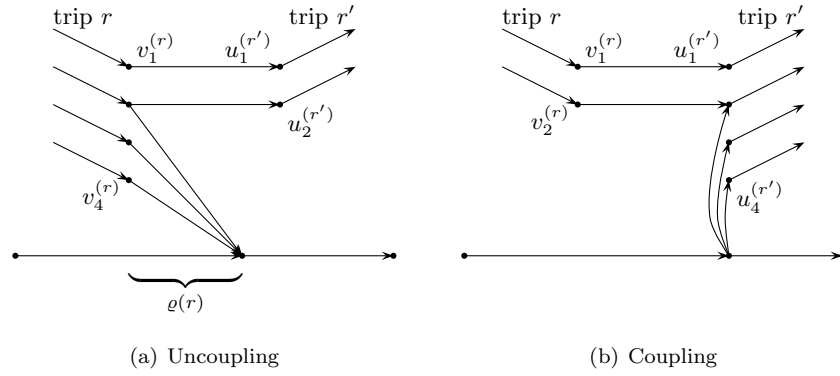


Figure 5.7: The graph representation of the case when units are uncoupled or coupled between trips r and r' in the input plan.

The case when units are added to the departing trip r' in the input plan is modelled similarly. Then the graph model does not include the possibility of uncoupling units after the arrival of trip r . An example is shown in Figure 5.7(b).

Finally, if trips r and r' have identical compositions in the input plan, then the graph model preserves the possibility of both coupling and uncoupling units. Suppose that uncoupling and coupling are possible at the left-hand side (the other cases being analogous). Then the first (i.e. left-most) $\ell_r - 1$ groups of trip r go unchanged over to the first $\ell_r - 1$ groups of trip r' . The last groups of trips r and r' , however, can be decreased and increased by uncoupling or coupling units of type m . An example is given in Figure 5.8.

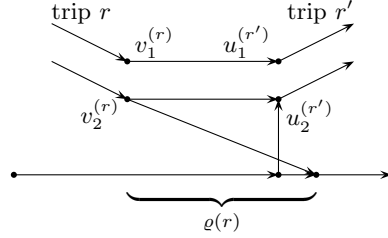


Figure 5.8: The graph representation of the case when r and r' have identical compositions.

We call an arc from a station node to a node $u_i^{(r)}$ a *coupling arc* and we call an arc from a node $v_i^{(r)}$ to a station node an *uncoupling arc* as they are intended to describe coupling and uncoupling of units.

b -Transshipments in the Graph

We use the concept of b -transshipments to describe the circulation of the units of type m . Let b be an integer-valued function on the nodes of a directed graph $G = (V, E)$ with

$$\sum_{v \in V} b(v) = 0.$$

Let $f: E \rightarrow \mathbb{Z} \cup \{-\infty\}$ and $g: E \rightarrow \mathbb{Z} \cup \{\infty\}$ be lower and upper capacity functions. An integral b -transshipment is a function $x: E \rightarrow \mathbb{Z}$ satisfying

$$f(a) \leq x(a) \leq g(a) \quad \forall a \in E$$

such that the net out-flow at each node v is equal to $b(v)$. That is,

$$\sum_{a \in \delta^{\text{out}}(v)} x(a) - \sum_{a \in \delta^{\text{in}}(v)} x(a) = b(v) \quad \forall v \in V. \quad (5.1)$$

Details about b -transshipments can be found in Schrijver (2003). We give a short reminder of the definitions and some main results in Appendix B.

The circulation of the units of type m in the input plan determines the b -transshipment x in the graph G as follows. The station nodes at the beginning of the planning period have positive b -values; for such a node v that corresponds to

station i , let $b(v)$ be the number of units that start at station i . Similarly, the station nodes at the end of the planning period have negative b -values; for such a node v that corresponds to station i , let $b(v)$ be the negative of the number of units that end up at station i . For all other nodes, let $b(v) = 0$. This defines the function b . The b -transshipment must be non-negative, the upper capacities are defined as follows. For trip r , let μ_r be the largest number such that increasing the train length by μ_r units of type m does not exceed the maximal allowed train length on trip r (denoted by $\mu_r^{\max}$). Then the upper capacity of a trip arc $u_i^{(r)} v_i^{(r)}$ is the size of the i^{th} group of trip r in the input plan plus μ_r . All other arcs have infinite upper capacity.

The x -value on trip arc $u_i^{(r)} v_i^{(r)}$ is the size of the i^{th} group of trip r in the input plan. The input plan explicitly says how many units of type m are coupled or uncoupled before or after each trip, these are x -values on the coupling and uncoupling arcs. The x -values on the remaining arcs are uniquely determined by constraints (5.1).

Example. Figure 5.9(a) shows a railway network of two trips between stations A and B as the usual time-space diagram; both stations have left-hand shunting side. The riding direction does not change between the two trips.

The input plan assigns a composition mm' for both trips, the types m and m' are represented by black and grey units. Then the graph is given in Figure 5.9(b). The function $b : V \rightarrow \mathbb{Z}$ is defined as

$$b(v) = \begin{cases} 1 & \text{if } v = s_A, \\ -1 & \text{if } v = t_A, \\ 0 & \text{otherwise.} \end{cases}$$

Bold arcs indicate the b -transshipment that arises from the input plan: Bold arcs have x -value 1, other arcs have x -value zero.

We have seen above that a rolling stock circulation defines a b -transshipment in the graph $G = (V, E)$. Conversely, consider any specification for the initial and final inventories and translate it into a function $b' : V \rightarrow \mathbb{Z}$. Then each b' -transshipment x' defines a rolling stock circulation. However, the obtained rolling stock circulation may violate some constraints of the problem specification. To avoid this, the following two additional side constraints have to be fulfilled:

- The train length on each trip r obeys the lower and upper bounds:

$$\mu_r^{\min} - L_r \leq \sum_i x' \left(u_i^{(r)} v_i^{(r)} \right) \leq \mu_r^{\max} - L_r \quad \forall \text{ trip } r \quad (5.2)$$

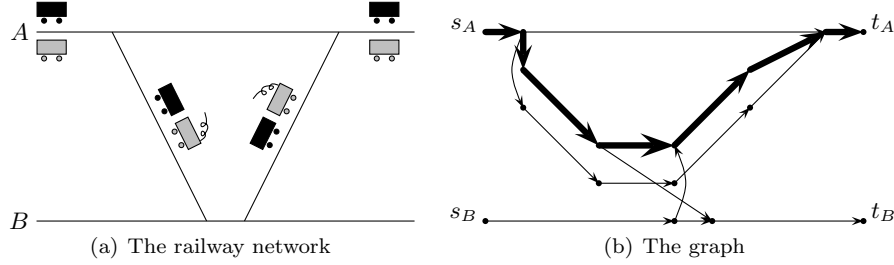


Figure 5.9: The graph representation of a small railway network. Bold arcs indicate the circulation of the black unit.

where L_r denotes the number of carriages in those units on trip r whose type differs from m .

- Coupling and uncoupling may not take place at the same time between a trip r and its successor r' :

$$\sum_i \sum_{\substack{a \in \delta^{\text{out}}(v_i^{(r)}): \\ a \text{ uncoupling arc}}} x'(a) = 0 \quad \text{or} \quad \sum_i \sum_{\substack{a \in \delta^{\text{in}}(u_i^{(r')}) : \\ a \text{ coupling arc}}} x'(a) = 0. \quad (5.3)$$

Note that constraint (5.3) is trivially fulfilled unless trips r and r' have the same composition in the input plan: this is the only case when the graph contains both uncoupling arcs from some of the nodes $v_i^{(r)}$ and coupling arcs to some of the nodes $u_i^{(r')}$.

The Heuristic Algorithm

Suppose first that there is a unique station node s that corresponds to a surplus in type m in the input plan and there is a unique station node t that corresponds to a deficit in type m in the input plan. Define the function $b': V \rightarrow \mathbb{Z}$ as follows:

$$b'(v) = \begin{cases} b(v) + 1 & \text{if } v = s, \\ b(v) - 1 & \text{if } v = t, \\ b(v) & \text{otherwise.} \end{cases} \quad (5.4)$$

Resolving an off-balance of one unit of type m amounts to finding a b' -transshipment x' in graph G that satisfies the side constraints (5.2) – (5.3). Indeed, when compared with x (i.e. with the input plan), the b' -transshipment x' decreases the surplus at s by one and decreases the deficit at t by one, without creating any new off-balance.

Unfortunately, this problem is NP-hard: The proof of Theorem 5.5 can be repeated without any change. In the proof of the theorem, an off-balance of one unit of type P has to be resolved, while the circulation of the other type Q is unaffected.

The heuristic algorithm looks for a b' -transshipment without taking the side constraints (5.2) – (5.3) into account, but afterwards it verifies whether (5.2) – (5.3) are satisfied.

Heuristic rebalancing for an off-balance of one unit {

Create the graph $G = (V, E)$.

Let s be the unique station node with a surplus.

Let t be the unique station node with a deficit.

Define b' as in (5.4).

Look for a b' -transshipment x' in G .

If x' exists and satisfies (5.2) and (5.3) **then**

Accept the solution.

Else

Report that no solution was found.

}

It is known in network flow theory that, given the nodes s and t , such a b' -transshipment can be found efficiently as follows. Define the auxiliary graph $\widehat{G} = (V, \widehat{E})$. Start from an empty graph on the node set V . For each arc $uv \in E$, add a *forward arc* from u to v if $x(uv)$ is smaller than the upper capacity of arc uv . For an arc $uv \in E$ with $x(uv) > 0$, add a *backward arc* from v to u . Let P be a directed $s - t$ -path in $\widehat{G}$. Then the function $x': E \rightarrow \mathbb{Z}_+$ given by

$$x'(uv) = \begin{cases} x(uv) + 1 & \text{if the forward arc } uv \text{ is used by path } P, \\ x(uv) - 1 & \text{if the backward arc } vu \text{ is used by path } P, \\ x(uv) & \text{otherwise} \end{cases} \quad (5.5)$$

is a b' -transshipment. Conversely, if there exists a b' -transshipment, then the auxiliary graph $\widehat{G}$ contains a directed $s - t$ -path. (For more details, see Appendix B.) Therefore, the b' -transshipment x' can be found in $O(|E|)$ time by simply finding a directed $s - t$ -path in the auxiliary graph.

In the general case, there may be several candidates for the nodes s and t . This can be handled in the same way as multiple-source multiple-sink single-commodity flow problems: Add two new nodes s^* and t^* to the graph $\widehat{G}$, insert an arc s^*v for each station node v that corresponds to a surplus of type m and insert an arc vt^* for

each station node v that corresponds to a deficit of type m . Then one has to look for an $s^* - t^*$ -path.

Example. Consider again the example in Figure 5.9. Using the notations of Figure 5.9(b), we have $b(s_A) = 1$, $b(t_A) = -1$ and $b(v) = 0$ for all other nodes. Suppose now that station A has a deficit of one unit at the beginning of the planning period, while station B has a surplus of one unit at the beginning of the planning period. That is, the function b' is given by $b'(s_B) = 1$, $b'(t_A) = -1$ and $b'(v) = 0$ for all other nodes. Moreover, $s = s_B$ and $t = s_A$. The auxiliary graph is shown in Figure 5.10. The bold arcs indicate a directed $s_B - s_A$ -path. The b' -transshipment and the corresponding rolling stock circulation of type m are shown in Figure 5.11.

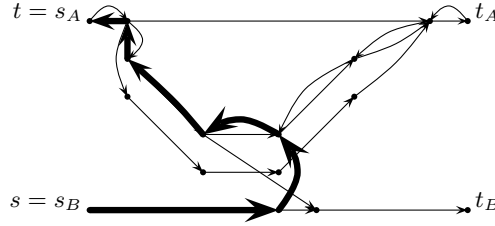


Figure 5.10: An $s - t$ -path in the auxiliary graph.

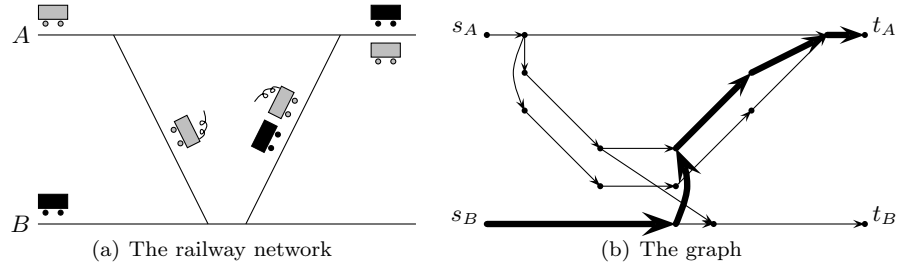


Figure 5.11: The off-balance of the black unit is resolved (cf. Figure 5.9 and 5.10).

Weighted Version

The problem formulation in Definition 5.1 contains the requirement that the number of trips that are modified during rebalancing and the number of trips whose train length in the output exceeds a certain limit $\lambda(r)$ are to be minimised. To comply with

this specification, we assign cost values to the arcs of the auxiliary graph. The output plan is obtained from the input plan by changing it along an $s^* - t^*$ -path. In the weighted version of the heuristic approach, we choose a minimum cost $s^* - t^*$ -path.

Let r be a trip whose composition length in the input plan is not longer than $\lambda(r)$ but by coupling a unit of type m , the limit $\lambda(r)$ is exceeded. As mentioned above, this means that coupling an additional unit to trip r requires an extra conductor. For such trips r , we assign a positive cost value to each forward arc $u_i^{(r)}v_i^{(r)}$ in the auxiliary graph. Observe that these arcs express the possibility of making the composition on trip r longer. The other cost component penalises the deviation from the input plan: An additional positive cost is assigned to each forward arc $u_i^{(r)}v_i^{(r)}$ and to each backward arc $v_i^{(r)}u_i^{(r)}$. Using such an arc in the $s^* - t^*$ -path indicates that the composition on trip r gets modified. Other arcs have zero cost.

Having defined these non-negative cost values, a minimum cost path can be found very efficiently in $O(|E| + |V| \log |V|)$ time by Dijkstra's algorithm using Fibonacci heaps.

As it is often the case when modelling real-life problems, there are several choices for the objective function. We point out here two possible adjustments for the case when the algorithm for resolving an off-balance of one unit is used in an iterative framework.

- The second cost component (that measures the deviation from the input plan) compares the output plan with the *last updated plan*. In other words, if the composition of a trip is adjusted in several iterations, the penalty is accounted for several times. However, the problem specification in Definition 5.1 instructs to minimise the deviation from the *intermediate plan*, i.e. from the plan at the beginning of the rebalancing phase. An alternative cost structure may penalise the modification of only those trips whose composition has not been adjusted in the previous iterations.
- Suppose that a composition of trip r is increased in an iteration by a unit of type m thereby requiring an extra conductor for trip r . In a later iteration, it may be possible to resolve an additional off-balance of type m by decreasing the composition of r . That is, after this latter iteration, no extra conductor is needed again for trip r . So, it is actually advantageous to make use of a backward arc $v_i^{(r)}u_i^{(r)}$. This can be expressed by assigning a negative cost to such backward trip arcs.

In our implementations, we do not take these two considerations into account. The reason for this is that the main emphasis of this research is on deciding whether

the heuristic approach can handle the rebalancing problem at all. To that end, we focus on the most important objective in rebalancing: to resolve as many off-balances as possible. We leave fine-tuning the heuristic algorithm to minimise the other two objective criteria for future research.

Off-balance of One unit Heuristically: Summary

We summarise the method for resolving an off-balance of one unit. The method tries to resolve an off-balance of one unit subject to the constraint that the circulation of all other rolling stock types remains unchanged.

First, create the graph $G = (V, E)$ and compute the b -transshipment x from the input plan as described above. Construct the auxiliary graph $\hat{G}$, add the nodes s^* and t^* and insert the arcs to and from the station nodes with a surplus and a deficit, respectively. Find a minimum cost directed $s^* - t^*$ -path P in $\hat{G}$ and construct the b' -transshipment x' according to (5.5).

We call a b' -transshipment *good* if it fulfils the side constraints (5.2) – (5.3), otherwise we call it *wrong*. The heuristic algorithm returns x' if it is a good b' -transshipment, otherwise it rejects x' reporting that it cannot resolve any off-balance of type m .

If the algorithm returns a solution, then the output is a rolling stock circulation that can be implemented in practise and that has one less off-balance than the input plan. However, if the algorithm does not return a solution, it may have two reasons: Either there does not exist any good b' -transshipment (in which case the outcome of the algorithm is correct), or there exist good b' -transshipments but the algorithm found a wrong one (in which case the outcome of the algorithm is incorrect). In our computational tests, each b' -transshipment found by the heuristic algorithm was good.

5.4.2 Arbitrary Off-balances Heuristically

For an instance of the rebalancing problem with arbitrary off-balances, we propose a method that is similar to the heuristic algorithm in Section 4.5.8 for solving the maintenance routing problem. We proceed iteratively, in each iteration resolving an off-balance of a single unit and updating the intermediate plan accordingly. The algorithm stops if it cannot resolve any more off-balances.

The basic structure of the algorithm is as follows:

```

Heuristic rebalancing algorithm {
  While there are off-balances {
     $M' :=$  the set of types with off-balances.
    Resolve an off-balance of one unit for an  $m \in M'$ .
    Stop if no off-balance could be resolved for any  $m \in M'$ .
  }
}

```

The number of iterations is at most the total number of off-balances. In each iteration, each type $m \in M'$ is tested at most once whether off-balances can be resolved. For such a check for a fixed type, the running time is dominated by the running time of Dijkstra's algorithm $O(|\widehat{E}| + |V| \log |V|)$, since every other step – such as building the graph, checking the side constraints and updating the rolling stock circulation – is accomplished in $O(|\widehat{E}|)$ time. The number of nodes and the number of edges in the graph $\widehat{G}$ are $O(\ell T)$ where T is the number of trips and ℓ is the largest possible number of units in a composition. Therefore, the overall running time of the heuristic rebalancing algorithm is $O(kM\ell T \log(\ell T))$ where M is the number of rolling stock types and k is the total number of off-balances. Note that different instances of NSR share the same (small) numbers ℓ and M . That is, the running time is essentially proportional to $kT \log T$.

Whenever the algorithm terminates, its output is a rolling stock circulation that satisfies all requirements. The question is, however, the quality of the solutions: How many of the off-balances can be resolved in such a greedy way. In addition, one may ask how many trips are modified in the rebalancing phase and how many trips require an extra conductor. To answer these questions, we carried out computational experiments. We report our results in the next section.

5.5 Computational Results

In this section we report our computational results on the rebalancing problem. As described above, the rebalancing problem arises in the second phase of operational rolling stock planning. The test instances used are not real-life problems of NSR. Instead, we take the tactical rolling stock circulation of the rolling stock type “Sprinter” on two consecutive days and assume that this is the intermediate plan. To obtain multiple types, we divide the available rolling stock into two types.

The test instances are generated from this intermediate plan as follows. We assume that the final inventories of the intermediate plan have no off-balance and we choose the off-balances of the initial inventories randomly. That is, we randomly select the stations to have a surplus or a deficit in the initial inventory; the amount of the surpluses and deficits is also chosen randomly. Of course, we take care that for each rolling stock type the sum of the surpluses is equal to the sum of the deficits.

We solve each test instance in two different ways. In one approach, we formulate it as an instance of the Composition Model and solve this model by the commercial MIP software CPLEX 9.0 (using the modelling software ILOG OPL Studio 3.7). We compare this with the other approach when we apply the heuristic rebalancing algorithm implemented in the programming language Perl. All computations have been carried out on a PC with a Pentium IV 3.0 GHz processor and 512 Mb internal memory under the operating system Windows 98.

5.5.1 Dimensions of the Problem

The timetable in the test instances contains 803 trips between 12 stations. There are 47 units available, divided into two types of 21 and 26 units. The instances concern an off-balance of 8 units: an off-balance of 4 units for both types. That is, the sum of the surpluses is 4 for both types; also, the sum of the deficits is also 4 for both types.

The Composition Model leads to a mixed integer program with 54,722 variables (among them 17,636 integer variables) and 26,594 constraints. The reduced mixed integer program has 53,753 variables, 25,888 constraints and 423,537 non-zeros in the coefficient matrix. The heuristic method requires finding b -transshipments in a graph with 4,200–4,400 nodes and 5,500–5,800 arcs. The essential part of the algorithm is to find a minimum cost path in the auxiliary graph with the same number of nodes and with about 7,300 arcs.

5.5.2 Objective Function

The objective function is a linear combination of the following three terms:

- 1000 times the amount of unresolved off-balances, summed for the two types,
- twice the number of trips that require an additional conductor,
- the number of trips whose composition is modified.

	z_0^{Heur}					Total number of instances
	0	1	2	3	4	
$z_0^{\text{CM}} = 0$	1,946	800	65	2	0	2,813
$z_0^{\text{CM}} = 1$		1,322	269	9	0	1,600
$z_0^{\text{CM}} = 2$			440	52	0	492
$z_0^{\text{CM}} = 3$				87	6	93
$z_0^{\text{CM}} = 4$					2	2
Total	1,946	2,122	774	150	8	5,000

Table 5.1: Number of instances with given z_0^{CM} and z_0^{Heur} .

Lacking information about the relative importance of the objective criteria, the coefficients 1000, 2 and 1 are chosen somewhat arbitrarily. In any case, the amount of unresolved off-balances dominates the objective function.

The objective value of the Composition Model and of the heuristic rebalancing method are denoted by z^{CM} and z^{Heur} , the amount of unresolved off-balances are denoted by z_0^{CM} and z_0^{Heur} .

5.5.3 Numerical Results

The first question is how many off-balances can be resolved by the heuristic method. To answer it, we group the test instances according to the number of unresolved off-balances, this gives Table 5.1. Table 5.2 shows the distribution of the value z_0^{Heur} for each fixed value z_0^{CM} . It turns out that in the vast majority of the instances, the heuristic method leaves at most one more unresolved off-balance than the Composition Model does. Actually, in about 75% of the instances (in 3,797 cases of 5,000) the two methods resolve the same amount of off-balances. It indicates that for most test instances, the smaller solution space of the heuristic approach still includes solutions with an (almost) minimal number of off-balances and that the very simple heuristic algorithm is able to find these solutions.

We compare the objective values of the instances for which the Composition Model and the heuristic approach resolved the same amount of off-balance. Table 5.3 contains the average and maximum values of z^{CM} as well as the average and maximum difference of $z^{\text{Heur}} - z^{\text{CM}}$. The averages and the maxima are computed separately for each given value z_0^{CM} . Note that the line for $z_0^{\text{CM}} = z_0^{\text{Heur}} = 4$ concerns only two instances.

For the number of trips that require an extra conductor and for the number of trips that are modified, solutions of the heuristic method have 50–100% higher

	z_0^{Heur}					
	0	1	2	3	4	
$z_0^{\text{CM}} = 0$	69.18%	28.44%	2.31%	0.07%	0.00%	100%
$z_0^{\text{CM}} = 1$		82.62%	16.81%	0.56%	0.00%	100%
$z_0^{\text{CM}} = 2$			89.43%	10.57%	0.00%	100%
$z_0^{\text{CM}} = 3$				93.55%	6.45%	100%
$z_0^{\text{CM}} = 4$					100.00%	100%

Table 5.2: Given the value z_0^{CM} , the percentage of instances with different values z_0^{Heur} .

z_0^{CM}	Number of instances	Average z^{CM}	Maximum z^{CM}	Average $z^{\text{Heur}} - z^{\text{CM}}$	Maximum $z^{\text{Heur}} - z^{\text{CM}}$
0	1,946	26	60	8.15	29
1	1,322	1,020	1,043	8.36	29
2	440	2,015	2,030	8.33	27
3	87	3,011	3,020	10.09	24
4	2	4,013	4,020	7.50	11

Table 5.3: Average and maximal values of z^{CM} and average and maximal values of $z^{\text{Heur}} - z^{\text{CM}}$ for instances with $z_0^{\text{CM}} = z_0^{\text{Heur}}$.

values on average than solutions of the Composition Model, although even these higher values are not extraordinarily high. In any case, further computational tests on real-life data and discussions with planners are still necessary to decide how far the solutions are acceptable in practise.

5.5.4 Solution Times

The mixed integer program that arises from the Composition Model can be solved to optimality quite quickly. The vast majority of the instances (4,784 instances of 5,000) requires less than 3 minutes of CPU time. Although CPLEX hits the time limit of 3 hours on a particular instance, all other instances are solved within 30 minutes. An almost optimal solution is found within 3 minutes in each case.

The heuristic approach has been implemented in the programming language Perl which provides a convenient environment for development. However, due to the fact that Perl is an interpreted script language, the running times are an order of magnitude higher than the running times of implementations in compiled languages such as C or C++. The heuristic algorithm requires 35–70 seconds. This means than

the same algorithm implemented in C++ would very likely achieve running times of a couple of seconds.

5.6 Conclusions and Future Work

In this chapter we discussed the operational rolling stock circulation problem. The methods described in Chapter 3 can be adapted to it. In particular, solving the Composition Model by commercial MIP software gives promising results in computational tests, although further investigation is needed concerning the performance of the method on operational rolling stock planning instances that are substantially larger than the test problems.

As an alternative approach, we considered a model-based version of the two-phase method that is applied currently at NSR. Focusing on the second phase only, we formulated the rebalancing problem, discussed its computational complexity and proposed a heuristic solution method. The Composition Model can be adapted to the rebalancing problem, too. This allows one to evaluate the performance of the heuristic approach by comparing its output with the optimal solutions of the Composition Model.

Computational experiments revealed that, despite its simplicity, the heuristic approach very often resolves the largest possible number of off-balances or just one less. Thus we can conclude that it is in fact plausible to use a heuristic algorithm of such a kind as a planning tool. The advantage of such a method is that its running time remains attractively low, even for problem sizes when the memory consumption and solution time of the Composition Model may become a bottleneck.

Our research on operational rolling stock planning problems is not completed with the results of this chapter. Concerning the two-phase approach, automating the process of creating the intermediate plan as well as extending the two-phase method for instances with splitting and combining is a subject of further research.

Moreover, the usefulness of any solution method is not only measured by the value of the objective criteria, but also by its flexibility and its actual contribution in speeding up the operational rolling stock planning process. The evaluation of both the Composition Model and the heuristic approach on real-life instances also belongs to our future plans.

Chapter 6

Conclusions and Future Work

In this thesis we considered various rolling stock planning problems that arise at the major Dutch passenger railway operator NSR. In Chapter 2, we gave an overview of the planning process. This describes the context for the particular planning problems studied in this thesis. The subsequent chapters are devoted to tactical, short-term and operational rolling stock planning problems.

6.1 Main Results

Tactical Rolling Stock Circulations

In Chapter 3 we studied the tactical rolling stock circulation problem where the available rolling stock is to be assigned to the trains such that various technical and market requirements are fulfilled. The objective is to find a good balance of efficiency, service quality and robustness.

We formulated the Composition Model which is a mixed integer programming model. Extensive computational tests show that by using commercial MIP software for solving it, the model provides good solutions within a couple of hours even for the most difficult instances of NSR; usual instances are solved to optimality within minutes. Planners at NSR agree that the solutions of this model are in any respect better than the manually created rolling stock circulations. From 2004 on, the Composition Model is used at NSR to specify the basic shape of the rolling stock plan. This allowed to decrease the number of carriage-kilometres significantly and thereby to reduce the annual rolling stock costs of NSR by a couple of millions of euros. As an additional benefit, tactical rolling stock planning at NSR became faster and

smoother. It is possible now to work out and compare rolling stock circulations with different characteristics when looking for the best balance of the objective criteria.

We also formulated the Job Model which is an alternative approach for tactical rolling stock circulations. This model has the drawback that it cannot handle splitting and combining of trains, a feature that occurs in some real-life instances. Also, computational tests revealed that the Composition Model clearly outperforms the Job Model.

We can conclude that operations research techniques can be used to model tactical rolling stock circulations. In particular, the Composition Model has proved its usefulness in practise.

Maintenance Routing

In Chapter 4 we described the maintenance routing problem of NSR. It arises in short-term planning, a couple of days before the rolling stock plans are to be carried out. Units that travelled a high number of kilometres since their last preventive maintenance check must be routed to the maintenance facilities by modifying the rolling stock plan of the forthcoming couple of days.

We formulated two multi-commodity flow type models for maintenance routing: the Interchange Model and the Transition Model. The Interchange Model needs a vast amount of input data about the shunting process which may not be available in practise; the Transition Model is a much simpler formulation. When one chooses appropriate objective functions, the Transition Model is a relaxation of the Interchange Model.

We proved the following complexity results. The feasibility problem of the Interchange Model is NP-complete if the number of urgent units is part of the input. With mild assumptions about the input, the optimisation version of the Interchange Model can be solved in polynomial time in the case of a single urgent unit. The feasibility of the Transition Model with a single urgent unit can be checked in polynomial time in graphs that arise in railway applications. However, the optimisation version of the Transition Model in such graphs is NP-hard.

We proposed a heuristic solution approach for the Interchange Model. In our computational experiments it turned out that the heuristic approach is able to solve all test instances. Easy-to-compute lower bounds verify that the heuristic solutions have a small absolute optimality gap. The integer programming formulation of the Interchange Model and the Transition Model can also be solved by commercial MIP solvers in reasonable time. When solving both models on the same instances, the optimal solution values of the two models did not differ much. This indicates that

the simpler Transition Model underestimates the solution costs of the Interchange Model only slightly.

We discussed a number of our solutions with maintenance routing planners; they found them satisfactory. Thus we can conclude that it is plausible to use any of the Interchange Model and the Transition Model in a decision support system for maintenance routing. Investigating the actual usefulness of the models in practise is a subject of further research.

Operational Planning: Rebalancing

In Chapter 5 we discussed operational rolling stock circulations. Operational planning takes the tactical rolling stock schedules as input and adjusts them for the forthcoming couple of weeks. The problem formulation is very close to that in tactical planning, but the objective differs. The main goal is that the operational plans (including the shunting plans) are accomplished as quickly as possible. Currently, a two-phase approach is applied at NSR. The first phase sets up an intermediate plan which may violate inventory specifications. In the second phase, these off-balances are resolved.

In this thesis we only focused on the second phase, called rebalancing. We proved that the feasibility of the rebalancing problem is NP-complete even for an off-balance of only one unit. Then we proposed an iterative heuristic rebalancing algorithm.

The Composition Model can easily be adapted for the rebalancing problem. So we considered rebalancing instances of NSR, formulated them as instances of the Composition Model and solved them using commercial MIP software. Also, we ran the heuristic algorithm on these test instances and compared the results of the two methods.

It turned out that on these instances the Composition Model can be solved to optimality within a couple of minutes. Despite its greedy character, for most test instances the heuristic approach is able to resolve the same number of off-balances as the Composition Model or just one less. We can conclude that it is plausible to apply the heuristic algorithm to the rebalancing problem.

Real-life operational planning instances can be significantly larger and more complex than the test instances we considered. The heuristic approach has the advantage that its running time is moderate, even on huge instances. Further research is needed, however, to examine the behaviour of both methods on large instances: the solution *time* of the Composition Model and the solution *quality* of the heuristic approach.

6.2 Future Work

The results of this thesis indicate that the models and solution methods have the potential to become the core of useful planning tools. However, many questions remain open, further investigations are necessary to answer them.

We have seen that the Composition Model is able to handle each tactical rolling stock circulation instance of NSR. However, these instances are obtained by decomposing the railway network into line groups and by considering one day only. A natural question is whether instances with several line groups on several consecutive days are still tractable and how much one can gain by planning the rolling stock in that way.

The practical value of the maintenance routing models has still to be verified. So far, we only had feedback from maintenance routing planners. Future research has to involve local planners into the evaluation of the models in order to decide whether the output of the models can be carried out in practise. In addition, systematic comparison of the output of our models with solutions of maintenance routing planners is needed to refine the models and to get an objective function that reflects reality as well as possible.

Our research on operational rolling stock planning has just started. In the context of the two-phase method of NSR, the first phase still needs to be addressed. Then the fully automated two-phase method is to be tested on real-life instances. Investigating the performance of the Composition Model on real-life operational planning problems also belongs to our future plans.

Besides the three rolling stock planning problems that we studied in this thesis, many other ones still wait for computer-aided tools. For example, there are barely tools for short-term planning, in particular for disruption management; this is quite in contrast with successful tactical planning applications and even with on-going research on operational planning. Further research has to explore solution methods for such railway problems.

So far we discussed results and directions for future work mainly from the operations research's point of view. We evaluated the solution quality and solving time of the models on real-life instances although we noted that the contribution of most of our models to railway planning still needs to be investigated. But even when the solutions of the models turn out to be the solutions of the railway problems, this is just the first step to build decision support tools. The solution methods have to be integrated into the planning process, this raises questions that reach beyond operations research.

Unified databases have to be built up so that the input of the models can be gathered automatically. This is especially important in short-term planning where the input has to reflect the very latest state of the railway system. The output of the models has to be evaluated by human planners. User-friendly interfaces are required to be able to interpret the output easily. A well-designed user interface can help operations research methods to gain acceptance in everyday railway planning.

Once these tasks are accomplished, the railway operators can take full benefit of the planning tools built upon operations research models. The case of tactical rolling stock circulations proves that there is a lot to gain by using such tools, not only financially but also in the flexibility and the speed of the planning process. We look forward to see a similarly successful application of our other models in railway planning in the near future.

Appendix A

Glossary

In this appendix we provide our understanding of railway terminology. The explanations below are intended to be short reminders. The terms are described more precisely in the previous sections. The index helps finding these definitions.

Unit: Two to six railway carriages attached to each other, supplied with own engines and having driver's cabin at both ends. Units cannot be split up during everyday operations.

Type: A group of *units* that have identical technical parameters and seat capacities. Physical units of the same type are only distinguished by their mileage.

Composition: *Units* attached to each other in a given order. In short-term planning, the members of a composition are the physical units. In tactical and operational planning, compositions are specified only by their type patterns.

Shunting: Movements of *units* inside the stations. Shunting also includes putting temporarily not used units to storage tracks and adjusting train lengths between an arrival and the subsequent departure by *coupling* or *uncoupling* some units.

Train: A movement of *units* that makes use of tracks between different stations. A train is characterised by the train number, the departure and arrival locations, the route through the railway infrastructure, the departure and arrival times and usually with the departure and arrival platforms.

Timetable: The description of all *trains*.

Timetable service: A *train* that may carry passengers.

Empty train: A *train* that does not carry passengers, used for repositioning *units* from one location to another.

Successor train of an arriving *train*: planned to be carried out by the *units* that served in the arriving train. Yet, minor composition changes may be possible. Usually, trains have zero or one successor. In case of *splitting*, a train has two successors. Note that the notion of successors also applies to *trips* and *tasks*.

Predecessor train: the opposite of successors. Note that the notion of predecessors also applies to *trips* and *tasks*.

Turn-around: the pair of events formed by the arrival of a *train* and the departure of its *successor train* together with the intermediate occupation of a platform track.

Coupling: Attaching *units* from the *inventory* to a departing *train*.

Uncoupling: Detaching *units* from an arriving *train* and adding them to the *inventory*.

Composition change: *Coupling* or *uncoupling units* during a *turn-around*.

Inventory: The number of *units* per *type* that are stored at a station at a given moment. The inventory contains only units that are ready to be used in any departing *train*. That is, we do not count those units that depart from a station soon after the arrival.

Splitting of *trains* occurs when an arriving train has two *successors*.

Combining of *trains* occurs when a departing train has two *predecessors*.

Train line: A series of *timetable services* that call at given stations. Train lines are operated with a given frequency, e.g. twice an hour.

Line group: A system of interconnected *train lines*. Most line groups have (almost) closed *rolling stock circulations*. That is, the daily workload of most *units* consists of train movements that belong to the same line group.

Trip: A sequence of train movements that has to be carried out without composition changes. Trips are the basic objects in tactical and operational planning. A timetable service that admits composition changes underway consists of several trips. The notion of *successor* and *predecessor trains* applies also to trips.

Tasks: The smallest indivisible piece of work to be carried out by a single *unit*. A task is characterised by a *trip* and a position in the *composition* that is assigned to the trip. The notion of *successor* and *predecessor trains* applies also to tasks.

Duty: The workload of an employee or of a *unit* for a certain period, usually a day. The duty of a rolling stock unit is a sequence of *tasks*.

Rolling stock circulation: The collection of the rolling stock *duties* for a certain period, e.g. for a day. It assigns *units* to the *trains* and describes the order of the units in the *compositions*. Rolling stock circulations of consecutive days must fit to each other: at each station, the *inventory* at the end of a day must be equal to the *inventory* at the beginning of the next day.

Appendix B

b -Transshipments

In this appendix we define the notion of b -transshipments and mention some important properties. The definitions and theorems below are well-known in the theory of network flows. For more details, we refer to Schrijver (2003). In this thesis we only consider integer-valued b -transshipments, therefore we restrict ourselves to the integer-valued cases.

B.1 Definition of a b -Transshipment

Let $G = (V, E)$ be a directed graph, let $f: E \rightarrow \mathbb{Z} \cup \{-\infty\}$ and let $g: E \rightarrow \mathbb{Z} \cup \{\infty\}$ be lower and upper capacity functions with $f(e) \leq g(e)$ for every $e \in E$. Let $b: V \rightarrow \mathbb{Z}$ with $\sum_{v \in V} b(v) = 0$.

Definition B.1. *The function $x: E \rightarrow \mathbb{Z}$ is a b -transshipment if the following two requirements hold:*

$$f(a) \leq x(a) \leq g(a) \quad \forall a \in E, \quad (\text{B.1})$$

$$\sum_{a \in \delta^{\text{out}}_v} x(a) - \sum_{a \in \delta^{\text{in}}_v} x(a) = b(v) \quad \forall v \in V. \quad (\text{B.2})$$

The latter constraint expresses that, using the terminology of network flows, the net out-flow of each node v is equal to $b(v)$.

For example, a network flow with one unit of flow from the source s to the sink t and with arc capacities $g: E \rightarrow \mathbb{Z} \cup \{\infty\}$ is a b -transshipment by setting $f(e) = 0$

for each arc e and by defining

$$b(v) = \begin{cases} 1 & \text{if } v = s, \\ -1 & \text{if } v = t, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{B.3})$$

In general, a b -transshipment can be considered as a multiple-source multiple-sink network flow problem: The nodes with $b(v) > 0$ are the sources with out-flow $b(v)$, while the nodes with $b(v) < 0$ are the sinks with in-flow $-b(v)$. This construction allows to carry over many results of network flows to b -transshipments.

B.2 Auxiliary Graph

The auxiliary graph $\widehat{G} = (V, \widehat{E})$ associated with a b -transshipment x is defined as follows. The graph $\widehat{G}$ has the same node set as the graph G . For $u, v \in V$, graph $\widehat{G}$ has a *forward arc* uv if $uv \in E$ and $x(uv) < g(uv)$. That is, if the value of x can be increased on arc uv without violating the upper capacities. For $u, v \in V$, graph $\widehat{G}$ has a *backward arc* vu if $uv \in E$ and $x(uv) > f(uv)$. That is, if the value of x can be decreased on arc uv without violating the lower capacities.

For any subset $F \subseteq \widehat{E}$, we define the function $\tilde{\chi}^F: E \rightarrow \mathbb{Z}$ as follows. For each arc $uv \in E$, let

$$\tilde{\chi}^F(uv) = \begin{cases} 1 & \text{if } uv \in \widehat{E} \text{ and } vu \notin \widehat{E}, \\ -1 & \text{if } vu \in \widehat{E} \text{ and } uv \notin \widehat{E}, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{B.4})$$

B.3 Results on b -Transshipments

In this thesis we use the following facts about b -transshipments. Let $s, t \in V$ and define $b': V \rightarrow \mathbb{Z}$ as

$$b'(v) = \begin{cases} 1 & \text{if } v = s, \\ -1 & \text{if } v = t, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{B.5})$$

Theorem B.2. *Let x be an integral b -transshipment and let P be a directed $s-t$ -path in the auxiliary graph $\widehat{G}$. Then*

$$x' = x + \tilde{\chi}^P \quad (\text{B.6})$$

is a b' -transshipment.

Theorem B.3. *Let x be an integral b -transshipment and let x' be an integral b' -transshipment. Then the auxiliary graph $\widehat{G}$ contains a directed $s - t$ -path P and directed circuits C_i such that*

$$x' = x + \tilde{\chi}^P + \sum_i \tilde{\chi}^{C_i}. \quad (\text{B.7})$$

Corollary B.4. *Let x be an integral b -transshipment and suppose that there exists a b' -transshipment. Then the auxiliary graph $\widehat{G}$ contains a directed $s - t$ -path.*

Appendix C

Notations

This appendix gathers all symbols used as parameters and variables for the models in this thesis.

C.1 Notations in Chapter 3

Parameters for Both Models

$\mathcal{M}$	the set of rolling stock types
n_m	number of available units of type m
c_m	number of carriages in a unit of type m
ℓ_m	length of a unit of type m (in metres)
$\mathcal{C}$	the set of service classes
$\kappa_{m,c}$	seat capacity of a unit of type m for service class c
$\mathcal{S}$	set of stations
$i_{s,m}^0$	wished initial inventory of station s in type m
$i_{s,m}^\infty$	wished final inventory of station s in type m
W_s	storage capacity of station s (in metres)
$\mathcal{T}$	the set of trips
$s_d(t)$	departure station of trip t
$\tau_d(t)$	departure time of trip t
$s_a(t)$	arrival station of trip t
$\tau_a(t)$	arrival time of trip t
d_t	length of trip t (in kilometres)
$\varrho(t)$	re-allocation time after trip t

$\mu_t^{\min}$	minimum number of carriages to be assigned to trip t
$\mu_t^{\max}$	maximum number of carriages to be assigned to trip t
$\delta_{t,c}$	passenger demand on trip t for service class c
$\sigma(t)$	successor of trip t
$\mathcal{T}_0$	the set of trips without predecessors
$\mathcal{T}_\infty$	the set of trips without predecessors
$\sigma^1(t), \sigma^2(t)$	successors of trip t in case of splitting
$\mathcal{T}^s$	the set of trips the are split
$\mathcal{T}^c$	the set of trips the are combined

Specific Parameters for the Composition Model

$ p $	total number of units in composition p
$\nu(p)_m$	number of units of type m in composition p
$\mathcal{P}_t$	the set of compositions that can be assigned to trip t
$\mathcal{B}_t$	the set of vectors $\nu(p)$ that can be assigned to trip t
$\mathcal{G}_t$	the set of allowed composition changes after trip t
$\mathcal{G}^s$	the set of allowed composition changes when trip t is split
$\mathcal{G}^c$	the set of allowed composition changes when trip t is combined

Specific Parameters for the Job Model

$\mathcal{J}$	the set of jobs
$\mathcal{R}_j$	the set of trips covered by job j
φ_j	first trip covered by job j
φ_j	last trip covered by job j
$s_d(j)$	departure station of job j
$s_d(j)$	time station of job j
$\tau_a(j)$	arrival station of job j
$\tau_a(j)$	arrival time of job j
d_j	length of job j (in kilometres)
$\hat{\mu}_j$	upper bound on the number of units that can be assigned to job j
μ_j	upper bound on the number of carriages that can be assigned to job j
$\gamma_d(t)$	coupling side of φ_j
$\gamma_a(t)$	uncoupling side of λ_j

Variables for the Composition Model

$X_{t,p}$	whether composition p is assigned to trip t
$Z_{t,p,p'}$	whether composition change $p \rightarrow p'$ occurs after trip t
$N_{t,m}$	number of units of type m assigned to trip t
$C_{t,m}$	number of units of type m coupled to trip t
$U_{t,m}$	number of units of type m uncoupled from trip t
$I_{s,m}^0$	initial inventory of station s in type m
$I_{s,m}^\infty$	final inventory of station s in type m
$I_{t,m}$	inventory of station $s_d(t)$ in type m right after trip t departs
$Y_{t,b}$	whether a combination $b \in \mathcal{B}_t$ is assigned to trip t
CKM	carriage-kilometres
SKM	seat-shortage kilometres
CCH	the number of composition changes
Z_{t,p,p_1,p_2}^s	whether trip t with composition p is split into compositions p_1 and p_2
Z_{t,p,p_1,p_2}^c	whether trip t with composition p is combined from compositions p_1 and p_2
$Z_{t,p,p_1,\dots,p_6}^{\text{Ut}}$	whether splitting-combining scenario $(p, p_1, p_2, p_3, p_4, p_5, p_6)$ is selected after trip t
C^{Gvc}	whether any unit is coupled in The Hague in the selected splitting-combining scenario
C^{Rtd}	whether any unit is coupled in Rotterdam in the selected splitting-combining scenario
U^{Gvc}	whether any unit is uncoupled in The Hague in the selected splitting-combining scenario
U^{Rtd}	whether any unit is uncoupled in Rotterdam in the selected splitting-combining scenario

Variables for the Composition Model

X_j	whether job j is selected
$N_{j,m}$	number of units of type m assigned to job j
H_t	whether a composition change takes place right before trip t
$I_{s,m}^0$	initial inventory of station s in type m
$I_{s,m}^\infty$	final inventory of station s in type m
$I_{j,m}$	inventory of station $s_d(j)$ in type m right after job j departs
$S_{t,c}$	seat-shortage of service class c on trip t

C.2 Notations in Chapter 4

Parameters for Both Models

$s_d(t)$	departure station of task t
$s_t(t)$	time station of task t
$\tau_a(t)$	arrival station of task t
$\tau_a(t)$	arrival time of task t
$\sigma(t)$	successor of task t
ϱ	buffer time
U	the set of urgent nodes
M	the set of maintenance nodes
M_u	the set of maintenance nodes that can be assigned to urgent node u

Parameters for the Interchange Model

D	number of available units
H	length of the planning horizon (in minutes)
$\mathcal{J}$	the set of interchanges
V_0	the set of first nodes
V_∞	the set of last nodes
$\text{stat}(v)$	station corresponding to node v of the graph representation
$\text{time}(v)$	time corresponding to node v of the graph representation
$\text{tail}(a)$	the tail node of a directed arc a
$\text{head}(a)$	the head node of a directed arc a
$\text{weight}(a)$	weight of an arc

Specific Parameters for the Transition Model

V_0	the set of source nodes of the acyclic graph
V_∞	the set of sink nodes of the acyclic graph
c	weight of a transition
$\tilde{c}$	weight of a transition derived from the Interchange Model

Variables for the Interchange Model

y_C	whether interchange C is selected
$z(a)$	network flow that indicates the paths of all units through the entire planning horizon
$x_u(a)$	a network flow that indicates the path of urgent unit u to a maintenance task

Variables for the Transition Model

$z(a)$	network flow that indicates the paths of all units through the entire planning horizon
$x_u(a)$	a network flow that indicates the path of urgent unit u to a maintenance task

C.3 Notations in Chapter 5

$s_d(r)$	departure station of trip r
$s_t(r)$	time station of trip r
$\tau_a(r)$	arrival station of trip r
$\tau_t(r)$	arrival time of trip r
$\varrho(r)$	re-allocation time after trip r
$\mu_r^{\min}$	minimum number of carriages to be assigned to trip r
$\mu_r^{\max}$	maximum number of carriages to be assigned to trip r
$\lambda(r)$	an extra conductor is needed for trip r if it is assigned more than $\lambda(r)$ carriages
μ_r	maximum number of units of type m that can be added to trip r without exceeding the maximal train length $\mu_r^{\max}$
L_r	the number of carriages in those units on trip r whose type differs from the given type m

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Samenvatting

Hoge servicekwaliteit en efficiency van moderne spoorwegbedrijven zijn onvoorstelbaar zonder zorgvuldige planning. De planningsproblemen betreffen de dienstregeling, het materieel en het personeel. Zij kunnen daarnaast ook op basis van hun planningshorizon geclassificeerd worden. Bij *strategische planning* is de lengte van de horizon enkele jaren, bij *tactische planning* een aantal maanden, en bij *operationele planning* een aantal weken. Daarnaast is er de *korte termijn planning*, dit betreft ook bijsturing en de planningshorizon is enkele dagen.

Dit proefschrift richt zich op tactische, operationele en korte termijn materieelplanningsproblemen van reizigersvervoerders op het spoor. Het doel van dit onderzoek is relevante planningsproblemen te identificeren en deze met wiskundige modellen te beschrijven, die vervolgens te analyseren om uiteindelijk oplosmethoden te ontwikkelen. Daarnaast is het doel te evalueren in hoeverre de methoden in de praktijk gebruikt kunnen worden. In dit onderzoek werden de methoden getest op real-life instanties van NS Reizigers (NSR). De modellen zijn dus speciaal ontwikkeld voor de planningsproblemen van NSR. Desondanks zijn wij van mening dat de in dit proefschrift beschreven methoden ook bij andere vervoerbedrijven toegepast kunnen worden.

Hoofdstuk 1 beschrijft het onderwerp van dit proefschrift en formuleert de onderzoeksvragen. In Hoofdstuk 2 wordt het planningsproces van NSR in detail bekeken en wat de positie en relevantie is van de in dit proefschrift behandelde materieelplanningsproblemen in het kader van dat planningsproces. Tegelijkertijd wordt een overzicht gegeven van de bestaande literatuur over spoorwegoptimalisatie.

In Hoofdstuk 3 worden materieelomlopen in de tactische planning onderzocht. De materieelinzet voor een generieke week wordt bepaald, de herhaling van dit basisplan geeft de materieelomloop voor de volgende maanden. Aan iedere trein wordt een aantal treinstellen in een bepaalde volgorde toegewezen. Dit plan geeft ook aan wanneer de samenstelling van een trein gewijzigd moet worden door het aan- of

afkoppelen van treinstellen. Het doel is het vinden van een goed evenwicht tussen operationele kosten, servicekwaliteit en robuustheid.

Er worden twee wiskundige modellen voor dit probleem beschreven en vergeleken. Het “Job Model” kan in redelijke oplostijd alleen voor kleine en eenvoudige instanties gebruikt worden. Het “Composition Model” is een veel efficiëntere formulering; hiermee kan zelfs voor de grootste en meest complexe instanties van NSR binnen een paar uur een oplossing van goede kwaliteit gevonden worden. Voor de meeste kleinere instanties vergt het oplossen zelfs maar enkele minuten. Sinds 2004 wordt dit model bij NSR gebruikt om een deel van de materieelomloop voor generieke weken te bepalen. Daardoor zijn jaarlijks enkele miljoenen euro’s bespaard.

Hoofdstuk 4 is gewijd aan de onderhoudsrouting, een onderdeel van de korte termijn planning. Treinstellen hebben regulier preventief onderhoud nodig. Een treinstel is *urgent* als het binnen drie werkdagen onderhoud behoeft. Het doel van de onderhoudsrouting is de materieelinzet voor de komende drie dagen zodanig te wijzigen dat de urgente treinstellen tijdig de werkplaatsen kunnen bereiken. De wijzigingen vereisen aanpassingen van de rangeerprocessen binnen de stations. Daarom moeten de voorgestelde wijzigingen door het rangeerpersoneel beoordeeld en goedgekeurd worden.

In dit proefschrift worden twee modellen voor de onderhoudsrouting beschreven. Het “Interchange Model” probeert zo veel mogelijk details van de werkelijkheid in acht te nemen, dit verhoogt de kans dat de voorgestelde wijzigingen daadwerkelijk uitgevoerd kunnen worden. Het leidt tot een groot integer multi-commodity flow model. Nadeel van deze aanpak is dat de benodigde invoergegevens moeilijk te verzamelen zijn. Dit motiveert het “Transition Model”, dat een veel simpelere beschrijving van het probleem geeft. Voor beide modellen worden theoretische complexiteitsresultaten afgeleid. Verder wordt een heuristisch algoritme voor het Interchange Model beschreven. Rekenexperimenten op instanties van NSR laten zien dat beide modellen binnen redelijk korte tijd opgelost kunnen worden door commerciële MIP solvers en dat de heuristische aanpak voor het Interchange Model snel goede oplossingen oplevert. Een aantal testinstanties is met planners van NSR besproken en zij vonden de oplossingen van beide modellen daarvoor bevredigend. Desondanks is nader onderzoek nodig om de modellen en hun oplossingen op real-life instanties te evalueren.

In Hoofdstuk 5 wordt de operationele materieelplanning behandeld. Hier wordt de materieelomloop voor de generieke weken aangepast aan de specifieke kalenderweken. Vaak zijn grootschalige aanpassingen van dienstregeling en materieelomloop nodig, met name als een deel van de spoorweginfrastructuur afgesloten is vanwege onder-

houdswerkzaamheden. Dit probleem lijkt sterk op dat van de tactische planning in Hoofdstuk 3, maar de doelstelling is verschillend. Vanwege de planningshorizon van maximaal enkele weken spelen de optimalisatiecriteria uit Hoofdstuk 3 een minder essentiële rol. Het is nu veel belangrijker snel een oplossing te vinden, ook als alles helemaal opnieuw gepland moet worden.

Hier wordt een twee-fase methode voor operationele materieelplanning beschreven waarvan de tweede fase in detail onderzocht wordt. De complexiteit van het probleem wordt geanalyseerd en een heuristisch algoritme wordt ontworpen. Ook worden er rekenexperimenten uitgevoerd. Verder onderzoek moet inzicht geven in hoeverre de oplosmethode daadwerkelijk op praktische instanties toegepast kan worden.

In Hoofdstuk 6 worden de resultaten van het proefschrift samengevat. De belangrijkste conclusie is dat besliskundige methoden inderdaad significant kunnen bijdragen aan een versnelling van het materieelplanningsproces, aan een verlaging van de kosten van de materieelinzet, en aan een verhoging van de servicekwaliteit voor de reizigers. Het “Composition Model” uit Hoofdstuk 3 vormt inmiddels de kern van een onmisbaar planningstool in de tactische planning bij NSR. Ondanks veelbelovende rekenresultaten moet de praktische waarde van de modellen en methoden voor de andere fasen van de materieelplanning echter nog bewezen worden.

Curriculum vitæ

Gábor Maróti was born on March 24th, 1976 in Szombathely, Hungary. He accomplished the basic and secondary school in Szombathely. In 1994 he started his studies in mathematics with additional training as specialised German translator at the Faculty of Sciences, Eötvös Loránd University in Budapest. He graduated in 1999 with his Master's thesis "Submodular functions and their polyhedra" written under the supervision of prof.dr. András Frank. Thereafter he went on as a Ph.D. student at the Department of Operations Research of the Eötvös Loránd University.

In February 2001 he became a Ph.D. student at the Centrum voor Wiskunde en Informatica in Amsterdam under the supervision of prof.dr.ir. Bert Gerards. His position was funded by the European Union's research training network "Algorithmic Methods for Optimizing the Railways in Europe" (AMORE), by NS Reizigers, and the Technische Universiteit Eindhoven. In his research, carried out also in cooperation with his second promotor prof.dr. Leo Kroon, he mainly focused on rolling stock planning problems arising at various stages of the railway planning process.

From October 2005 on he is working at the Rotterdam School of Management, Erasmus University Rotterdam, and since February 2006, he is involved in the European Union's research programme "Algorithms for Robust and Online Railway Optimization: Improving the Validity and Reliability of Large-scale Systems" (ARRIVAL).